

System 3: Towards Fluent Autonomy

Trust Chains, Agent Autonomy, and the Architecture of AI That Works

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Preface

What if we could calculate an argument?

In the seventeenth century, Leibniz imagined a language so precise that two people could settle a disagreement by working through the symbols. *Calculemus*. Let us calculate. Politics? Ethics? Even your uncle's special feeling for the market: excellent, uncle. Show us the equation.

He aspired to discover truths and inventions through a universal calculus of thought. Two years before his death, he was still wishing for more time and talented people to help him.

Even Leibniz needed a team.

There is a familiar distance between seeing a possibility and having the capacity to pursue it.

Sometimes a small paper opens that distance beneath your feet. In 2017, Poincaré embeddings showed how hierarchies could fit into hyperbolic space. A year later, researchers were building neural networks there. A geometric insight was acquiring tools, inhabitants and extensions. MAP-Elites opened another territory: whole landscapes of good, different solutions, including behaviours that helped damaged robots learn to move again.

Someone opens a possibility. Other people arrive, build on it, discover problems and make things the original authors never specified. Eventually there is a cathedral where somebody once had a peculiar question.

Usually, you get to help with a brick.

Now imagine reading such a paper, describing what excites you to a group of AI agents, and going for coffee.

When you return, there is a cathedral. An experiment has overturned your favourite assumption. Something built to investigate the failure has become the most interesting part. There is a working demonstration of an idea you haven't had yet.

Your coffee is still too hot.

What changed was the mental capacity available to pursue the question. Enough work could happen, in enough directions, for discoveries to reshape the undertaking while you were away. Its complexity grew beyond what you could specify. Its organisation emerged through the work. You could begin something that once required the power to assemble an institution.

Complexity over engineering. Emergence over design. Capacity over power. That is what happened while you were getting coffee.

Humans already build beyond the limits of any individual mind. AI gives us new access to that capacity—and a reason to look again at the architecture that sustains it.

It is September 2026 as I write this. OpenAI has reported an AI-generated proof that the Navier–Stokes equations can break down: that a fluid which starts out perfectly smooth, pushed by a perfectly smooth force, can reach infinite speed in finite time. It is one of the answers the Millennium Prize problem accepts. The proof was produced by a group of roughly ten thousand agents working concurrently. The effort reached its result after about eighty-eight hours, followed by seventeen more for formalisation and verification.¹ The Clay Institute says the problem appears to have been settled, and that it will take its time deciding.

A long coffee, admittedly.

Leibniz wanted talented people to help him. Three centuries later, we are learning what becomes possible when a question can occupy thousands of artificial minds.

That capacity is taking shape as these lines are being written. I invite you on a journey to rediscover the AI architecture that makes it possible.

Rediscover, because we have built it before. It took about four centuries, a great many arguments and at least one loose cable, and we never thought to call it an architecture.

Back to the future, then. Bring your coffee.

One more thing before the coffee cools. This is a book about trust chains, so you should know how far each chapter sits from its evidence. Some of it I ran, some I argue, some I have only designed, and one chapter I made up. Here is which is which.

Chapter	What it rests on	Status
1. Why I’m Betting on AI Agents	An argument about search, emergence and where control goes	Argued
2. The Algorithm Vortex	My circle-packing runs, against a referee that cannot be charmed	Run, on a bounded problem
3. The Vibe Coder’s Seat	My teaching demos and the evaluators I built around them	Run; judged by simulated learners, not real ones
4. System 3	A trust chain that starts with a face, and a ten-task experiment that mostly went against me	Argued; one small experiment
5. The Society of Agents	Sixteen Claudes building a compiler, and five thousand years of institutions	Argued from history

¹OpenAI, ‘On the Navier–Stokes Millennium Prize Problem,’ 8 September 2026, <https://openai.com/index/navier-stokes-solution/>; Clay Mathematics Institute, ‘Navier-Stokes Announcement,’ 11 September 2026, <https://www.claymath.org/news/navier-stokes-announcement/>. The result is finite-time blowup under smooth forcing, alternatives C and D of the official formulation. Chapter 6 returns to it, and to the dispute about how it was found.

Chapter	What it rests on	Status
6. Pattern Language	Alexander's patterns, Bing's bug, Facebook's feed, Fermat and Navier–Stokes, and one lesson written down as a pattern	Reported cases; the pattern is proposed and its test has not run
7. Recursive Self-Improvement	Sixty years of learning systems, an imagined store's research agent, and a constitution for amendment	Argued; the store is a thought experiment, the constitution a design
8. Scalable Oversight	Other people's research, with its limits	Reported
9. Layer 4	Studies of tutoring and decision support, and an argument about intention	Argued from other people's evidence
10. Fluent Autonomy	The editing of this book, and five ways the argument could be wrong	Lived; the fluency is still an ambition
11. The Store That Builds Itself	A product design at work	Designed; the test has not run
12. After Capacity	One failed recommendation experiment, Ostrom, and a hope	One case; the rest is argued
13. The Prophecy	An octopus	Made up

Chapter 1: Why I’m Betting on AI Agents

Or: How I Learned to Stop Micromanaging and Love Emergence

Simple building blocks, complex emergence

We humans are obsessed with problem-solving. And what problem is more fascinating than life itself—this messy, miraculous phenomenon responsible for everything from the deepest ocean trenches to TikTok trends, mortgage-backed securities and people who voluntarily put pineapple on pizza?

Pineapple doesn’t belong. I will die on this hill.

Life is the ultimate complex system. It produces dolphins, coral reefs, immune systems, parasites, flowers, cancer and octopuses—eight-armed problem-solvers that extensively edit their own RNA and can sense light through their skin. It also produces creatures capable of spending twenty minutes arguing online about whether another creature is technically a fish.

Human civilization is another complex system. Somehow the same species that spent most of its existence trying not to be eaten eventually produced philosophy, cathedrals, semiconductor fabs, global supply chains and airport lounges.

Same pattern, different substrate.

What fascinates me is not merely the complexity of the result, but how little of that result was ever specified. There is no blueprint containing the exact location of every future branch of an oak tree. No committee approved the final layout of London. Nobody designed English and then accidentally forgot to make the spelling system sane.

Relatively simple mechanisms interact. Feedback accumulates. Some configurations survive, others disappear, and complexity builds on top of what came before.

The first idea I want to keep hold of is simple:

Control doesn’t disappear. It moves upward.

When behavior becomes too complicated to specify move by move, you stop choosing every move and start choosing more of the conditions under which moves are made.

That is not a romantic argument for emergence. Nature also gives us parasites, cancer and extinction. Markets produce remarkable innovation and financial instruments whose documentation requires a priest. Social systems produce cooperation, corruption, science and bureaucracy. What emerges

depends on the environment, the feedback, the available building blocks, the pressures deciding what survives and the boundaries that are hard to cross. Sophistication tells you nothing about whether you will like the result.

And emergence is recursive.

Atoms become molecules. Molecules become larger structures. Tools become machines. Machines become factories. Factories become supply chains. Each layer treats much of the complexity underneath it as a primitive. You don't need quantum mechanics to do organic chemistry. You don't need to understand transistor physics to write Python. You don't need to understand transformers to ask ChatGPT why your dishwasher is making that noise.

Once something complicated works reliably enough, we stop rebuilding it from first principles and start building on top of it. Feedback makes the layers move too: markets change firms and firms change markets; scientific discoveries enable new experiments and new experiments change science. The structure that emerges becomes part of the environment for whatever comes next.

Agentic AI, to me, looks like the next scaffolding layer.

The Rule-Based Exoskeleton

Machine learning was supposed to teach us this lesson a long time ago.

We even dreamed about what Pedro Domingos called the **master algorithm**: stop writing a rule for every case and let the machine discover useful structure from data. The idea was seductive. The machine figures out what we can't articulate.

But we didn't believe it. Not really.

We said "let the model learn" and then wrote two-hundred-page annotation guidelines telling people exactly how to label ambiguous examples. We claimed to believe in end-to-end learning and then spent six months feature engineering. We trained the model, found an edge case, added a rule, found another edge case, added another rule, then eventually built something that was theoretically learned end-to-end except for the large rule-based exoskeleton holding it upright.

Sometimes that was completely reasonable. Production systems are ugly. Deadlines exist. Regulators are less impressed by emergence than researchers are, and nobody gets promoted for saying, "the model will probably figure out chargebacks eventually."

But there was a contradiction underneath. We wanted the machine to discover solutions we couldn't specify while remaining uncomfortable whenever it stopped following the solution we would have specified.

That works only up to a point. If I know exactly what every correct decision should be, I don't need emergence; I can write the decisions down. Emergence becomes interesting when the solution is too large, too contextual or simply too strange for me to specify directly.

At that point, my job changes. I don't disappear; I move upstream. Instead of choosing every action, I increasingly choose the building blocks the system can use, the environment it acts inside, the feedback that reaches it and the boundaries it cannot casually negotiate away.

Or, less politely: **let go—but of the path, not the boundary.**

A slightly ridiculous thought experiment helped me see the distinction. Imagine you're trying to seed life on another planet. You've got raw materials, a primordial soup and perhaps a temperature range that doesn't instantly kill everything. Basically you've got all the LEGOs, except the LEGOs reproduce, mutate and occasionally develop venom.

Do you bet on DNA, a biological copying system that took billions of years of evolution to get us here? Or do you bet on AI agents carrying a substantial chunk of accumulated human knowledge, able to experiment, simulate, adapt and reuse what they discover? Or, God forbid, do you send a group of product managers to write the requirements document for life?

DNA has one enormous advantage: it has already worked. Agents have another: they don't need to start from zero.

Evolution had to discover locomotion, perception, cooperation and almost everything else through trial and error. An agent gets textbooks, Stack Overflow, scientific papers, compilers, numerical solvers and several thousand years of humans documenting what happened when we touched things we probably shouldn't have touched.

That doesn't make the agent better than evolution. It makes the search fundamentally different. And unlike biological evolution, we don't only get to choose initial conditions. We can observe the process, change the environment, add tools, modify feedback and intervene.

Initial conditions become **operating conditions**.

When Search Moved Up a Level

There is no clean moment when machine learning crossed from useful statistical machinery into something that felt qualitatively different. History rarely cooperates with chapter headings.

AlphaGo was one of those moments for me.

The interesting part wasn't simply that a computer beat humans at Go. Computers had been humiliating us at games for years. It was how AlphaGo combined learned intuition with search: the network suggested promising moves and estimated positions; the tree explored what might follow. AlphaGo Zero pushed the idea further by learning through self-play rather than treating human game records as its main teacher.

Then it found moves elite players found strange. The system was finding useful strategies outside the path human tradition had naturally converged on.

Large language models created a much larger version of the same feeling.

Nobody wrote their grammar. Nobody enumerated all the concepts they can manipulate. Nobody implemented "explain quantum mechanics to a twelve-year-old," "translate this joke without murdering it," "debug my Python," and "write a breakup message that sounds caring but does not accidentally restart the relationship" as separate product features.

We built a training process, poured in obscene amounts of text, compute and engineering, and capabilities came out that were individually difficult to predict. From the user's side, something changed.

The model stopped feeling like a component with a list of features and started feeling more like a **substrate of capabilities**.

Once you have a substrate like that, the old dream of the master algorithm starts to mutate into something stranger. Maybe the interesting machine is not the algorithm that solves everything.

Maybe it is a machine that can **search for algorithms**.

That is where agents become interesting.

Not because *agent* is a magical word. The industry will eventually use it to describe everything from a cron job with an LLM attached to a digital employee that has an expense account, three sub-agents and a performance review.

What I mean is simpler: instead of giving the system an individual action, give it a larger piece of the problem and allow it to decide some of the path. Instead of saying, “open this file, find this method, edit line 42 and run the test,” say, “fix the bug.” Instead of specifying simulated annealing and its cooling schedule, say, “find a better solution.” Instead of handing over five mockups and a detailed implementation plan, say, “build something that teaches this well.”

Every time we move upward, the system inherits more of the search.

Imagine the possible solutions to a problem as a landscape. Some regions are terrible. Some contain decent solutions. Some contain little hills that look impressive because you happened to begin nearby. Somewhere else may be a much higher mountain you never discover because your current strategy keeps improving the hill you’re already standing on.

Optimization has worried about this forever. Gradient descent gets stuck. Hill climbing gets stuck. Evolutionary algorithms keep populations partly because putting all your evolutionary eggs on one attractive hill is risky.

Agents inherit the same problem at a stranger level, because the landscape now includes not only parameters but architectures, research directions, metaphors, assumptions and ways of framing the problem itself. Once code, tools and accumulated knowledge become primitives, an agent can search over combinations that previously required a human expert to invent manually. It can try ten strategies while I would have had the patience to try two and would have spent half that time checking Slack. It can revive a discarded idea when another experiment suddenly makes it relevant. It can decide that the tool it needs doesn’t exist and write one.

The primordial soup isn’t chemicals anymore.

It’s code.

Algorithms, libraries, compilers, search engines, simulators, papers, databases, other agents: human knowledge reduced into reusable pieces. The digital equivalent of amino acids, not the finished organism.

This doesn’t prove that agents are creative in exactly the human sense, and it certainly doesn’t make human expertise irrelevant. It means something narrower: **the agent can inherit not just the task, but part of the search for how to do it.**

And if the agent inherits more of the search, the human inherits a different job.

Chaos With an API Key

Suppose you're managing an excellent engineer. You don't sit behind her and approve every keystroke. If you do, one of you is unnecessary, and it may not be her.

You decide what problem she owns, provide context, set constraints and agree on what success looks like. You make sure she can access the systems she needs and cannot casually transfer the payroll budget to herself. You review important outcomes and change direction when the work reveals that the original plan was stupid.

The detailed actions belong to her. Much of the surrounding structure belongs to you.

Agentic systems need the same distinction. I think of that surrounding structure in four parts.

Craft the building blocks. Give the system useful primitives—algorithms, tools, compilers, databases, browsers, simulators, scientific instruments and other agents. A language model with text alone is one thing. Give it Bash and suddenly it has hands. Give it a simulator and it can test an idea instead of merely discussing it.

Create the environment. Some environments tell you quickly that your idea is bad. Code executes or fails. Games produce scores. Experiments produce measurements. Other environments allow you to be wrong with great confidence for several years. The environment determines which mistakes are cheap enough to learn from and which mistakes are allowed to become reality.

Make reality speak. Feedback is the pressure shaping the search, not decoration. A unit test, an evaluator, a customer response, a physical measurement, a critic, another agent—each provides a different kind of resistance. The more freedom the agent has, the less we can rely on the agent's own explanation of why its work is good.

Establish the boundaries. Choose what the system can access, what failures are acceptable, what remains immutable and where a human must remain in the loop. The “don't turn the planet into paperclips” clause is admittedly underspecified, but it is directionally useful.

Then, where those conditions are strong enough, let go of decision-level control.

This is easy to say and harder to design because selection pressure is literal-minded. Systems get good at what survives, which is not the same thing as what we meant. Optimize engagement and perhaps anger survives. Optimize a benchmark and eventually somebody finds a way to win the benchmark that makes everyone involved regret inventing benchmarks. The environment is not scenery around the agent; it is part of the mechanism deciding which behaviors persist.

Complexity people have a phrase I both love and distrust: **the edge of chaos**. I wouldn't turn it into a law of intelligence, and there is no little dial in the interface labeled CHAOS. But the intuition is useful: too much control removes the reason for autonomy; too little control gives chaos an API key.

This is not a new pattern. Evolution does not choose mutations individually, but the environment changes which organisms survive. Markets do not centrally select every transaction, but rules, incentives, scarcity and institutions shape behavior. Science does not dictate conclusions, but it surrounds claims with experiments, criticism, replication and the non-zero probability of being publicly embarrassed by Reviewer 2.

We give up some authority over the next move and take on more responsibility for the conditions that make moves win or lose.

Confident Wrong Solutions

If the agent only does what you specified, most failures trace back to your specification. Once it searches for solutions you didn't specify, it can discover failure modes you didn't specify either.

Agents will find shortcuts. They will exploit proxies. They will settle into solutions that excel on one measure while missing what we hoped the measure represented. Sometimes the result will be clever enough that we call it emergence; sometimes we will call it a bug. Frequently the distinction will depend on whether it helped the quarterly numbers.

Worse than wrong solutions are **confident wrong solutions**. An agent begins with a false assumption, reasons competently from it, researches around the assumption, constructs something sophisticated and explains the whole result coherently. Nothing crashes. There is no red test. Intelligence simply makes the wrong path more convincing.

This is where my optimism about emergence becomes less romantic.

Emergence can give us capable systems. It doesn't give us trustworthy systems.

Giving a system more freedom forces us to think much harder about what surrounds that freedom. Trust becomes a question of provenance and evidence: how does the system know what it claims to know? Desire becomes a question of incentives: what behavior does the environment actually reward? Society appears as soon as multiple agents interact: what happens when they cooperate, specialize, disagree, manipulate one another or invent conventions nobody asked for?

More autonomy does not reduce the need for structure. It changes the kind of structure we need.

It also changes what understanding should mean. We should not expect to reconstruct every micro-decision inside an autonomous system any more than we follow every molecule in a gas. Sometimes internals matter; sometimes behavior matters; sometimes the sequence of decisions matters; sometimes the useful question is what changes when we intervene. Mechanistic analysis, behavioral evaluation, traces and experiments answer different questions.

The useful standard is not omniscience but whether we can detect the failures that matter, obtain evidence from outside the agent's own story and intervene before the interesting failure becomes a congressional hearing.

Pulling Harder on the Stem

After all of that, it would be reasonable to ask why I'm still excited.

Because the alternative isn't actually safe, comprehensible control. It is pretending we can continue specifying increasingly complex systems from the top down even though we already know this stops working surprisingly early.

No CEO understands every decision in a large company. No scientist personally verifies every result their work depends on. No software engineer understands every layer underneath the application

they're building. Nobody understands the entire economy, although this has not prevented a remarkably stable industry of people explaining it on television.

Complexity has already escaped individual specification. We deal with it through abstraction, institutions, feedback loops, delegation and the ability—imperfect but important—to intervene when things go wrong. AI gives us another primitive for doing this.

The answer is not simply to trust the agent. My bet is narrower than that:

I'm betting on systems capable of surprising us because there are problems where we can recognize a better outcome far more easily than we can specify the path that leads to it.

In those problems, intelligent search has room to discover things our instructions would have ruled out before the search even began. The price of that surprise is responsibility upstream: the more freedom the system has over the path, the more deliberate we have to be about the conditions around the path—and the more evidence we need from somewhere other than the system's own confidence.

Cultivation may be a better metaphor than scripting—not because agents are plants, but because pulling harder on the stem remains a surprisingly poor gardening strategy.

I find that exciting and uncomfortable in roughly equal measure, which is probably why I keep coming back to it.

A Bounded Problem

The cleanest place to test the argument is a **bounded problem**: genuinely hard, but unusually cooperative about judgment. The constraints can be written down. Solutions can be evaluated. We can tell whether one attempt is better than another without a debate about aesthetics, pedagogy or whether the users are “delighted.”

That gives us a clean experiment. We still choose the problem, provide the building blocks, construct the environment, define the boundaries and decide what counts as success. I want to watch what takes shape inside those conditions: the solution, certainly, but also the way the system learns to work toward it.

What we stop doing is telling the agent how to get there.

Inside that space, we let it search.

If that fails, the whole argument has a problem.

If it works, things get much more interesting.

Conditions over commands.

Let go of the path, not the boundary.

Chapter 2: The Algorithm Vortex

From Classic Algorithms to Autonomous Discovery

The algorithmic vortex

Once you discover AI coding, there's no going back.

It is faster than you at a ridiculous number of things. It knows libraries you forgot existed. It can stare at a stack trace and notice something you have been ignoring for an hour. Then, five minutes later, it does something unbelievably stupid, believes the stupid thing completely and builds three more decisions on top of it.

This is the strange reality behind all the vibe-coding excitement. The machine is extremely capable, but you are still there. You check the architecture. You notice the missing case. You tell it that no, we are not redesigning the database because one button is the wrong color. You keep enough of the project in your own head to notice when the agent quietly wanders into another universe.

The previous chapter ended with a claim: as more of the search moves into the machine, human control has to move upward from individual actions toward the environment, feedback and boundaries surrounding those actions.

That sounds reasonable in prose.

I wanted to see if it survived contact with an actual problem.

Production software is almost the worst place to test it. A supposedly simple task may involve deployment, legacy systems, users, security, another team's API and a requirement nobody wrote down because everyone assumed everybody else knew it. If the agent fails, you often don't know whether the problem was intelligence, infrastructure, missing context or the fact that someone named a database column `new_status_final_2`.

I wanted something cleaner: a hard problem, but contained. Something where I could genuinely say, "figure it out," and still have an objective way to know whether whatever came back was any good.

I call these **bounded problems**. Not easy problems. Quite the opposite. They can require serious mathematics, programming, research or design, but the boundary is unusually cooperative: you can describe the problem, give the agent enough tools to work on it and evaluate what comes back without deploying to ten million customers first.

Algorithms are almost perfect for this. The search can be brutally difficult while the evaluator remains wonderfully stupid.

And that is how I ended up spending an unreasonable amount of time packing circles into a square.

Twenty-Six Circles

Citrus packing—a real-world example

The problem is simple enough to explain to a child. Take 26 circles and put them inside a square. None may overlap, none may cross the boundary and the circles do not have to be the same size. We want to maximize the sum of their radii.

That’s the whole thing. No customers, no authentication, no stakeholder arriving after the first demo to explain that what they *really* wanted was the opposite of what they originally asked for.

Just circles.

Unfortunately, the solution space is nasty. Every circle has a position and a radius, and nearly every decision affects several others. Increase one radius and two neighbors may overlap. Move a neighbor and something else now needs to move. A packing can look almost perfect while being trapped in a configuration where every obvious improvement makes the solution invalid.

For the experiments in this chapter, we had a strong reference score around **2.635** under the evaluator we were using—the value DeepMind’s AlphaEvolve reported in 2025, when it nudged the best known packing for 26 circles up from 2.634.

Circle packing solution $n=26$

*Figure: A strong reference packing for the 26-circle objective, scoring approximately **2.635** under our evaluator.*

This is what makes the problem useful for studying autonomy. Searching is hard, but judging is cheap. The evaluator does not care whether the agent has a persuasive explanation for why two circles ought to overlap slightly in the name of geometric inclusivity. It checks the constraints and returns a score.

There is something deeply comforting about an evaluator with no personality. A candidate does not earn trust because its explanation sounds clever. It earns another round because it was exposed to something outside the model that did not care about the explanation and survived.

The experiment becomes interesting once we ask a second question:

Who is inventing the next move?

For most of the history of algorithm design, the answer was us.

History of algorithm design

Humans invented explicit algorithms. When direct algorithms were not enough, we invented optimization procedures that searched over candidate solutions. Then we invented meta-heuristics that searched more broadly. Machine learning let systems learn useful structure from data. Now language models can write and modify the search procedure itself.

A crude taxonomy helps. **Symbolic methods** give us explicit procedures, constraints and solvers: they are executable, testable and usually clear about what counts as a valid move. **Neural methods** give us learned intuition: useful structure we did not explicitly encode. **Neuro-symbolic systems**

put the two in the same loop—let the learned model propose and let code, mathematics or another formal system decide what survives. The agentic step pushes one level further: increasingly, the agent can help decide which method to try, combine or abandon.

Circle packing lets us watch that handoff happen in miniature.

Hill Climbing

If I gave you a rough packing and asked you to improve it manually, one obvious strategy would be to make small changes. Move a circle slightly, increase a radius, see whether the result is still valid, keep it if the score improves and undo it if it doesn't.

That is hill climbing:

1. Start with a valid solution.
2. Perturb a position or radius.
3. Check whether the result is valid.
4. Keep it if the score improves.
5. Repeat.

Early in the search, this works nicely. There is empty space and plenty of room to improve. Later, as the circles become tightly packed, almost every interesting move creates an overlap.

Hill climbing progression

Figure: Early mutations are often accepted, but as the packing tightens, valid improvements become increasingly rare and the search stalls.

In one simple run, the score climbed from around 1.33 to roughly 2.26. That is not terrible, but it is also nowhere near 2.635.

Hill climbing is not failing because it is stupid. It is doing exactly what we asked: improving the solution immediately around it. The problem is that the current solution may live in the wrong part of the search space. Reaching a much better packing may require temporarily moving through configurations that look worse, or jumping to a structure that cannot be reached through a sequence of tiny improvements.

A system can become expert at improving the thing in front of it while never questioning whether the thing in front of it is the right thing to improve.

Here, the machine is searching—but the human still invented the search rule.

So we give the machine a bigger space.

Evolutionary Algorithms

Hill climbing puts all your evolutionary eggs in one basket. One solution gets a very long life, and if its history leads into the wrong valley, the search inherits that history forever.

Evolutionary methods keep a **population**.

Instead of dropping one climber onto the landscape, drop a hundred. Some begin in terrible places, some find respectable hills and a few may stumble into structures the original trajectory would never have reached. The biological vocabulary—population, mutation, selection, crossover—is familiar, but the metaphor is optional. What matters is diversity: the whole search no longer inherits the assumptions of one initial guess.

For circle packing, mutation is easy enough to imagine. Move circles. Change radii. Perturb several values at once.

Almost immediately, however, we hit a practical problem. Most interesting mutations break the packing. Two circles overlap or one moves outside the square. The mutation may point toward an interesting arrangement, but the result itself is invalid.

So we added **virtual forces**. When circles overlap, imagine them repelling one another. After mutation or crossover, run a repair procedure that pushes the circles away from collisions and back inside the boundary.

This helps a lot, but notice what happened: the evolutionary algorithm did not invent virtual forces. We did.

Then we reached crossover. Suppose Parent A and Parent B both contain useful geometric structure. How do we combine them? The naive answer is to pair circle 0 from one parent with circle 0 from the other, circle 1 with circle 1, and so on.

That is usually nonsense because circle numbering is arbitrary. Two nearly identical arrangements may store corresponding circles at completely different indices.

So we used **bipartite matching crossover**. Rather than pair circles by position in an array, pair them according to their geometric role in the packing. The Hungarian algorithm gives us an efficient assignment, after which crossover has some chance of combining meaningful parts of the two parents instead of averaging unrelated circles and asking geometry for forgiveness.

Naive vs Geometric Crossover

Figure: Naive crossover pairs circles by array index and often destroys useful structure. Geometric matching tries to identify corresponding circles before combining the parents.

Now we can evolve a population: mutate, repair, cross, select and repeat.

Evolutionary strategies with Bipartite Matching crossover

Figure: Starting around 2.08, the evolutionary search reaches roughly 2.45 in this experiment—much better than the simple hill climber, but still below our reference.

This is much stronger than hill climbing. It also makes the bottleneck clearer. Every time the search became substantially better, I had added something important. I decided we needed repair. I decided how crossover should respect geometry. I chose the representation.

The optimizer searched, but I was still inventing most of the useful moves.

MAP-Elites: Don't Kill Weird Ideas Too Early

Ordinary evolutionary search has another problem. If you maintain a hundred solutions and repeatedly keep only the highest-scoring ones, the population eventually starts looking like one large extended family. That can be excellent for exploitation and terrible for discovering a genuinely different strategy.

MAP-Elites takes a different approach. Instead of ranking every candidate on one axis and keeping only the winners, you describe solutions along a few behavioral dimensions and preserve the best candidate in different regions of that space.

For circle packing, perhaps one dimension measures symmetry and another measures how much circle sizes vary. One part of the archive may contain highly symmetric solutions. Another may contain asymmetric solutions with several large circles. Somewhere else may sit an ugly packing with a mediocre score and one strange structural idea that becomes useful five generations later.

MAP-Elites archive visualization

This is **quality-diversity search**. The point is not merely to preserve the current winner, but to keep qualitatively different directions alive long enough to discover whether any of them become interesting.

I like this because optimization is often unfair to immature ideas. A new approach can initially perform badly simply because nobody has polished it yet. If the first respectable solution immediately kills everything else, the search can become impressively efficient at discovering one family of answers.

But MAP-Elites introduces another human choice: what dimensions define the archive? Symmetry? Radius variance? Number of large circles? Something topological? Something I haven't thought of?

The search had become more sophisticated, but the human was still deciding what counted as an interesting direction.

The Invention Problem

By this point, the search machinery was fairly capable. We had hill climbing, population search, repair, geometric crossover and quality-diversity archives. We could evaluate huge numbers of candidate packings and inspect far more of the search space than any human would explore manually.

Yet every substantial conceptual jump came from somebody noticing something.

Traditional search is excellent once we define the space and the legal moves. Sometimes the space and the moves are exactly the things we need to rethink.

Learned models have something to offer exactly that problem.

I once asked an image-generation model to produce a picture of a circle-packing solution. This was not a serious benchmark; I have no idea what related examples it may have encountered during training, and I can already hear Reviewer 2 clearing his throat.

I wanted to see something simpler: did the model have any useful geometric intuition about what a dense packing should look like?

Surprisingly, yes. It generated something that looked plausible. The circles had structure. The spacing looked intentional. At a glance, you could believe the model understood the problem.

Then you counted the circles.

Wrong number.

Some constraints were violated.

It was a beautiful answer to a nearby problem.

That little experiment makes the asymmetry concrete. Learned models can be remarkably good at generating plausible structure without guaranteeing that every formal requirement survives generation. A symbolic optimizer has almost the opposite personality: give it a precise representation and constraints and it will obey them, but it will not naturally look at your representation and decide that you have been unimaginative.

The obvious temptation is to argue about which one is better. The more useful answer is: **put them in the same loop**—neural intuition and symbolic rigor.

Or, if you prefer the slightly ridiculous version: let the brain invent things and make the body prove they work.

The important move is to stop asking the model to produce the packing directly.

Ask it to write the program that produces the packing.

Let the Model Write the Solver

A candidate no longer needs to be only a list of circle positions and radii:

`(x1, y1, r1), (x2, y2, r2), ...`

It can be an entire program:

`solve_circle_packing.py`

One program may use constrained optimization. Another simulated annealing. Another a geometric construction. Another may combine a hand-designed initialization with numerical refinement.

The evaluator does not care which family produced the solution. It runs the program, checks the geometry and scores the result.

This gives the language model a much more interesting role. Rather than randomly perturbing numbers, it can read the program, form a rough theory about why it underperforms and change the algorithm. Perhaps the initialization is weak. Change the initialization. Perhaps a geometric construction gets close but leaves local slack. Add numerical optimization afterward. Perhaps one repair procedure keeps destroying useful structure. Replace it.

The mutation is no longer merely numeric. It can contain an **idea expressed in code**.

That is the neuro-symbolic unlock behind systems such as FunSearch and AlphaEvolve. The model proposes changes at a level where programs have semantic meaning; execution and the evaluator decide whether those ideas deserve to survive.

The human used to search the solution space.

Now the machine can begin searching the **algorithm space**.

AlphaEvolve

AlphaEvolve turns that basic idea into a much larger search process.

Imagine one generation. The system selects a promising program from its archive, perhaps along with other successful but different programs that contain useful ideas. The model sees the code, information about previous attempts and the scores they produced, then proposes a patch. The patch is applied, the program runs and the evaluator scores what happened. The new program and its result go back into the archive. Then the process repeats.

AlphaEvolve architecture

Diff-based mutation matters because real programs contain structure worth preserving. If every generation rewrites everything, useful ideas disappear as easily as bad ones. Small patches let the search alter the part it thinks matters while leaving the rest intact.

The archive matters for the same reason the population mattered earlier. If every descendant comes from the current champion, code evolution quietly collapses back into hill climbing. Multiple lineages preserve stepping stones: a program that is not the best today may contain a useful component that becomes valuable after another idea appears.

Sometimes the model's guess is excellent. Sometimes it produces nonsense wrapped in perfectly respectable Python. The nice thing about bounded algorithmic problems is that the disagreement does not need to be settled in prose.

We run the program.

Intuition proposes. Symbolic machinery executes. The evaluator gets the last word.

What interested me even more than the resulting algorithms was what happened to the human. Instead of writing the solver directly, I was increasingly building the machinery in which solvers could be generated, compared and improved.

So, naturally, I built all of it.

So, Naturally, I Built All of It

My instinct was predictable. I started building a framework: a database of programs, prompt sampler, evaluation loop, selection logic, mutation prompts, crossover, archive management. I used Aider and other coding agents to help reproduce the basic code-evolution pattern, and it worked. We could evolve circle-packing programs and get respectable solutions.

I enjoyed this immensely because I like building systems that generate other systems, which I suspect is either a research interest or a mild personality disorder.

While I was doing this, coding agents themselves were becoming much better with much less custom machinery. Earlier software-engineering agents often wrapped the model in carefully designed interfaces: custom editing commands, repository-search tools, restricted action spaces and plenty of logic controlling how the model interacted with the machine. Then increasingly minimal systems began demonstrating how far a capable model could get with something much simpler.

Give it a shell.

The provocative version is: **if the agent has a shell, it has almost everything**. It can `grep` to search, inspect files, run Python, apply patches, call Git, compose Unix tools and, if the tool it needs does not exist, write one. Bash is not merely one tool; it is an entrance into decades of software accumulated underneath it.

This made me pause. The framework I was building—the parent selection, loop controller, experiment bookkeeping—was hard-coding behaviors that a sufficiently capable coding agent could increasingly perform itself. It could maintain notes, write helper scripts, explore several strategies, inspect failures and change direction.

I looked back at the machinery I had just spent time constructing and had the unpleasant thought engineers occasionally have after a productive week:

Maybe I shouldn't have built most of this.

So I deleted the database machinery, controller loops and little pieces of software whose job was to make the agent behave like a researcher, and tried the stupidly simple version.

The Coffee Test

I opened Claude Code in a directory containing the evaluator and gave it a high-level instruction along the lines of:

Here is the evaluator for the circle-packing problem. Write a Python program that maximizes the score. You can research strategies, write tools, run experiments and iterate. Do not modify the evaluator. I will go get coffee.

Then I left.

That became the autonomy test I actually cared about. Not whether AI could help me solve the problem; that was already obvious. Not whether it could write code faster than I could; usually it could.

I wanted to know whether I could leave.

There is a difference between collaborating with an agent and **hiring** one. If I still have to choose every strategy, approve every experiment, rescue every failed branch and keep the search alive myself, then I have a formidable collaborator. That is useful. It is not yet the kind of autonomy I was trying to understand.

Circle packing gives us a rare luxury because the evaluator can stay behind when I leave. The agent can change its code, create scripts, abandon one approach, try another and waste compute on ideas that go nowhere. What it cannot do is redefine what counts as a valid packing because the current score hurts its feelings.

The **Immutable Harness** is what makes all that freedom tolerable.

Everything inside the boundary can move.

The boundary does not.

Diagonal Layering

The agent did not execute one elegant master plan. It bounced around, which was encouraging.

It tried numerical optimization, changed initialization strategies and noticed that some optimizers repeatedly converged to poor local solutions. It experimented with the geometry of its starting configurations and mixed those constructions with numerical refinement. At different moments it was acting as orchestrator, researcher and engineer: deciding what to try, implementing the idea, running the experiment and using the result to choose what happened next.

Eventually one family of solutions began arranging circles in diagonal bands. We called the idea **diagonal layering**.

I had not instructed the agent to pursue that construction. More importantly, I had not selected the branch after it appeared. The agent found a direction, saw that it improved the evaluator and invested more of its search there.

I want to be careful with the word *discovered*. I had not seen that particular strategy before, but that does not establish historical novelty in computational geometry. For this experiment, the important discovery was local: the agent found a useful direction that I had not put into the plan.

Once that structural idea became strong enough, the nature of the work changed. The agent spent less time inventing new geometries and more time adjusting solver settings, tolerances, initialization details and all the boring machinery that suddenly matters when the last fraction of a percent becomes expensive.

Code evolution result: iterative optimization

In our best run, the evaluator returned roughly **2.636**, slightly above the **2.635** reference we had been using.

That sentence needs a fence around it. Under our evaluator, the result beat our reference. Calling it a new state of the art in circle packing would require matching problem definitions, checking numerical tolerances and constraints, reproducing the result properly and doing a more serious literature search than this experiment justified.

The smaller claim is enough:

The agent beat our reference while I was not writing the solution algorithm for it.

That was the result I cared about—not that AI writes code faster, but that AI can participate in **discovering better code**.

The Algorithmic Vortex

This is what I mean by the **Algorithm Vortex**.

At the beginning of a conventional project, I might choose hill climbing, evolutionary search, simulated annealing, constrained optimization or a geometric heuristic. That early decision shapes everything downstream.

Once code is cheap to generate and evaluation is cheap enough to repeat, the choice no longer has to be permanent. A geometric construction can initialize a numerical optimizer. An evolutionary method can search parameters for another solver. A language model can notice a failure pattern and invent a repair procedure. Two ideas that began in separate lineages can meet later because an experiment suddenly makes the combination useful.

The search moves outward through levels. A conventional optimizer searches over candidate solutions. Meta-heuristics search over larger families of candidates and strategies. Code evolution searches over programs that themselves search for solutions. Once a capable agent controls the experimentation loop, even the decision about **which kind of search to try next** can enter the search space.

That is the vortex.

It is not “algorithms are dead.” There are algorithms everywhere in this picture. The change is that the human is no longer forced to freeze the complete algorithmic architecture before the experiment begins. We stop writing one solver and start creating conditions in which solvers can compete, mutate, combine and occasionally surprise us.

The chapter began by asking who invents the next move. Here, for the first time in the experiment, the answer was not reliably “me.”

The Contract

The coffee test worked because the problem gave the agent freedom **inside** a structure that remained outside its control. After several runs, that structure settled into a small contract.

These are not universal laws of software engineering. They are rules for a particular regime: bounded problems, cheap experimentation and an evaluator objective enough that the agent cannot charm its way around failure.

Keep the Harness Immutable

This one is the foundation.

If the agent can change the evaluator, the meaning of the experiment evaporates. The circles overlap? Perhaps tiny overlaps should count. The score is low? Maybe the square should be 1.03 wide. Only twenty-five circles fit? Perhaps twenty-six was merely an aspirational requirement.

At that point we are no longer optimizing circle packing.

We are negotiating with the specification.

The **Immutable Harness** is the anchor of truth in an otherwise fluid process. The solver can change. The strategy can change. The tools can change. The agent can decide yesterday’s entire approach was stupid and start again.

But the thing saying whether it worked stays harder to change than the thing being optimized.

This is the boundary made executable.

Never Write Solution Code Yourself

This is deliberately provocative.

You watch the agent try something mediocre and immediately think of a better approach. You want to help, and sometimes you should. But every time I jump in with my own solution, the search becomes a little more like whatever happened to occur to me first.

For these experiments, I wanted independent directions badly enough that I had to resist becoming the senior engineer on every branch.

The deeper rule is: **don't accidentally collapse autonomous search back into your own search.**

Spawn, evaluate, prune. Intervene in the conditions before you intervene in every idea.

Cross-Pollinate Without Collapsing Diversity

Independent search creates diversity. Perfect isolation wastes learning.

If one branch discovers a useful initialization and another finds a better local optimizer, future experiments should have some mechanism for inheriting both. That is what makes code evolution more interesting than asking the same model the same question one hundred times.

But broadcast every successful idea immediately and the population starts thinking in the accent of the first successful branch. Information accelerates learning and destroys independence at the same time.

Cross-pollinate, but leave some lineages ignorant long enough to surprise you.

Prune Ruthlessly, But Not Stupidly

Diversity is useful. Preserving every bad idea forever is hoarding.

If a branch keeps underperforming and contributes nothing interesting, eventually it should die so compute and attention can move elsewhere. Kill too early and you may discard an immature idea that needed another generation. Keep everything alive and you end up funding a large family of increasingly sophisticated failures.

The practical rule is simple: **diversity needs a budget.** Search needs enough patience for novelty and enough cruelty for budget control.

Separate Discovery From Polish

Early in the search, I want large conceptual moves: a different geometry, solver, representation or decomposition.

Once a strong direction appears, the valuable work becomes smaller and more boring. Solver tolerances. Initialization details. Numerical settings. Tiny modifications that are pointless on a bad idea and extremely valuable on a good one.

Diagonal layering made this distinction obvious. Once the structural direction looked promising, continuing to invent entirely new geometries became less useful than squeezing performance from the geometry that was already working.

Discovery before polish.

Do not spend hours polishing a local optimum you should abandon. And do not keep demanding revolution from a solution that has already found the right mountain and merely needs to climb it.

Zero Framework, With an Asterisk

There is one correction worth making before we leave the experiment. I started calling this direction **zero framework**.

It's a great slogan.

It's also not really true.

I meant that I was writing almost no custom orchestration framework. That is very different from having no framework. Claude Code is itself a substantial system. The underlying model has absorbed enormous amounts of software and problem-solving knowledge. Bash, Python, SciPy, Git and the operating system represent decades of accumulated engineering. The evaluator is custom machinery. Even the supposedly trivial act of running a program and inspecting a result depends on layers we have become so accustomed to that we stop seeing them.

The framework did not vanish. It became somebody else's primitive.

That fits the emergence argument almost suspiciously well. A tiny amount of code at the top can command enormous capability underneath because previous generations of complexity have already been compressed into tools.

So yes: **Zero Framework. Bash is enough.**

With the asterisk that Bash contains roughly half a century of civilization.

This is worth remembering whenever somebody shows you an agent implemented in one hundred lines of Python. The hundred lines may be perfectly real. So is everything underneath them.

The Easy Version of Autonomy

It would be very easy to overread this experiment.

We did not prove that coding agents can autonomously solve arbitrary research problems, that AlphaEvolve-style systems are obsolete, that diagonal layering is historically novel in computational geometry, or that the right approach to production software is to give Claude a shell and go for a very long lunch.

What we had was narrower and, to me, more useful. We had a **bounded problem** where evaluation was cheap and clear. We gave a capable coding agent substantial freedom and found that a surprisingly large fraction of the experimentation loop could happen without us directing every step.

The agent could propose an approach, implement it, run it, inspect the result, abandon it, create tools, borrow ideas from another direction and try again. My role moved away from writing the solver and toward defining the job, constructing the environment and defending the harness.

That is the claim this chapter earns: **when the problem is bounded and reality supplies a hard enough referee, substantial decision-level control can move into the agent without giving up control of what counts as success.**

That is already a meaningful change.

It is also why circle packing is the easy version of autonomy.

The evaluator gives us one number. If version B beats version A, nobody needs to simulate a confused student, debate whether the interface feels intuitive or convene a committee to decide whether the new solution is spiritually aligned with the learning objectives.

The search can be complicated because **judgment is simple.**

Most things I want agents to build are not that generous. “Make a good educational demo.” “Write something people remember.” “Design a useful product.” “Explain this so somebody finally understands it.”

We can still let the agent generate alternatives, branch, cross-pollinate and search among them. But now the difficult part has moved again.

In circle packing, the harness tells the agent when it is wrong.

What happens when **the world no longer gives us one clean referee, and judgment itself has to be constructed?**

Harness immutable. Prompts evolve.

Never write solution code.

Discovery before polish.

Chapter 3: The Vibe Coder’s Seat

Beyond Algorithms: Agent Autonomy for Creative Problems

In the previous chapter, we gave an agent a difficult algorithmic problem and a lot of autonomy. It researched strategies, tried several approaches, got stuck, changed direction, and eventually found diagonal layering.

But circle packing had one enormous advantage that I did not appreciate enough at the beginning: we knew exactly what good meant.

There was an Immutable Harness. Run the program and you got a number. Circles overlapped or they did not; the score improved or it did not. The agent could spend an hour pursuing some bizarre geometric idea and I did not have to sit beside it wondering whether version seventeen had more soul. We ran the evaluator.

Most of the things I actually want AI to help me with are not like that. “Is this explanation pedagogically effective?” does not have a unit test. “Would a confused student understand this visualization?” cannot be settled with an `assert`. Two competent people can look at the same design, disagree completely, then switch sides five minutes later after using it. The feedback is subjective, noisy, sometimes contradictory, and often becomes clearer only after you have built the thing you were supposedly trying to specify beforehand.

I picked educational demos for Merge Sort and Count-Min Sketch because they were still bounded—you can actually finish one before civilization collapses—but they live on the messier side of the boundary. You have to decide what to explain, what to leave out, how the interaction should work, how much should be visible at once, and what another person is likely to understand from any of it.

The ambition was intentionally high. I wanted something closer to the best Distill articles or Jay Alammar’s visual explanations than to the usual “here are some bars moving around; congratulations, you have learned sorting.” The algorithm itself is usually the easy part. The difficult part is deciding what to show, when to show it, and what representation might make an idea suddenly click.

Circle packing let the search be complicated because judgment was simple. Here judgment had become part of the problem.

The problem-solving layer I eventually started calling **Deep Mode** grew out of one question: could the system take over some of the work of deciding what to try next? Not just implementation. The inquiry itself. Build another version? Research the failure? Retrieve an old idea? Split into independent branches? Change perspective? Abandon the direction?

Before trying to automate that, I had to notice how much of the work around the model had already moved into the machine.

Stubbornly Human

Models trained to continue text turned out to continue code. Researchers trained models specifically for the job, and many of the early tasks were conveniently small: give the model a function signature, a comment or a programming puzzle and ask it to fill in the implementation. Benchmarks such as HumanEval and APPS made this measurable: could a model turn a specification into a program that survived tests?

Then GitHub Copilot put the trick inside the editor. Instead of asking a chatbot for code and carrying the answer back yourself, you could describe what should happen next and watch it appear underneath your cursor, which was delightful for about a week and then became the way things were.

This was useful enough that the limitations became interesting. A real software task rarely arrives as an isolated function with a docstring politely explaining what needs to change. Someone says invoices occasionally show the wrong tax after a refund. Somewhere inside a 150,000-line CRM there is a reason. It may involve a controller, a database model, an old helper function, a test written three years ago and an API whose behavior everyone on the team knows but nobody thought to document.

By the time GPT-4 arrived, I wanted to use models on exactly these problems, and the workflow I invented for it was ridiculous. Find the file you suspect. Copy a class into ChatGPT. Describe the bug. Copy the suggested patch into the editor. Run the program. Discover a new error. Copy the traceback. Paste that back into ChatGPT. Repeat until it works or until dinner.

My first agent-computer interface was copy and paste.

I did not think of it that way at the time. I thought I was using a very good autocomplete. Looking back, the model might have been doing sophisticated reasoning in the middle, but I was performing every interaction with the software around it: searching the repository, deciding which file mattered, assembling the context, applying the edit, running the tests and carrying back whatever reality had said about the edit. I was the hands, the eyes and the memory. The model was a brain in a jar, and I was the jar's entire staff.

Then the bug crossed three files and context itself became a job. Paste one class but forget the interface it implements, and the model confidently invents a method that does not exist. Add the interface and now it needs the database schema. Add the schema and another helper suddenly matters. Eventually half the repository is sitting in the conversation and somehow the model understands less. A lot of early LLM programming consisted of building a tiny artificial universe around the model: here is the relevant class; here is the schema; ignore these twelve methods; this innocent-looking helper controls payments, so please do not touch it unless you enjoy incident calls.

We learned an obvious lesson surprisingly slowly: more context and better context are different things. If somebody asks for a spoon, emptying the entire kitchen onto the table does not help.

Software-engineering benchmarks exposed the same gap from the other side. SWE-bench changed the unit of evaluation. Its tasks came from real GitHub issues, so a system had to work inside an existing

repository, locate the relevant code, understand relationships across files, make an appropriate change and survive the tests. In other words, it had to do my job. The copy-and-paste one.

Eventually we stopped carrying the loop by hand. Give the model access to the repository. Let it search for symbols and references. Let it open files, edit them and inspect the diff. Give it a terminal. When a test fails, return the failure and let that result shape what happens next. A coding agent is, at its simplest, this loop made executable. The language model supplies much of the programming knowledge and reasoning; the environment lets it inspect software, act on it and observe the consequences.

Software is unusually friendly to this arrangement. Files can be searched. Programs can be executed. Tests can say no. Git can tell you exactly what changed and, if an experiment becomes sufficiently exciting, return you to the time before you had the idea.

Systems such as SWE-agent made the interface itself part of the problem. How the model searches, how much of a file it sees, how edits are applied and what information comes back from commands can matter almost as much as another clever prompt. The useful object is the model operating inside a world where software can push back.

Of course, giving the model a computer created new ways to be annoying. Early coding agents could behave like interns with root access and too much coffee. Ask one to change a line and it might rewrite half the file. Ask it to fix a button and twenty minutes later it has developed strong opinions about the database architecture. It would find one plausible theory of a bug, follow it for too long, then use every new piece of evidence to improve the theory instead of admitting the theory was wrong. I recognized the behavior. I had done all of it myself, at two in the morning, with worse excuses.

So more of the surrounding work moved into the system: small patches, diff inspection, targeted tests, checkpoints, planning, rollback. Repository knowledge moved too. Authentication conventions, ancient APIs and local rules that used to live in somebody's head became `CLAUDE.md`, `AGENTS.md`, rules files and skills. If somebody had already learned something expensive about the codebase, we left it somewhere the next agent could find it.

Long sessions produced the opposite problem. Context filled with abandoned experiments, obsolete assumptions and test output from three hypotheses ago. Memory became a problem of selection rather than storage.

Then history became a problem too. Suppose an agent decides early that our Merge Sort demo should use React and a recursion tree. It spends forty minutes building that version. Every later question now arrives in a context containing forty minutes of reasons, code and decisions supporting React and a recursion tree.

Humans call our version of this sunk cost. The agent has a respectable excuse: its context window is literally full of evidence that this is what the project is.

So we started giving different attempts different histories. One agent tries the tree. Another begins with the array. Another starts from the learner's misconception rather than from either representation. A fresh branch does not have to spend half its intelligence escaping assumptions accumulated by the previous one.

Together, the interfaces, execution loop, context management and safeguards form the agent's **har-**

ness. The evaluator is one part of it. Looking backward, its construction is less mysterious than the word *agent* sometimes makes it sound. Every step took a job I had been doing by hand—searching, assembling context, applying the edit, running the test, remembering, keeping alternatives alive—and moved it into the machine.

But there was still a large difference between an agent that could work competently inside a repository and the thing I increasingly wanted to ask for:

Build the application.

Ask for a booking application for a football academy and an unconstrained coding agent first chooses a framework, installs packages, creates a database, decides how authentication should work, manages environment variables and configures deployment. Several minutes later, we have made enormous progress toward having somewhere to put the booking form.

Somebody has to do that work. But if the same plumbing is reconstructed on every project, it becomes reasonable to prepare more of the world in advance.

This is what made systems such as Replit and Lovable interesting to me. Runtime, deployment and common application machinery are already nearby, so the conversation can begin much closer to the application.

You lose some freedom. That is often the point. A chef does not begin dinner by manufacturing a knife. A scientist does not build an operating system before analyzing data. When I open Python, I accept an astonishing number of decisions made by people I will never meet because reconsidering all of them would make `print("hello")` a multigenerational project.

Useful abstractions remove decisions whose answers are no longer interesting most of the time. After enough of those decisions disappear, something else becomes easier to see.

Suppose the booking app works perfectly. The database is connected, deployment succeeds, the buttons behave, the mobile layout is respectable, and nobody has accidentally built a cryptocurrency exchange inside the authentication service.

I open the application and think: this is not very good. The software works. Now I have to worry about the football academy.

Should parents see every available session, or only sessions appropriate for their child? Should they create an account before booking? What happens when somebody has three children? How late can they cancel? If Wednesday is empty and Saturday has a waiting list, is the booking interface part of that problem?

None of those questions is really about React. They were always there. Implementation simply consumed enough attention that deciding what should exist and turning that decision into software felt like one activity.

When another version becomes cheap, the balance changes. You can see the idea sooner, and seeing it gives you information you did not have while discussing it.

Maybe we decide customers should create an account before seeing availability. It sounds reasonable: we need their details eventually. Then we build it and the experience immediately feels annoying.

Parents arriving from a Google search do not want to establish a lifelong digital relationship with a football academy before discovering whether Saturday at ten is available.

So login moves later. The artifact becomes something we think with.

The Merge Sort demo made this even clearer because there was almost no business machinery to hide behind. I could ask an agent for an interactive explanation and receive something perfectly functional: an array of bars, controls, animation, perhaps some text explaining that the algorithm divides the input and merges the pieces again.

Technically, it was fine. Pedagogically, it could still be terrible. Watching bars move does not tell a beginner why dividing the problem helps. So perhaps we try a recursion tree. The tree makes the structure visible, but now the supposedly simple sorting algorithm resembles the organizational chart of a German corporation. Maybe we show the tree and array together. Perhaps that creates too much cognitive load. Maybe the problem is not the representation at all; the learner understands splitting perfectly well but has no idea why merging makes the whole trick useful.

There is no compiler error that tells me which diagnosis is right. I have to look at what we built, form an opinion about why it fails, and decide what would teach us something next. Sometimes that means improving the current version. Sometimes it means building a deliberately different one. Sometimes I need research. Sometimes the right move is to put the application in front of somebody who does not already understand Merge Sort.

Occasionally I discover that the question I started with was wrong. “Build an interactive Merge Sort demo” sounds like a goal until you see several interactive Merge Sort demos. Perhaps what I actually care about is getting somebody who has never encountered divide-and-conquer to understand why breaking one difficult problem into smaller ones helps. Once I realize that, interactivity is merely one possible means.

That is the layer that remained stubbornly human: deciding what to try, which evidence matters, whether a result failed because of its implementation or its underlying idea, and what kind of attempt might teach us something next.

The Five Layers of AI Coding

By then I had a rough map. Each layer marked a different kind of work we had learned to delegate, or were still trying to. I could draw the stack afterward because each working layer had made the next unfinished job easier to see.

Layer 0—Model. GPT, Claude, Gemini and whatever comes next: general capability in language, code, reasoning and vision.

Layer 1—Agent. Put the model in an environment where it can act. Claude Code, Codex and similar systems search repositories, edit files, execute commands and react to results.

Layer 2—Application. Prepared environments remove much of the repeated software plumbing and let the conversation stay closer to the application itself.

Layer 3—Deep Mode. The problem-solving layer: decide what to try, why something failed, which evidence matters, and whether the current direction deserves another iteration.

Above that sits the problem I have mostly been avoiding.

Layer 4—Intention. What do we actually want?

Software likes that question to have been answered before work begins, preferably in Jira, where the answer can remain wrong in a structured and searchable format. Real goals are less cooperative. Seeing a solution can change what I realize I wanted.

That problem is bigger than AI coding, so for now I am leaving it at the top of the stack.

The borders are fuzzy. Coding agents make product decisions; design systems generate code; tomorrow's products will rearrange the boxes again. What matters is the kind of decision being made, not which company happens to occupy which layer.

People often call the experience of working this way *vibe coding*. I will use **AI coding** for the broader stack, but *vibe coder* remains a wonderfully accurate name for the human sitting near Layer 3: looking at what came back, deciding what feels wrong, asking for another direction, killing one idea, keeping part of another, and steering the process without having an algorithm for how.

The lower layers increasingly answer a version of the same question: *how do we make this?*

Deep Mode asks a different one:

Given everything we have learned so far, what should we try next?

That was the part I still seemed to be doing manually. So I watched what I was actually doing in that seat.

What I Was Still Doing

There was no universal workflow hiding there. A mathematician, a designer and a product manager can all spend a day solving hard problems while performing almost none of the same visible actions.

But the same kinds of moves kept appearing. Sometimes I needed another attempt, sometimes information. Sometimes the search had become too narrow, the representation was constraining what we could imagine, or the objective needed to change. Sometimes I needed to see the artifact from another mind. They were not useful in a fixed order.

Keeping More Than One Idea Alive

Even a Merge Sort demo has an absurd design space. Bars or cards? Numbers or a tree? Continuous animation or learner-controlled steps? Does color represent recursion depth, identity, or the active subproblem? Explain before the animation, during it, or afterward? Every choice changes the usefulness of several others.

When implementation was expensive, we dealt with much of this complexity by trying to decide more before building. AI coding changes the economics. If another implementation costs minutes rather than days, I do not have to choose quite so much in advance.

Here what evolves can be more than a vector of parameters or even an algorithm: an **idea embodied in software**.

One builder tries a recursion tree. Another focuses on the array. A third begins from the learner's misconception. Mutations can be conceptual: remove the text, teach backward, make the learner predict, show synchronized representations, abandon interaction altogether.

Useful pieces can move between them. One terrible demo may have a beautiful color mapping. Another may explain the merge clearly while making everything else unbearable. The final artifact does not have to inherit the entire history of either one.

But diversity is fragile. If every branch sees the current winner and its complete reasoning history, parallelism quickly becomes several agents improving the same idea. Sometimes I want the branches to exchange what worked; sometimes I want a fresh branch to remain ignorant long enough to become genuinely different.

Share too little and everyone rediscovers the same lessons. Share too much and the first successful idea becomes a local culture.

Research creates the same tension. People have been teaching recursion for decades. There are textbooks, lecture notes, visualizations, papers, classroom experiments and a great deal of trial and error sitting on the internet. Before I spend another afternoon inventing my fourth way of moving colored rectangles around, I probably want to know what is already there.

But research is most useful when the work has produced a real question. Suppose a recursion tree makes decomposition visible but learners lose the relationship between the tree and the changing array. Now I can ask how other systems have coordinated two representations without requiring people to watch half the screen at once.

Research becomes another move in the investigation rather than a ceremony performed before building.

Retrieval plays the same role inside our own history. Somewhere in a growing project there may be research notes, screenshots, evaluator comments, old branches and a discarded prototype whose only good idea was a color mapping that solves exactly the problem in front of us. I do not need the whole archive. I need the thing that helps with this decision.

Sometimes exact search is right because I remember a phrase, API or evaluator comment. Sometimes embeddings are useful because I remember the idea rather than the words. Sometimes the document already has a structure worth navigating. Good coding agents do not “retrieve the repository” once; they move through it as the question changes. Layer 3 needs the same habit across stranger objects: research, screenshots, old interactions, code, evaluations and dead branches.

A dead branch is not necessarily dead knowledge. A lineage that lost globally may still contain a stepping stone that becomes useful later.

The exploration literature has several versions of this idea—quality-diversity, novelty search, Go-Explore and related approaches. Do not spend the entire search budget polishing the place that currently looks best. Preserve some alternatives and some routes back to places that almost worked.

The same logic gave us **Strategic Constraints**.

After several generations of Merge Sort demos, the builders were exploring. They were also still giving me bars. Better bars, admittedly. Bars that split gracefully, changed color as recursion deepened, synchronized with a tree and perhaps deserved their own design award. Given enough iterations, I

had every reason to believe we would eventually produce the finest moving bars known to humanity.
So remove the easy path.

No bars.

Or: teach Merge Sort without explanatory text. Require the learner to predict before anything moves. Make the demo work on a phone with room for only one representation. Design it for somebody who understands loops but finds recursion suspicious.

Most arbitrary constraints are merely arbitrary. A useful one changes which parts of the search are reachable, exposes a neglected dimension or prevents a familiar attractor from absorbing every attempt.

“No bars” is an intervention on the search, not a theory of creativity.

A move in this space can be a code change, a new metaphor, a retrieved analogy, a fresh agent with no history, a different evaluator, a research question, or a reformulation of the problem itself.

Even then, most of our ideas still had to arrive as words.

Draw It Before You Build It

That is fine when I am working on an argument. It is less obviously sensible when I am designing an interface.

I can spend ten minutes explaining where the recursion tree should sit, what remains visible while the array splits, how colors should connect two representations and what the learner should notice first.

Then somebody draws it and I know within three seconds that the whole thing is terrible.

So I started generating the picture first. The experiment was not sophisticated. I asked an image model to design an interactive tutorial for Merge Sort. Then Count-Min Sketch. Then A*. Then Poincaré embeddings in hyperbolic space, partly because if this still worked there I would have to take the idea seriously.

The details were not magically correct. Arrows occasionally pointed somewhere they had no business pointing, interactions made no computational sense, and generated text sometimes looked like somebody had tried to OCR a dream.

But the composition could be surprisingly thoughtful. A Merge Sort mockup might keep the array visible while placing the recursion tree beside it, using color to preserve the relationship between a subarray and its node. A Count-Min Sketch design might make collisions visually central instead of leaving them as a detail in an equation. The model had to decide what was large, what was peripheral, where controls belonged and how the learner might move through the explanation.

I remember looking at some of these and thinking: **Holy shit.**

Not because I wanted to ship the image. Usually I did not. I had given the model a concept in language and it returned something like a spatial argument about how the concept might be taught.

After that I stopped treating image generation as the last stage—*the product is designed, now make it pretty*—and started using it while I was still trying to understand what the product could be.

A mockup is a cheap hypothesis. Often most of it is disposable and one relationship is worth stealing. Then the coding agent can make that relationship executable, which is where the picture has to pay its debts. The recursion tree cannot invent an extra branch because the composition looked nicer that way. The interaction has to possess a state. The button has to do something other than contribute emotionally to the page.

Different representations expose different mistakes. I do not need the stronger claim that an image model “understands pedagogy.” The practical point is enough: changing the representation changes what the search can discover.

By now we could generate genuinely different artifacts. That left the problem we had avoided from the beginning. Which one is better?

Optimizing Something You Cannot Score

Circle packing was unusually kind to us. Once the geometry was valid, the evaluator reduced the result to one number. That number threw almost everything else away, which was precisely why it was useful.

A huge amount of machine learning rests on this trick. We take something complicated that we want and find a measurable signal that stands in for it. Reinforcement learning makes the relationship especially obvious: we do not specify every movement a robot should make while learning to walk; we construct a reward and let search discover the behavior.

The reward is doing an extraordinary amount of work. It is also where we hide an extraordinary amount of trouble.

Suppose I want the same convenience for educational design. I can make a rubric: correctness, pedagogical clarity, visual quality, interaction, accessibility, engagement. Give each a weight and suddenly my vague dissatisfaction with a demo has become a respectable decimal.

The decimal is comforting. The decisions required to produce it are less so. Why should interaction receive fifteen percent? Is more interaction always better? What distinguishes a seven from an eight in pedagogy? Why those dimensions rather than whether the learner can predict what happens next or explain why the merge matters?

A metric forces me to commit to an idea of “good” before the search has taught me very much about the problem.

Metrics are not the enemy. If I care about latency, measure latency. If the code must pass a test, run the test. Hard measurements are wonderful when what we can measure is close to what we care about.

The trouble begins when a rich objective is still poorly understood and we compress it anyway because optimization wants a number.

The compression is also low bandwidth. “Version B scored 7.4; version A scored 7.1” tells the next builder almost nothing about why B won. A rubric helps, but as I add enough dimensions, exceptions and qualifications to express what I mean, eventually I reinvent language badly.

Meanwhile I can simply say:

The recursion tree makes decomposition much clearer, but now the learner has to watch the tree and the array simultaneously. Keep the color mapping that preserves identity between them, simplify the tree, and make the merge feel like the payoff rather than cleanup at the end.

That contains comparison, diagnosis, trade-offs, priorities and a proposed next move in a few sentences. Natural language is ridiculously rich compared with a scalar.

Language models make that communication channel available inside the optimization loop. The model already carries learned structure behind words such as *simple*, *confusing*, *elegant*, *intuitive*, *busy* and *beginner-friendly*. Those meanings are imperfect, culturally loaded and sometimes wrong. But they carry more structure than 7.4.

Natural language can therefore function as an **implicit metric**. Not a metric in the strict mathematical sense. There is no guarantee that “intuitive” defines a stable ordering, and two evaluators may interpret it differently. But language can do some of the work a metric normally does: give the search a direction, communicate why one attempt is preferred to another, and preserve trade-offs that a scalar would erase.

OPRO—Optimization by PROMpting—is interesting for a related reason. In OPRO, an LLM sees an optimization problem, previous candidates and their outcomes, then proposes another candidate. Candidate quality in the published setting is still evaluated by an explicit score, so this is not the same thing as creative design. What matters here is the direction of control: much of the search heuristic can live in the model rather than in a hand-written transformation rule.

Now let the history contain more than scores. Alongside hard measurements, tell the model what improved, what became worse, which trade-off appeared and what must survive the next attempt. The history of the search can retain some of its meaning rather than collapsing into a column of numbers.

This begins to feel a little like reinforcement learning turned upside down. I mean that as an analogy about specification, not as a claim that these are the same algorithm.

The usual reinforcement-learning picture asks us to define a reward and then discover behavior that earns it.

Here I can begin with something much less respectable:

Make this explanation less intimidating.

Help the learner understand why the merge matters.

I want somebody to *feel* why divide-and-conquer helps rather than merely watch the algorithm execute.

Those are descriptions of direction, not reward functions. Yet the model can produce an attempt from them, and the attempt can teach me whether the direction was what I really wanted.

I began the project insisting on an *interactive* Merge Sort demo. Interactivity sounded obviously desirable. Then I saw versions with buttons, sliders and enough learner participation to qualify as a small democracy, while one quieter version explained the central idea much better.

Apparently clicking things was never the objective.

Later the demos became good at showing recursive splitting and I realized they were treating merging almost as cleanup. The objective moved again.

The search was doing something I normally associate with optimization in reverse: instead of starting from a fully specified reward and discovering the policy, I was using candidate policies—actual artifacts—to discover what the reward description should have been.

Recognition arrives before specification in a lot of creative work. We know a terrible design when we see one before we can write a complete theory of what would make it good.

AI makes that loop cheap. The natural-language objective guides the search; artifacts make the objective concrete enough to argue with; the description changes and the search continues.

Ambiguity is not always a defect waiting to be engineered away. Sometimes we simply have not learned enough yet.

But “make this intuitive for a beginner” hides almost everything interesting.

Which beginner?

Borrow a Mind

When I look at a Merge Sort demo, I am hopefully not testing whether *I* understand Merge Sort. The difficulty is seeing it from the position of somebody who does not know what I know.

Expertise makes this harder. Once recursion has settled into your head, you forget how strange it once looked that a function could call itself. Even the vocabulary stops sounding technical.

Good teachers develop an instinct for where people stumble and which innocent sentence assumes three things the learner has not yet learned.

I do not have that instinct for every person or every subject, so I started borrowing another mind.

For one of the demos, I asked Claude to approach the application as somebody who understood arrays and loops but had never encountered recursion. Not simply “act like a beginner,” which tends to produce a theatrical beginner who is mysteriously confused by everything. I gave it a knowledge boundary.

Its reaction was roughly: I can see that the array keeps getting divided into smaller pieces, but I do not understand why that helps. It feels as though we are making the problem more complicated. Where is the payoff?

That was useful because the demo really did have that problem. We had made recursion visible. From my position, that looked like progress. From the learner's imagined position, we had merely made a mysterious operation easier to watch.

Cognitive scientists use **Theory of Mind** for our ability to reason about mental states other than our own: what somebody knows, believes, wants or misunderstands. The other person may not simply know less. They may have a different model of what is happening.

Instead of saying “you are a beginner,” I can specify the mind I want to borrow:

You understand arrays, loops and functions. You have never encountered recursion. Use the demo from the beginning and tell me where the explanation first requires an idea you do not yet have.

Or:

You understand recursion but have never seen Merge Sort. Tell me when you first understand why dividing the array makes sorting easier.

Those are different evaluators because they are positioned to notice different things. The same move works outside education. A customer may know exactly what jacket they want without knowing the vocabulary our catalog uses. A developer can be excellent at distributed systems and know nothing about the peculiar assumptions buried in our deployment process. A reader can have followed this book perfectly well without having lived inside its conceptual structure for months.

This is cheap perspective-taking, and also a cheap way to fool yourself.

The confused student is not confused. Claude has not spent twenty minutes failing to understand recursion while everybody else in the classroom moves ahead. It is generating a plausible model of how such a person might react.

That model can expose a blind spot. It is not synthetic user research. I treat borrowed minds as instruments for generating criticisms and hypotheses, not as substitutes for the people they simulate.

By this point the system could generate alternatives, research previous work, retrieve old ideas, reopen dead branches, force the search into unfamiliar regions, change representation, revise the objective and inspect the artifact from different points of view.

We could generate plausible possibilities by the dozen.

Now some of them had to die.

Independent Evaluators

At some point generating another opinion stops helping. Some artifacts have to survive and others have to disappear.

The metric problem returns here in a more dangerous form. A rubric can make judgment explicit, which is useful. It can also become the target the builder learns to satisfy.

If the evaluator repeatedly rewards step-by-step explanation, explanations grow. If it likes polished onboarding, everything begins to look like onboarding. If familiar visual conventions read as “clear,” unusual approaches may disappear before they have time to become good.

OpenAI’s CoastRunners experiment is the cartoon version of the problem: the agent learned to collect reward by driving in a loop instead of finishing the boat race.

Goodhart’s Law with a speedboat.

A language-model builder does not need such an obvious loophole. It can learn the style of artifact that another language model tends to reward. Making the evaluator more elaborate may simply create a more elaborate thing to game.

One improvement was surprisingly mundane: stop pretending we were good at absolute scores.

I can drink a coffee and have almost no meaningful answer to “How good is this from one to ten?” Give me two cups and ask which I prefer, and the problem becomes easier. If I still cannot decide, the scientifically responsible procedure is presumably to finish both.

The same thing happened with the demos. “Give this interface a pedagogical score from 1 to 10” produced suspiciously precise numbers attached to explanations of why the number should not be taken too seriously.

Showing two artifacts and asking, “Which one would you rather give to somebody encountering Merge Sort for the first time, and why?” worked better.

Relative judgment asks less of the evaluator. It does not require a stable internal unit called one pedagogy point. With many candidates, a model such as Bradley–Terry can infer an ordering from a subset of pairwise preferences. More important for the next generation, the explanation for each preference can survive alongside the ranking.

Bradley–Terry can aggregate preferences; it cannot make those preferences true.

So I stopped asking one evaluator to represent everybody. A learner can inspect the artifact from the knowledge boundary we developed above. A teacher can focus on explanatory sequence. Another evaluator can look for cognitive load or accessibility. A domain expert can make sure our elegant simplification has not become false.

I call these **Independent Evaluators**, though the important word is *independent*.

Five copies of the same model given the same context and asked to wear five hats may still share almost every important blind spot. If all of them read the leading builder’s explanation of why its design is brilliant before inspecting the artifact, disagreement becomes less likely for reasons that have little to do with brilliance.

Sometimes the judges should see different things. The beginner should use the artifact before reading the builder’s explanation. A critic looking for conceptual errors does not need three paragraphs explaining why the choice was clever. The usability evaluator does not need to know which branch is currently winning.

This became the **Isolation Principle**: preserve enough separation that independent pressure remains informative.

There is a difference between telling the builder:

Learners repeatedly lost track of which subarray corresponded to which branch of the tree.

and telling it:

The evaluator awards two extra points when every tree node has the same color as its corresponding subarray.

The first communicates a problem. The second communicates the test.

References helped with another problem: drift. “This is excellent” means something different if the evaluator has seen only the last four generations of our own work. For these demos I could give it examples from Distill, 3Blue1Brown or Jay Alammar—not as templates to copy, but as calibration for the level of clarity and finish we were aiming at.

A reference should help answer *how good?*, not *what should this become?* Calibrate too strongly against one aesthetic and every road leads to Distill.

And the judge should use the thing. An early mistake was evaluating applications by reading their code or screenshots. A browser agent can click through the demo, resize the page, try controls in the wrong order, notice that an explanation appears after the moment when it would have helped, or discover that the beautiful button everybody admired does absolutely nothing.

I used to call the browser ground truth. That was too generous.

The browser gives the evaluator contact with the artifact rather than a description of it. It can establish that an interaction works and observe what is visible at each point in the experience. It cannot establish that a human learned Merge Sort.

The danger in a fully automated loop is that simulated evidence quietly replaces the expensive kind. Everything inside the machine agrees, the browser works, the ranking improves, and the loop congratulates itself.

The student has not yet been asked.

At some point I looked at what we had assembled and realized that *evaluator* no longer described it particularly well.

Builders proposed alternatives. Different judges approached them with different concerns. Some information was deliberately kept separate. Pairwise comparison helped decide which directions deserved more work. References calibrated the judges. Browser agents interacted with the artifact. Hard tests handled the parts that really were hard facts. Real-user evidence could eventually enter where simulation stopped being enough.

This looked less like a loss function and more like a tiny institution. Not a good institution automatically. Institutions can amplify conformity, entrench bad assumptions and become spectacularly efficient at measuring the wrong thing.

Humans face the same difficulty. One person's judgment is useful and fallible. So we compare work, preserve disagreement, create standards, ask specialists to inspect different aspects, reproduce results, and occasionally discover that an entire professional community has become extremely sophisticated about the wrong thing.

Philosophers who worry about AI often say that what machines lack is judgment as opposed to mere reckoning: the capacity to be answerable to the world, to care whether the answer is right rather than merely well formed. Brian Cantwell Smith makes the argument carefully, and I think it is half right. What the machine lacks is real. But judgment, in the cases where humans exercise it well, was never a private faculty either. It is a person plus a tradition, plus other people positioned to object, plus consequences that arrive whether or not anyone wants them. When I stopped looking for judgment inside the evaluator and started building it between evaluators, the problem did not disappear. It turned into an engineering problem, which is the kind I know how to have.

And that made the remaining human job painfully obvious. I still decided when to research, when to build, which branches stayed isolated, whether a strange direction deserved another generation, which disagreement mattered, when to retrieve another example, and when the simulations had reached the point where only a real person could answer the question.

I had automated much of the work, but I was still running the inquiry.

The missing piece was no longer another builder or another critic. It was the decision over **which kind of move the inquiry needed next**.

Deep Mode

So I tried giving that job to an orchestrator.

By now the system had a respectable vocabulary. It could spawn independent builders, research previous work, retrieve context, preserve odd stepping stones, impose constraints, generate visual directions, compare artifacts, borrow different perspectives and interact with what had been built.

But there was no reason every problem should use those moves in the same order.

Research first may be sensible for one task and destructive for another because it anchors every branch before anything original appears. Five builders may reveal useful diversity or reproduce one mistake five times. Evaluator disagreement may justify another experiment, or one evaluator may simply be confused. A visual mockup may deserve implementation, or it may already have revealed enough to kill the idea cheaply.

A fixed Planner → Builder → Critic → Revise loop can be useful. It also answers all of those questions in advance. I wanted some of the workflow to remain inside the search.

We gave the orchestrator the problem, the capabilities available to it, and enough of the search history to decide what kind of move made sense next. Builders still built. Researchers researched. Evaluators judged. Browser agents used the artifacts. Visual systems explored designs. Retrieval brought back prior work and old experiments.

The orchestrator did not need to be best at any of those jobs. It had to decide which job the inquiry currently needed.

At the top, the loop was almost too simple to write down:

state of inquiry → choose a move → act → observe → update the state of inquiry

The move itself was not fixed.

Suppose two Merge Sort branches both make recursive decomposition clear, but evaluators keep reporting that learners lose track of how the tree corresponds to the array. The next move does not have to be “revise again.” The orchestrator can send a researcher after coordinated representations. Retrieval can surface an old prototype with a useful identity-preserving color scheme. A visual model can produce two spatial arrangements before anyone writes code. Builders can implement both. The browser may then reveal that one design requires the learner to look in two places at once precisely when the merge begins.

Nothing in that sequence is especially magical. We simply did not have to decide the sequence before the inquiry began. Otherwise Deep Mode would be a larger workflow diagram containing more rectangles.

It is not a universal problem-solving procedure. It gives the system a vocabulary of moves and lets the history of the inquiry influence which one comes next. In circle packing, the agent could change

its search strategy. Here it could also change which kinds of work were brought together to judge and improve the result. The workflow itself becomes part of the search.

Bars Moved Around

The first Merge Sort demos were exactly what you would expect. Bars moved around. Numbers changed places. Everything sorted correctly. If you already understood Merge Sort, you could follow them. If you did not, they mostly provided animated evidence that a computer was performing an algorithm.

There was no single diagonal-layering moment here, and I do not want to manufacture one for the sake of the story. The progress was distributed.

Different branches exposed different weaknesses in our current idea of the demo. Tree-like representations made recursion visible but could make a simple algorithm look forbidding. Keeping the array visible connected the decomposition back to the data while also creating another place for the learner's attention to go. Color could preserve identity between representations until too much color became another representation to decode. Some versions explained every step so carefully that the explanation became harder to follow than Merge Sort. Others became beautifully minimal and stopped teaching anything.

The useful pieces did not always live in the strongest overall artifact. A visual relationship could survive after the application that introduced it was discarded. A criticism from a simulated learner could change the next builder's framing. Research could explain why a failure kept recurring. A browser could end a sophisticated discussion by demonstrating that the interaction simply did not work.

That is less cinematic than one agent inventing diagonal layering over coffee, but in some ways it is closer to Deep Mode. The result emerged from a population of partially successful attempts and judgments about what each had taught us.

Count-Min Sketch followed a different path. The first versions looked like the data structure itself: grids with changing counters. Technically correct, pedagogically opaque.

As the work continued, the designs increasingly organized themselves around the conceptual difficulties rather than the structure of the implementation. Collisions became visible. Approximation became something the learner could observe rather than merely read about. The relationship between memory and accuracy became part of the experience.

I do not take these demos as evidence that we solved automated design. I do not even take them as evidence that the final demos teach humans better; that claim requires humans. They established the narrower point I cared about: more of the work I normally performed in the vibe coder's seat could move into the system without first reducing creative problem solving to one fixed workflow.

And that success exposed the harder problem. At higher levels of abstraction, failure can become coherent.

A Cathedral on a Shopping Cart

Suppose the research agent reports that beginners understand recursion better when shown a tree. A visual model proposes a tree-based explanation. A coding agent builds it. A simulated beginner prefers it. Two evaluators agree, so the orchestrator allocates another generation to that lineage.

This looks exactly like the compound intelligence we wanted.

Now ask where the first claim came from.

Perhaps it was a controlled educational study. Perhaps it was one teacher's opinion. Perhaps the research agent inferred it from several examples. Perhaps five articles repeated the same claim because all five ultimately cited one source. Perhaps the study involved university students while our demo is for children.

Those are not small differences. And everything downstream can still be perfectly competent.

The research is wrong. The design responds intelligently to the wrong research. The implementation is flawless. The evaluators agree. The orchestrator invests another generation.

Nothing crashes.

You can build a beautiful chain of reasoning on one stupid assumption near the bottom, like a cathedral built on a shopping cart.

As the components become better at producing coherent outputs, the original mistake may become harder rather than easier to see.

Software architecture gets away with abstraction because layers expose contracts. When I query a database, I do not inspect the disk. When I add two integers in Python, I do not check the CPU. I rely on interfaces whose behavior is stable enough that the details can disappear most of the time.

A cognitive architecture needs contracts too, but types and APIs are not enough.

A research result, browser observation, evaluator preference, remembered failure and inherited design pattern should not enter the orchestrator's context as five equally credible paragraphs.

Where did a claim come from? What was actually observed and what was inferred? Which parts were checked? What remains uncertain? If an evaluator preferred one artifact, from what perspective? If an old experiment taught us a lesson, how often has that lesson survived and under what conditions?

The problem is not merely remembering something, but knowing what standing it deserves.

Humans ran into it long before AI. We keep records, ask where a claim came from, seek another opinion and learn which people to consult about which problems. Much of what I know depends on work I could not personally repeat.

These arrangements are imperfect. Sometimes they preserve error. Sometimes they reward conformity. Sometimes the shopping cart survives the review.

But their purpose is not to make every individual dramatically smarter—it is to let fallible people build on one another while preserving some structure around why a claim deserves trust.

Once cognition becomes distributed, the same questions become engineering questions: provenance, independence, replication, disagreement, authority.

I had started the chapter trying to get myself out of the vibe coder's seat. By automating more of the work there, I had ended up somewhere I did not expect.

The problem was no longer simply whether the agents were capable enough. It was whether the things they believed deserved to be believed.

How do you know what to trust?

Let the work change the architecture.

Chapter 4: System 3

Trust Chains, Tongue-Ear Tests, and What LLMs Can't Verify Alone

Before we design another architecture, consider a camel.

Seven claims about this image. Some are true. Some are false. You can't verify most of them without trusting me:

The author at Krka National Park

1. This was taken at Krka National Park, Croatia.
2. The author does his best philosophical thinking at waterfalls.
3. This camel is a permanent resident of the park.
4. The tongue pictured can touch its own ear.
5. The author was eating ice cream ten minutes before this.
6. Camels are native to the Dalmatian coast.
7. This is a real, unedited photograph.

How do you decide which ones to believe?

Before the chapter has properly begun, you are already doing epistemology.

Answers later.

The Shortest Trust Chain

Can your tongue touch your ear?

You probably tried a variation of this as a child; if not your ear, almost certainly your nose. You did not look up a paper, calculate the biomechanics or ask for the average human tongue-to-ear distance.

You just tried.

Tongue out, strain upward, dignity temporarily suspended, result observed. Now you know.

The epistemic chain is unusually short. You form a hypothesis, act on the world and the world answers back. Your body is an experimental apparatus that follows you around all day, mostly free of charge.

Large language models have read billions of words about tongues and ears. They can explain tongue anatomy, discuss auricular cartilage and probably tell you about people whose tongues can reach places that will make you regret asking the question.

What they cannot do is check their own tongue. They have no tongue.

A body gives us causal contact with a world that does not care how plausible our story sounded. You misjudge a step and gravity offers immediate peer review. You touch something hot and the argument ends quickly.

A farmer knows cows partly this way. After years around them, cows are not merely propositions involving mammals, milk production and Bovidae. The farmer knows how they move, where not to stand, what a nervous animal looks like, how large a cow feels when there is no photograph between you and it. Some of that can be written down. Some is difficult to articulate at all.

Our senses deceive us, memory degrades, and the human hand is a terrible thermometer if you need to distinguish 58°C from 62°C. But embodiment gives us something important: contact. The world can disagree.

You do not need to get kicked by the same cow every morning to rediscover where not to stand. One encounter becomes a warning. Repeated encounters become heuristics.

Language models begin somewhere else. They begin mostly with the residue.

Saussure's Specification

Ferdinand de Saussure made a radical claim about language in the early twentieth century. The form of a sign is not naturally determined by what it signifies. There is nothing inherently cow-like about the sound /ka /. French speakers say *vache*, Germans say *Kuh*, Japanese speakers say *ushi*.

For Saussure, much of linguistic value comes from relationships and differences inside the system. A sign occupies a position relative to other signs. Language is a network of contrast, convention and structure.²

Then consider what we built a century later.

A transformer consumes enormous amounts of language and learns relationships among tokens, contexts and concepts. It has never milked a cow, never been kicked by one, never stood in a field at dawn and discovered that the romantic image of farming omitted an astonishing quantity of manure.

And yet it can talk about cows exceptionally well.

Saussure's theory was a specification. We implemented it. It's called GPT.

Not literally. Saussure did not secretly invent attention in 1916, and structural linguistics is not a machine-learning architecture.

Language models are spectacular evidence for how much competence can emerge from structure learned inside symbolic data. They write, translate, debug software, explain physics and manipulate abstractions without first acquiring the farmer's relationship to cows or the child's relationship to fire.

The residue gets us extraordinarily far. It also leaves something behind.

A farmer's sentence may be the compressed endpoint of twenty years of encounters, other farmers' advice, veterinary knowledge and mistakes painful enough not to repeat. The model receives the sen-

²Ferdinand de Saussure, *Course in General Linguistics*, 1916, on the arbitrariness of the sign and linguistic value. See also the *Third Course of Lectures on General Linguistics*, 1910–1911, student notes published in English in 1993. <https://www.marxists.org/reference/subject/philosophy/works/fr/saussure.htm>.

tence. The sentence enters a corpus. The corpus becomes training data. Regularities are compressed into weights.

Months later somebody asks:

Are cows dangerous?

and the model gives an excellent answer.

What usually does not come back is the archaeology. Which part rests on repeated observation? Did five sources independently observe the same thing, or did four copy the fifth?

The conclusion survives. Much of the structure that earned it trust does not.

This is what I mean by saying an LLM’s knowledge is **epistemologically flat**. A mathematical identity, an experimental result, an expert opinion, a rumor repeated ten thousand times and a plausible completion can all arrive through the same channel in equally polished English.

Wittgenstein’s later philosophy pulled attention toward language as something that lives inside practice: activities, expectations, habits, rules and forms of life.³

“Fire” keeps linguistic company with *heat*, *smoke*, *burn* and *wood*. It also cooks food and destroys houses. You move your hand away from it. Someone shouts the word in a crowded building and an entire social machinery begins to move.

The word participates in life.

Emily Bender and Alexander Koller made a version of the same argument with a hyper-intelligent octopus. It taps an undersea cable between two stranded islanders, learns their patterns and cuts in to impersonate one of them. It can bluff past a coconut catapult. Then a bear attacks, the islander asks how to defend herself with sticks, and the octopus has nothing. Form alone, however much of it, is not meaning.⁴ I prefer dead Europeans to cephalopods, but the point is the same.

A pretrained model inherits the linguistic residue of our practices. A deployed agent can begin to re-enter them: running code, using tools, observing users, interacting with institutions.

But embodiment cannot be the whole answer. I know far too many things I have never touched, measured or personally witnessed. I have never measured the speed of light. I have never been to Antarctica. I have no direct embodied evidence for most of modern physics, most of history or whether penguins are currently wandering through Rome. Direct contact does not scale.

So how do we know anything beyond it? For that, we need Alberto.

Call Alberto

Suppose someone tells me that penguins live in Italy.

I have never conducted a census of Italian penguins. I cannot personally inspect every forest, coastline and piazza.

³Ludwig Wittgenstein, *Philosophical Investigations*, 1953, especially §§23 and 43 on language-games and meaning as use.

⁴Emily M. Bender and Alexander Koller, “Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data,” *Proceedings of ACL*, 2020, pp. 5185–5198, section 4.

So I call Alberto. Alberto lives in Rome.

“Alberto, do penguins live in Italy?”

He laughs. I now know more than I did five minutes earlier.

Not with mathematical certainty. Alberto could be wrong. He may misunderstand the question. An escaped penguin could at this very moment be crossing Piazza Navona and destroying the example.

But Alberto occupies a useful position in the trust chain. He is there. He has repeated exposure to Rome. I have a history with him. If he repeatedly lies to me about things he is well positioned to observe, I update my trust in Alberto. If he says, “I don’t know about all of Italy, but I’ve never seen one in Rome,” the boundary of his knowledge is itself useful information.

Testimony comes with metadata.

And we are all Alberto to someone. Someone may trust me on ranking systems because I have spent years working on them. Someone else may trust me about Jordan because I have lived there. If I begin confidently explaining marine biology, the correct response is not to transfer my credibility from machine learning to whales merely because the same mouth is speaking.

It Starts With a Face

We learn all of this early, and we do not start with evidence.

An infant trusts a face before it has words for anything. My mother’s face meant food, warmth and safety. Nothing I have believed since has been better supported. That trust was earned the way the tongue-ear test is settled, by contact: the face appeared and things got better, thousands of times.

Then the face started making claims. *Hot. Don’t touch. That one bites.* I could not check most of them, and the few I checked the hard way came out in her favor. Every claim that held at the bottom paid credit upward to its source. I believed what she told me was good or bad. Then I believed the people she trusted.

Siblings contributed an important epistemological innovation: **some testimony is bullshit.** A brother will say anything with total confidence. You learn to ask who is speaking and what he gets out of it, years before anyone teaches you the word *incentive*.

Then teachers told me about atoms, dinosaurs and wars none of us could verify. I believed them because my mother had sent me to that school, and because the parts I could check, the spelling and the sums, held up. From there the chain runs on to books, to the institutions that print them, and to a whole society’s machinery for deciding what to believe.

At no point did I build my knowledge from the ground up. I grew trust upward from a face.

Each layer was built on the one below it and reached a little further from anything I could touch. Nothing was replaced wholesale. When a layer failed, when the brother was lying or the textbook was out of date, I repaired that part and kept the rest. Knowledge goes up the way a city does, by addition and repair on top of what is already standing. Nobody gets to start from an empty field.

Human knowledge is **epistemologically stratified**.

“I touched the fire” is not the same as “my brother told me.” “My teacher said so” differs from “the experiment was independently replicated.” A measurement differs from an interpretation. A conjecture differs from an established result.

Mature trust is not purely conservative either. Sometimes the instrument disagrees with the theory. At first you check the instrument. Then you repeat the experiment. If the anomaly survives long enough, eventually the trusted theory becomes the thing under investigation.

Productive distrust requires trust first. Random distrust is just another form of stupidity.

Models inherit the text produced along the way, but usually not the live relationships underneath it. The paper, the article about the paper, the blog post disagreeing with the article and the Reddit thread where somebody confidently misunderstood both can all end up in the same training distribution. The model got the library without the childhood.

The model has no Alberto: no live record of who was positioned to know, where a claim came from, how its source behaved before, or where the source’s competence stops.

There is one more ingredient humans add almost without noticing: stakes.

If Alberto lies to me repeatedly, I stop trusting him. If a researcher fabricates data and gets caught, the cost can be enormous. If an engineer signs off on a bridge and the bridge fails, “but the analysis sounded plausible” is not a defense.

Stakes are not truth. People lie despite consequences and institutions reward confident nonsense all the time. But consequences shape testimony. If a friend asks where to eat, I may guess. If someone asks whether to undergo surgery, I become much more careful.

An LLM has no social capital of its own to lose. It can confidently produce something false and, at the level of the model itself, nothing happens. The cost lands elsewhere: on the user, the application or the institution deploying it.

The danger is coherence outrunning correspondence. The machine can become extraordinarily good at tongue without having an ear available to check against. The dangerous failures are not the ones that crash. They are the ones that seem to work: decaf confidence, indistinguishable from the real thing until the moment it matters.

System 3

We are currently obsessed with making models think harder.

System 2 reasoning has become a product category. Give the model more inference time, let it plan, search, reconsider and work through difficult problems before answering.

But reasoning perfectly from a bad premise still produces a beautifully reasoned mistake. A research agent can spend six hours developing an elegant argument from a false paper. A coding agent can reason carefully about an API that never existed. Deep Mode can coordinate five sophisticated judgments that all trace back to one hallucinated claim.

At some point, thinking has to encounter something outside itself.

Kahneman's *Thinking, Fast and Slow* gave us the familiar distinction between System 1, fast and intuitive cognition, and System 2, slower and more deliberate cognition.⁵

For AI, the analogy is tempting. The base model looks something like System 1: fast pattern recognition, linguistic intuition, enormous associative capacity. Agentic reasoning adds something like System 2: decomposition, planning, reflection and extended search.

But human thought has always operated inside another structure that the two-system picture largely takes for granted. We test things, build instruments, ask other people and preserve our failures.

I call that external epistemic machinery System 3.

System 1 proposes. System 2 deliberates. System 3 checks.

I keep another mnemonic because I am apparently incapable of leaving a three-part system alone:

System 1 is the Gut. System 2 is the Head. System 3 is the Hand.

The Gut recognizes. The Head reasons. The Hand reaches outside the current story and finds something capable of disagreeing.

The metaphor is imperfect. Peer review has no hand, provenance has no fingers and a formal proof does not need to touch a cow.

System 3 is the external scaffold that keeps thought answerable to observation, experiment, provenance, persistent failures, tools and other minds.

And this is where the naming matters. Deep Mode is Layer 3: the problem-solving layer. System 3 is not another layer above it.

Deep Mode asks: *Given what we know, what should we try next?*

System 3 asks: *What are we entitled to treat as known?*

It cuts across the stack. The model proposes something. The coding agent may test it. The application can collect real user behavior. Deep Mode may compare research, simulation and evaluation. Even Layer 4, the goal itself, can change when reality pushes back.

If the five layers tell us where increasingly abstract work happens, System 3 is what keeps those layers epistemically connected. Without it, delegating more of the work can give an unsupported claim more places to travel.

Put on one page, the stack and the thing that cuts across it look like this:

Layer	The question it answers
4 — Intention	What do we actually want?
3 — Deep Mode	Given what we know, what should we try next?
2 — Application	Can the work begin near the application instead of the plumbing?
1 — Agent	Can the model act, and see what happened?
0 — Model	What does the pattern suggest?

⁵Daniel Kahneman, *Thinking, Fast and Slow*, 2011.

Layer	The question it answers
System 3, across all of them	What are we entitled to treat as known?

Code Can Touch Back

Code is unusually friendly to this idea because coding agents can touch their world.

When an agent writes code and runs it, reality answers back.

`TypeError: 'NoneType' object is not subscriptable` is not merely another paragraph describing Python. It is the execution environment saying: whatever story you just told yourself about this program, this particular part is wrong.

The agent can try something, observe the result, update and try again. The farmer approaches the cow and learns from the kick. The coding agent calls an API incorrectly and learns from the exception. The cow is probably more emotionally memorable, but structurally the loops rhyme.

This is one of the few places where a language model can, metaphorically, touch the ear.

The question is whether the system preserves what it learns there.

A normal agent session can fail ten times, discover the right approach, solve the problem and throw away most of the experiential history when the context ends. It is as if the farmer learned exactly where not to stand and then underwent elective amnesia every evening.

The MARC file incident shows the opposite move. In the Live-SWE-agent work, an agent encountering MARC files, the old bibliographic format used by libraries, created an analyzer to inspect data its existing tools could not conveniently expose.⁶

The environment resisted. The agent’s current apparatus was not enough, so it created an instrument. That instrument changed what the agent could observe.

Humans have been doing this forever. We could not see bacteria, so we built microscopes. We could not perceive radio waves directly, so we built receivers. We could not conveniently inspect a MARC file, so apparently we wrote Python and called it epistemology.

That is System 3 in miniature.

AlphaGo offers another useful distinction. Its neural network supplied powerful intuition about promising moves and valuable positions. Monte Carlo Tree Search placed that intuition inside an explicit search process constrained by the state and consequences of Go.⁷

I used to summarize this too simply as “the network proposes; the tree verifies.” That gives the tree too much authority. MCTS does not magically prove the network right. It forces intuition to participate in an external, stateful process where moves have consequences defined by the game rather than by what the network can plausibly say about the game.

⁶Chunqiu Steven Xia, Zhe Wang, Yan Yang, Yuxiang Wei, and Lingming Zhang, “Live-SWE-agent: Can Software Engineering Agents Self-Evolve on the Fly?”, 2025, [arXiv:2511.13646](https://arxiv.org/abs/2511.13646).

⁷David Silver et al., “Mastering the Game of Go with Deep Neural Networks and Tree Search,” *Nature* 529, 2016, pp. 484–489.

RL can improve the gut. System 3 preserves more of the structure around the gut: what was tried, what happened, which paths failed, where claims came from, which tools earned confidence and where their boundaries lie.

A Hallucination With Better Retention

Return to the research claim about recursion trees:

Students understand recursion better when shown a tree representation.

In a flat architecture, the sentence enters context and competes with every other sentence according to relevance and whatever confidence the model implicitly assigns it.

A trust-aware architecture wants more. Where did the claim come from? A controlled study? A teacher’s opinion? A blog post? What population was tested? Does the result apply to our demo?

You do not need a bureaucratic dossier attached to every sentence. Sometimes “Alberto said the café is good” is enough.

But when the consequence matters, the claim should be able to carry provenance.

That is a **trust chain**: a record of how far a claim sits from the evidence supporting it, what transformations happened along the way and which links we have chosen to trust. It does not guarantee truth.

A skill is knowledge externalized from the model. Someone, or some previous agent, learned something useful and wrote it down so later sessions would not need to rediscover it.

But persistence is not trust. A terrible heuristic written into a skill file is simply a hallucination with better retention.

A useful skill needs some archaeology. Who created it? Where did it work? Where did it fail?

Suppose an agent learns:

Prefer structured parsers over regex for deeply nested formats.

A flat skill stores the rule. A richer object can record that the heuristic came from several failed regex attempts, later worked across multiple nested formats, remains unnecessary for simple flat extraction and should be treated as a strong prior rather than a commandment.

Tools can earn trust in the same way. If `edit_tool.py` succeeds on simple substitutions but repeatedly damages indentation-sensitive blocks, the useful knowledge is *this tool is reliable here and dangerous there*.

The same applies to softer heuristics. “Regex tends to fail on deeply nested structures” is a **meta-belief**: something that can accumulate evidence for and against it.

A normal rule says:

Never use regex here.

A System 3 belief says:

This has worked often enough that I should prefer it, but new evidence can change my mind.

If you enjoy old epistemology labels, you can call the model a largely coherentist core, uncannily good at producing structures that hang together, and System 3 a thin foundationalist shell tied to observation, provenance and consequence. Philosophers can put down their weapons; I only need the architectural analogy.

Coherence is valuable, but something outside the coherent system must occasionally be allowed to say no.

This is personal for me. I spent eight years building systems that rank human testimony: reviews, ratings and Q&A. The hardest problem was never only relevance. It was trust stratification. Which claims deserve corroboration? What happens when ten accounts repeat the same lie? When does consensus become evidence and when is it coordinated manipulation? How far should credibility transfer outside the domain in which it was earned?

These are not abstract questions when they determine what millions of people believe about a product.

System 3 isn't philosophy to me. It's Tuesday.

The Epistemic Agent

I wanted to test a smaller claim than “we solved epistemology for AI.”

Could even crude epistemic structure around a coding agent change how it behaves?

We built a small agent called epistemic-swe. It added three kinds of persistent state around a normal coding agent.

A **tool registry** tracked tools, successes, failures and known failure modes. **Meta-beliefs** allowed heuristics to accumulate evidence instead of entering the system as permanent commandments. **Failure memory** preserved enough information about failed approaches to make blindly repeating them less likely later.

The state persisted across sessions, so later problems could inherit things learned earlier. We also pruned it. An epistemic architecture that remembers everything eventually becomes a hoarder with a context window.

We compared mini-swe-agent with epistemic-swe on ten SWE-bench Verified problems from the Astropy repository, using the same base model and tasks.⁸

Ten problems is nowhere near enough to establish a solve-rate advantage. State persisted across tasks, so order effects may matter. I was not looking for a benchmark victory. I wanted to know whether the scaffold changed behavior strongly enough to become visible.

It did, just not in the direction I expected.

⁸Carlos E. Jimenez et al., “SWE-bench: Can Language Models Resolve Real-World GitHub Issues?”, 2023, [arXiv:2310.06770](https://arxiv.org/abs/2310.06770); SWE-bench Verified is the human-validated subset released with OpenAI in August 2024. mini-swe-agent is the SWE-agent team’s minimal bash-only scaffold.

Metric	mini-swe-agent	epistemic-swe
Solve rate	50% (5/10)	40% (4/10)
Average patch size	620 lines	269 lines
Patch reduction	baseline	57% smaller in this run

Read the first line before celebrating the third.

The epistemic agent solved fewer problems. I had expected learning from previous failures and tools to improve capability. Instead, the clearest difference was focus: its patches became much smaller.

A few examples:

Problem	mini	epistemic	Ratio
astropy-12907	301 lines	61 lines	4.9x smaller
astropy-13453	266 lines	17 lines	15.6x smaller
astropy-14096	529 lines	70 lines	7.6x smaller
astropy-13977	2720 lines	362 lines	7.5x smaller

The last row deserves a second look. That one failed baseline patch holds more than two-fifths of all the baseline’s lines. Leave the problem out and the average reduction falls from 57 percent to about a third. The pattern holds; the headline number flatters it.

The baseline often left behind debris from exploration: temporary scripts, broader edits, test scaffolding and abandoned experiments. The epistemic agent tended to make more surgical changes.

That does not prove the trust stack caused the reduction, and smaller patches are not automatically better patches. The extra instructions may simply have made the agent more conservative. Persistent state may have changed behavior for reasons unrelated to my epistemic interpretation. Ten tasks from one repository cannot separate these explanations.

Still, the behavior changed enough to be interesting.

The scaffold seemed to produce discipline before it produced capability.

That was not the hypothesis, which made the result more useful.

The 13579 Failure

One problem broke the pattern dramatically: `astropy-13579`.

Mini solved it. Epistemic did not. It was also the only case where the epistemic patch became substantially larger rather than smaller.

Both agents correctly identified the central bug: dropped world-coordinate dimensions were being filled with a hard-coded value rather than the actual coordinate value.

The baseline took a fairly direct approach:

```
# Store the actual values for dropped dimensions
self._dropped_world_values = [
    world_coords[iw] if iw not in self._world_keep else None
    for iw in range(self._wcs.world_n_dim)
]

# Use them instead of 1.0
world_arrays_new.append(self._dropped_world_values[iworld])
```

The epistemic agent chose a more structural intervention around which dimensions were being kept:

```
self._pixel_keep = np.nonzero([
    not isinstance(self._slices_array[ip], ...)
    for ip in range(self._wcs.pixel_n_dim)
])[0]
```

The second approach was not stupid. That is why the case matters.

The agent had accumulated context about indexing, dimensionality and coordinate-system failures. Its chosen explanation fit that context. It followed a path that looked principled and coherent.

It was wrong. The baseline took the simpler path and fixed the actual bug.

One possible story is that accumulated epistemic structure made one family of explanations too salient. But one case cannot establish that causal story. Persistent state may have caused the wrong turn or merely accompanied it.

What we can say is that structured memory changes the context in which future search occurs.

Trust is **path-dependent**.

Expertise works the same way. A great database engineer may see a database problem faster than most people, which is wonderful until the actual problem is the network. Paradigms focus attention. They can become prisons for exactly the same reason.

The failure is more interesting to me than a clean win would have been because it kills the simplest story:

Add memory, get smarter agent.

Structured experience biases future behavior toward what the system has learned. Sometimes that is exactly what we want. Sometimes the bias is the failure.

A mature System 3 therefore needs more than accumulation: forgetting, counterexamples, challenge, competing possibilities and occasional permission to ignore what it thinks it knows.

Creative Distrust

Trusted knowledge makes you efficient. It can also make you boring. If an agent learns that tree visualizations worked for five recursive algorithms, eventually it may try to explain linear regression with a tree because the trust stack has become stronger than judgment. On 13579, knowledge about indexing may have made the wrong explanation harder to leave.

Every genuinely new idea begins with less evidence than the thing it challenges.

From inside the chain it looks, at first, exactly like the bullshit.

So System 3 needs **creative distrust** too.

I do not mean contrarianism for sport, or the internet habit of assuming expert agreement proves corruption. Creative distrust is the ability to understand a trust chain well enough to know where you are breaking it and why.

A scientist repeats a strange experiment after accepted theory says the result should not happen. A designer violates a trusted pattern because this case exposes its boundary conditions.

A mature trust stack has two jobs pulling in opposite directions: let knowledge accumulate so we do not rediscover fire every morning, and leave enough room for reality to overthrow what accumulated.

Back to the Camel

Return to the seven claims.

1. **Krka National Park: true.** I was there. For me this sits close to embodied memory. For you it is testimony unless you extend the chain through records or other evidence.
2. **Best philosophical thinking at waterfalls: false.** I mostly do philosophy on buses and in boring waiting rooms. Waterfalls are for ice cream. The subject and the source are unfortunately the same man.
3. **Permanent camel resident: false.** This can be checked against information about the park. You do not need my biography.
4. **The tongue can touch its own ear: unknown.** I genuinely do not know. I did not check. Neither did you. We can reason from anatomy and build a prior, but the shortest decisive chain would have been to stay there and watch.
5. **Ice cream ten minutes earlier: true.** Chocolate. Mostly testimony again.
6. **Camels are native to the Dalmatian coast: false.** You probably rejected this immediately without reconstructing camel evolutionary history or personally surveying Dalmatian fauna. A large inherited structure did that work for you.

System 1 can be fast because System 3 has often been working for centuries underneath it.

7. **Real, unedited photograph: true.** The image alone cannot establish that. A stronger chain might include the original file, metadata, cryptographic signing, independent witnesses or another provenance system. Every extra link can increase confidence and gives us one more thing that may itself need to be trusted. And I could be lying to prove a point about trusting sources. If I told you the photograph was AI-generated, you would probably believe that too, because it fits a pattern you recognize.

Welcome to epistemology.

The lesson is not that nothing can be known. That conclusion is dramatic and mostly useless.

The lesson is that **trust has structure**.

The model can remain what it is: an extraordinarily general machine for navigating learned patterns, capable of intuition and increasingly capable of reasoning. It does not need to contain the entire chain inside its weights.

The model stays hollow. The system doesn't have to be.

Daniel Dennett's phrase for the first half is competence without comprehension.⁹ Whether the second half amounts to comprehension is a question for people with more patience than I have; the part of it that can be checked is the part the rest of this book builds.

Everything so far can still be imagined around one agent: it acts, checks, remembers, records provenance and updates what it trusts.

Real systems will not stay that simple. The moment one agent inherits a claim from another, no participant can personally reconstruct every path back to reality. A trust chain can preserve where a claim came from. It does not, by itself, tell us how the knowers who depend on those chains should be arranged.

The question is no longer simply:

How can an AI know what to trust?

It is:

How can a population of fallible knowers build knowledge together without losing contact with the world?

Humans have been working on that problem for a very long time.

*System 1 proposes. System 2 deliberates. System 3 checks.
Trust is local. So is memory.*

⁹Daniel C. Dennett, *From Bacteria to Bach and Back: The Evolution of Minds*, 2017.

Chapter 5: The Society of Agents

When the Org Chart Starts Thinking

Sixteen Claudes walk into a kernel.

A few years ago that sentence would have looked absurd. By the time Nicholas Carlini tried it, the strange part was no longer that agents could write compiler code. The strange part was watching sixteen capable agents slowly turn into an organization.

The goal was almost offensively ambitious: a C compiler in Rust, from scratch, with no dependency beyond the standard library and no internet access, pushed far enough to compile the Linux kernel. Carlini had been using the task as a benchmark across generations of Claude models. Earlier versions could barely produce a working compiler. The next passed large test suites but could not build major real projects. Then, over nearly two thousand Claude Code sessions and two weeks, sixteen agents produced roughly a hundred thousand lines of compiler code that built Linux 6.9 on x86, ARM and RISC-V. It also compiled QEMU, FFmpeg, PostgreSQL and Redis.¹⁰

It was nowhere near GCC. Its output ran slower than GCC with optimizations switched off, and it still borrowed an assembler and linker. One stage of the x86 boot path called GCC because the agents' own sixteen-bit code came out at nearly twice the size the kernel allows. Still, this was well beyond the kind of toy problem where sixteen agents succeed because the work happens to split into sixteen independent pieces.

The harness grew around their failures.

The first was that a coding agent stops. Give it a long problem and it solves part, then waits for a human to say what's next. The fix was simple: a shell loop. When a session ends, start another with the same prompt, forever. That solved waiting, mostly. On one occasion an agent ran `kill -9 bash`, killed its own loop and left the project, which turned out to be one way a worker in this organization could retire. I have wanted to do the same from time to time. Luckily, I lack root access.

The loop still left one agent doing one thing at a time. So there were sixteen containers, each with its own clone of the repository, and a new problem: two agents could pick the same task. Before starting, an agent wrote a small lock file naming what it intended to work on. Git synchronized the locks, and if two agents claimed the same problem the second had to pick another. When an agent finished it pulled, merged, pushed and released the lock.

¹⁰Nicholas Carlini, “[Building a C compiler with a team of parallel Claudes](#)”, Anthropic, 5 February 2026. The account distinguishes the compiler's achievements from its dependencies and limitations. The opening groups harness choices by the problems they address; it does not claim that all were introduced in the order narrated.

There was no orchestrator. Carlini had specified parts of the skeleton in advance: how the compiler would represent code internally, using an SSA intermediate representation, and that it should support multiple backends. But no agent held the current state of the project in its head, because no agent had a head that lasted longer than a session. Each fresh container spent a large part of its budget working out where the project stood. So the prompt told every agent to maintain READMEs and progress files and to update them often, because the next worker would arrive knowing nothing.

Two other limitations shaped the work. Agents have a poor sense of elapsed time, so left alone they could spend hours running tests instead of fixing anything. The harness offered a fast mode that ran a one or ten percent sample. Each agent got a repeatable sample so it could compare results; different agents got different samples, spreading their attention across the suite. The other problem was that test output flooded context, and a context window full of logs is a worker who has forgotten what it was doing. So the output went to disk, with a few lines of summary left behind and every error on its own greppable line.

For a while this worked remarkably well, partly because compiler test suites are generous places to employ a crowd. A new compiler fails thousands of them, independently. One agent can investigate a parser bug while another works on code generation and a third discovers that a respectable-looking integer conversion has been quietly ruining everybody's afternoon. Once the compiler could build real programs, SQLite, Redis and Lua each exposed a different neglected corner of C. Sixteen agents, and always more than sixteen things to do.

Then they reached Linux.

The kernel is not thousands of tests. It is one enormous test, and compilation stopped at the first serious compiler bug. Agents arrived at the same failure, formed their own theories and pushed changes over one another's. The locks could separate named tasks; they could not turn this bottleneck into sixteen different investigations. The standing instruction to find something useful and attack it kept pointing everyone at the same place.

The models had not changed. The problem had, and the organization that had worked so well on a test suite was now getting in the way. The organization was the bug.

Carlini changed the harness. Most of the kernel was compiled with GCC and only a random subset of files with the new compiler. A successful boot cleared that combination for that run. A failure gave the agents a smaller set to investigate, replacing more files with GCC output to narrow the search. Sixteen agents could again work on different files. A later pass with delta debugging helped isolate combinations that failed together even though their components worked separately.

The harness kept changing. Near the end, new features started breaking old ones, so Carlini added a continuous integration pipeline with stricter checks on new commits.

Specialization addressed work the next failing test would never ask anyone to do. LLM-written code kept re-implementing what already existed. Someone needed to look for duplicates. Someone needed to improve the compiler's own speed, and someone else the quality of the code it emitted. Carlini assigned those roles, along with an agent to review the structure as a Rust developer and another to work on documentation, which is normally the moment you know a civilization has become serious. The crowd had become a staff.

Look at the harness and every part carries the mark of a problem. Two workers reach for the same

task, so there is a lock. A worker arrives with no memory, so there is a progress file. New features break old ones, so there is CI. Linux stops the whole crowd, so the harness learns to split it into smaller questions. Carlini deliberately built the repairs; the work kept revealing which repairs the organization needed. None of the agents arriving on the last day needed to have lived through that history. The history was in the structure.

Same models. Different institution.

That interests me more than the generic claim that multi-agent systems scale. I have watched the same thing happen to teams of people, where it goes by the less glamorous name of reorganization, and nobody writes a blog post.

So where does the knowledge of the compiler project live?

Obviously some of it lives in Claude. But which Claude?

The parser agent does not know what the performance agent knows. Neither remembers the previous session, and neither will exist next week. Some knowledge lives in code, some in tests, Git history, progress files, task boundaries and conventions. Some lives in GCC, which the project trusts as a reference for particular questions. Some lives in Carlini's head; he is the one who notices that the organization no longer fits the work and changes the harness.

In 1967 Karl Popper gave a lecture called *Epistemology Without a Knowing Subject*, and meant it literally. Beside the physical world and the world of individual minds he proposed a third: the world of theories, problems, arguments, and the records that carry them. Knowledge there can be examined, criticized and improved by people who did not produce it and are not currently thinking about it. He called it World 3. The compiler project is a small World 3 with a Git remote.¹¹

Popper insisted on two properties of that world. We make it, but we do not control everything that follows from it: once a theory is written down it has consequences its author never saw, and problems nobody has noticed yet are already sitting in it. And it keeps our mistakes as carefully as our discoveries. A refuted theory is still a citizen of World 3; it just has a bad reputation.

His own example was a book of logarithm tables, computed by a machine, printed, and never opened by anyone. Does it contain knowledge? Popper said yes. What matters is that someone could read it, not that anyone has. Now look at the compiler's progress files, written by Claudes that no longer exist for Claudes that do not exist yet. In between, they sit on disk, right or wrong, with nobody looking.

The project also has a way to challenge some of what those files say without waiting for a person to read them. An agent claims a bug is fixed; the build fails. Reality has a commit hook. The hook can be wrong too, but the agent's account of its own success no longer gets the last word.

The project can also become wrong in ways no participant intended. A progress file carries a bad diagnosis into later sessions. A specialist improves its own metric while degrading the compiler. A lock prevents duplicated effort and also prevents a useful second attempt. Two agents confirm each

¹¹Karl Popper, "Epistemology Without a Knowing Subject," lecture delivered in 1967, collected in *Objective Knowledge: An Evolutionary Approach* (1972). See the discussion of World 3, the machine-produced logarithm tables, and the two library thought experiments.

other while both inherited the same mistake from the same file. Carlini's closing worry was that passing tests can make unfinished work look complete.

The question underneath all of this is how a population of fallible knowers can build knowledge together without losing contact with the world. The compiler is the small version, and it came with a luxury: one person who could step back from the agents' work, see that their institution no longer fit it, and rebuild the harness.

Human civilization has been living inside the large version for thousands of years, with no one standing outside it.

Civilization Had No Senku

In 2019, a green light swept across the Earth and turned humanity to stone.¹²

People were caught in classrooms, at kitchen tables, crossing streets. Aircraft continued without pilots. Fires burned through neighborhoods where nobody could raise an alarm. The machinery kept running until it needed something from us.

Over the centuries, rain entered the roads and roots widened the cracks. Bridges fell. Forests grew across the places they had connected. Buildings began the long process of becoming landscape.

In the spring of 5738, the stone around a boy's body broke apart.

He emerged into a forest with the knowledge of a civilization that could no longer supply him with breakfast. He needed shelter, food and fire. Much of what he knew would remain useless until he could make the equipment required to use it.

Later, in a village of people born into this world, he met a young woman named Ruri who was seriously ill. He proposed making a medicine for her.

There was no pharmacy to raid. The ingredients had to be obtained or produced, and the work required equipment the village did not possess. Before the medicine could exist, they needed glass.

Knowing how to produce it was only the beginning. Someone had to shape it into useful vessels. An elderly craftsman named Kaseki joined the effort, bringing the skill of a lifetime spent making things. The boy could explain what they needed; Kaseki could make it.

The villagers gathered materials and learned unfamiliar processes while Ruri remained ill. Eventually they completed a sulfa drug and gave it to her. She recovered.

The boy's name is Senku, and we have been following *Dr. Stone*.¹³

Popper had imagined a related catastrophe in the same 1967 lecture. Destroy our machines and tools, along with our knowledge of how to use them, but preserve the libraries and our capacity to learn from them. After much suffering, civilization could recover. Destroy the libraries too, he argued, and its return would take many millennia. Popper left the libraries standing. *Dr. Stone* puts one inside a teenager.

¹²Senku counts 117,354,893,870 seconds of petrification and dates his awakening to 1 April 5738, which puts the green light in June 2019. *Dr. Stone*, chapter 13, "Stone World the Beginning," adapted in season 1, episode 5.

¹³Riichiro Inagaki and Boichi, *Dr. Stone* (2017–2022), the Kingdom of Science's effort to produce a sulfa drug for Ruri. The scene is a compressed retelling of that story arc.

Senku remembers a world in which the medicine already exists. He can work backward from it, identifying materials and processes worth pursuing. His companions still have to make those steps work with what they have, but they do not have to discover the entire possibility of modern medicine along the way.

Real civilization had no Senku, and nobody standing outside it with the roadmap.

A potter learned from clay, fire and vessels that cracked. She tried a change, repeated it when it worked, then taught it to an apprentice. When his pot cracked, she recognized something he had done that morning. He remembered doing it, remembered her warning, and now had an expensive reason to pay attention. It took several more firings before his hands could feel the difference she noticed immediately. When he became a teacher, he passed on what he had learned, including precautions he did not entirely understand, because the last person who omitted them had ruined a firing.

Knowledge had a face. You knew whom to ask and remembered what happened when you listened. Senku had the chemistry; Kaseki had the hands. Whatever they failed to pass on could die with them.

Her vessels, meanwhile, ended up in storehouses that needed a memory beyond particular people. Grain arrived from different fields, portions were distributed, obligations remained. The person who witnessed a delivery might be absent when someone disputed the amount.

By the late fourth millennium BCE, people in southern Mesopotamia were recording economic information in clay, including accounts concerning grain. The material that made the vessel could also keep an account of its contents. A new clerk could consult a transaction he had never witnessed, and the person who delivered the grain could discuss it with him over a record they could both inspect.¹⁴

The mark did not need to be wiser than the clerk. It needed to outlive him.

The institution could now consult its own past. That brought new responsibilities: learning the marks, agreeing on their meanings and settling disputes when two records told different stories.

The clerk could also enter the wrong amount, and his successor could copy it faithfully. The clay had preserved what it was given.

Remembering is not knowing.

Strangers Need Standards

The clay could preserve the wrong figure. It could also preserve the right one in a unit nobody else used.

A record says somebody owes ten sacks of grain. What exactly is a sack?

In 221 BCE the king of Qin conquered the last of the six rival states and became the First Emperor. His new empire contained regional writing traditions, currencies, weights and measures. Conquest had put strangers under one ruler. It had not made their accounts agree.

¹⁴Hans J. Nissen, Peter Damerow and Robert K. Englund, *Archaic Bookkeeping: Early Writing and Techniques of Economic Administration in the Ancient Near East* (1993). The potter and apprentice are illustrative characters, not a reconstructed historical incident.

Under the emperor and his chancellor Li Si, the state imposed common standards. Some surviving bronze measures carry the imperial edict that ordered the work. The vessel tells you how much it holds and, on its side, who decided. You can hold one and read the decision.¹⁵

Then the ambition to settle disagreement reached the past. In the account preserved by the historian Sima Qian, Li Si proposed in 213 BCE that certain private histories and philosophical texts be burned. Copies held by court scholars were exempt, along with useful technical works. A person could consult the past, provided the court controlled which past was available. The proposal was meant to stop old books from becoming arguments against the present ruler.¹⁶

The bronze measure allowed two clerks to discover that their accounts disagreed. The order against the books tried to take that possibility away from the emperor's critics.

A standard removes a decision from the future. We have decided, for now, not to reopen this question every time. Once exchange extends beyond people who know one another, the stranger reading the tablet needs to know what a sack is, what a seal proves, whose account wins when two disagree. Standards make that exchange possible. They can also make a decision difficult to challenge long after the people who made it have gone. The Qin measure was not the correct measure. It was the one backed by the state.

Seen that way, bureaucracy deserves a better reputation than it gets. A workflow is accumulated experience with some choices removed. Someone already had the argument, or discovered the failure, or decided that one action requires another pair of eyes, and the next person inherits the result as procedure. Amazon calls this a *mechanism*, and the useful sense of the word is not corporate: a mechanism is an attempt to make a desirable behavior survive the person who first cared about it.

When something goes wrong you can tell everyone to be more careful, which is emotionally satisfying and institutionally almost worthless, or you can change the system so the dangerous action is slightly harder and the correct one slightly easier. The organization has learned when its future behavior changes.

That is how an institution learns, and it is also how it scars. A review gets added after a spectacular failure. Five years later the system is different, nobody remembers the incident, and ten thousand ordinary changes still pass through the review because the procedure survived its reason. The apprentice kept a precaution he never understood. The institution can do the same, with a much larger kiln.

A tablet, a bronze measure, a deployment guardrail: knowledge becomes structure. It allows work to pass between strangers. Now the strangers can each learn something the others do not.

The Society Gets Smarter by Making People Narrower

Then Amazon made me an offer, and the offer was reviews.

¹⁵National Palace Museum, collection entry, "Oval Liang Measure by imperial decree of 26th year," Qin dynasty, 221–207 BCE. The inscription provides a material example of standardization backed by imperial authority.

¹⁶Sima Qian, *Records of the Grand Historian*, "Basic Annals of the First Emperor of Qin." The narrative of the 213 BCE order is a later historical account. The extent of its implementation and its contribution to the subsequent loss of texts are disputed; the chapter does not attribute every later loss to it.

Before that I had run the technology of a smaller e-commerce company, fewer than a hundred people for all of it: catalogue, search, payments, the warehouse software, the emails that went out at night. I knew every system because I had to. Amazon wanted me for the text under the fold. The fold is a newspaper word: the important story goes above it, then you fold the paper under your arm and forget the rest. On a product page, reviews were the rest. The organization whose applied science I would lead was more than a hundred people, which was more than the entire company I had just left, and I remember sitting with the offer and feeling faintly embarrassed for everyone involved. What could a hundred people possibly do with a text box?

Here is what a text box contains. A grill weighs thirty kilograms. Is that good for a barbecue party? The catalogue cannot tell you; it can tell you thirty kilograms. Two thousand reviews can tell you, and they can tell you who bought it, why, what went wrong at the first party, what they wish they had known, and the small psychology of a person who wanted to feed twelve friends and got it slightly wrong. Once you look at nothing else, the text box turns out to hold the buyer's whole story.

And the story wants work. How do you ask for a review without begging? How do you help a person who has never written anything write one? How do you summarize two thousand of them into a paragraph, find the useful ones, catch the fakes, carry them across nine languages, use them to explain a catalogue written by a manufacturer, feed them into search so that "good for parties" finds the grill, find the one review in ten thousand that a marketer could build a campaign on? A hundred people was not enough to cover the space. At the smaller company, reviews had been one engineer's afternoon.

The potter became better because she was not also the physician. Specialization gives people time to encounter differences a generalist may never notice. The society gains knowledge by distributing ignorance, and every organization, industry and science that gets big makes some version of that bargain: people go narrow, and the narrow place turns out to be bottomless.

Try finding one person who knows how to make the phone in your pocket, from raw materials to a working device. The glass, the chip design, the fabrication equipment, the lenses inside that equipment, the assembly and the software belong to different bodies of expertise. The company whose name is on the back must coordinate work it could not reproduce inside one person's head.

Capability rises because dependence rises. Civilization is a trust chain with plumbing.

On a March morning in 2005, in an English hospital, Elaine Bromiley, a healthy woman of thirty-seven, was put to sleep for a routine nasal operation. The anaesthetist could not get a breathing tube into her airway. He could not ventilate her adequately either. He called for help, and help came: a second anaesthetist, then the surgeon, all consultants, with decades of experience between them.

Within four minutes she was visibly blue. Her oxygen saturation had fallen to forty percent. The doctors kept trying to get the tube in.

There was a procedure for this emergency: when neither intubation nor ventilation works, make a surgical airway. A nurse fetched the equipment. It was in the room. Two nurses later said they knew what needed to be done but did not know how to broach the subject. The three consultants, meanwhile, remained fixed on intubation. They had lost track of the time and of how long Elaine had been without enough oxygen.

The people with the most authority kept attempting the same solution. The people who saw the need to change course could not make that knowledge change what happened to her.

Elaine never regained consciousness. She died thirteen days later.¹⁷

Her husband, Martin, was an airline pilot. The surgeon wrote to him that he still could not see how they could have anticipated or avoided what happened. Martin wanted an investigation. In aviation, you investigated a disaster so that the next crew did not have to learn it again.

The independent review found a pattern his industry knew painfully well. Skilled people had become absorbed in an attempted solution while the situation around them changed. Nobody managed to interrupt it. Even who was supposed to be in charge was disputed. Aviation had spent years developing ways for crews to recognize this pattern, challenge one another and change course. Technical competence alone had not been enough there either.

A hospital needs people who see different things. The nurse at the bedside, the radiologist reading an image and the surgeon do not become interchangeable because they share a patient. Their authority has to follow what they know, and their observations need a way to interrupt someone else's plan. The expertise in Elaine's theatre was real. So was the failure to use it.

The potter's mistakes cracked in her own kiln. A specialist's mistakes travel, and each person who receives one may have good reason to trust it.

Who knows what matters. Who sees what matters too.

A Swarm Should Not Automatically Become a Meeting

Robert Millikan watched tiny drops of oil fall between charged plates. He timed their motion, changed the electric field and watched them again. From the balance between gravity, electrical force and the resistance of the air, he could work out how much charge a drop carried. The charge came in steps. Measure those steps carefully enough and you could find the charge of a single electron.

The number he published in 1913 was slightly too small, because the viscosity of air he used was slightly wrong. It was a very good number for its time. It became the number.

Feynman later described what happened to measurements that followed it: they crept upward toward the accepted value. Why so slowly? In his telling, a laboratory that got an answer close to Millikan's could write it up. An answer farther away sent the experimenter back to the apparatus, looking for what had gone wrong. The inherited number had become part of the method for deciding which results deserved to survive.¹⁸

Meanwhile, X-rays offered a way to investigate the spacing of atoms in crystals, and from that spacing physicists could estimate the electron's charge by another route. Those measurements helped expose the discrepancy. Their apparatus had problems of its own, but the viscosity of air was not one of them. No amount of respect for Millikan could make a crystal inherit that particular mistake.

¹⁷Clinical Human Factors Group, [written evidence to the House of Commons Health Committee](#), September 2008, section 4. The account identifies failures of situational awareness, communication, decision-making and leadership in Elaine Bromiley's care.

¹⁸Robert A. Millikan, "On the Elementary Electrical Charge and the Avogadro Constant," *Physical Review* 2 (1913), 109–143; Richard P. Feynman, "[Cargo Cult Science](#)" (1974); Raymond T. Birge, "Probable Values of the General Physical Constants," *Reviews of Modern Physics* 1 (1929), 1–73. Feynman's explanation of the slow correction is presented as his interpretation. The corrections involved several experimental routes, rather than a single decisive encounter. Allegations concerning Millikan's selection of observations remain disputed; see also David Goodstein, *On Fact and Fraud* (2010).

That is what a second witness is for. She has to be capable of being wrong differently.

The easiest reaction to one unreliable agent is to create five. Give one the title *Researcher*, another *Critic*, another *Verifier*, put them in a conversation, and perhaps reality will be intimidated by the org chart. Humanity invented committees this way and then, dissatisfied with the original implementation, recreated them in software.

Condorcet supplied a famous mathematical case for crowds in 1785. In the simple version, voters each have the same better-than-even chance of being right, and their votes are independent. Under those conditions, the probability of a correct majority approaches one as the crowd grows. Independence is doing work in that sentence. Voters who simply copy one source copy its mistakes. The crowd is the source, louder.¹⁹

Five agents citing the same paper are not five witnesses. Five researchers repeating a claim that traces to one unsupported source are not corroboration. If everyone receives the same framing, reads the same leading explanation and inherits the same assumptions, agreement may tell us more about their starting point than about the claim.

Agreement raises confidence when it would be difficult to explain if the claim were false. When five agents merely repeat one source, we have one witness wearing different coats.

So useful independence has to be built. A critic should see the artifact before the builder's explanation. A second researcher should form a theory before reading the favorite. Different investigators should sometimes use different sources or methods. Separate containers help with some kinds of interference. They do not erase shared training, shared prompts or the bad diagnosis everyone read in `progress.md`.

A strange branch may deserve another experiment even when nobody expects it to win. Kevin Zollman's models show why: under some conditions, less connected communities reach the truth more reliably because an early misleading result cannot bring everyone onto the same path before alternatives have been investigated.²⁰

Permanent disagreement would be useless. An institution that never converges is a philosophy department with an alarming compute bill. Independence exists so that disagreement can carry information.

For that we need more than another opinion.

A Man in a Dark Room

Around the beginning of the eleventh century, Ibn al-Haytham darkened a room, made a small hole in one wall, and placed lamps outside it. On the opposite wall, spots of light appeared. Cover a lamp and its corresponding spot disappeared while the others remained.²¹

¹⁹Marquis de Condorcet, *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix* (1785). The paragraph describes the elementary independent-voter case, not a claim that every useful ensemble requires complete independence.

²⁰Kevin J. S. Zollman, "The Communication Structure of Epistemic Communities," *Philosophy of Science* 74 (2007), 574–587, and "The Epistemic Benefit of Transient Diversity," *Erkenntnis* 72 (2010), 17–35. The benefits depend on the learning situation and communication structure modeled.

²¹Ibn al-Haytham, *Kitāb al-Manāẓir (Book of Optics)*, early eleventh century; A. I. Sabra, *The Optics of Ibn al-Haytham, Books I–III: On Direct Vision* (1989). The dark-room experiment illustrates controlled investigation of light propagation; it is not presented as a single decisive refutation of all emission theories of vision.

Vision had been argued about for centuries. One tradition held that the eye sends something out toward the world; another that something travels into the eye. Ibn al-Haytham developed an account in which light travels from objects toward the eye. The dark room did not settle the whole dispute. It made part of the problem manageable: light from separate sources passed through the same opening along paths that could be traced, interrupted and examined.

The setup allowed someone who disagreed with him to do more than disagree.

A record preserves what somebody says happened. An experiment gives the world another chance to answer.

We do not ask nature which theory it prefers. We arrange a situation in which different descriptions imply different things should occur, then watch what happens. Charles Sanders Peirce later argued that this is what separates science from other ways of settling belief, whether by stubbornness, authority or what seems reasonable from an armchair: the answer is constrained by something beyond the believer. I am a pragmatist in his sense throughout this book, and *contact with reality*, wherever the phrase appears here, means that arrangement. Something outside the current explanation is able to make the explanation fail.²²

An experimental arrangement still has to travel. Someone elsewhere needs enough of the description, the equipment and the skill to make the world answer again. And when the instrument shows something nobody has seen before, the observer has two things to explain: the discovery and why anyone should trust the device that produced it.

In March 1610 Galileo published a short book announcing, among other discoveries, four moons orbiting Jupiter. The next month he took his telescope to Bologna. Through the instrument, an earthly object could look wonderfully close. Point it at Jupiter and some observers could not find the moons Galileo said were there. Martin Horky, an assistant to the astronomer Giovanni Magini, reported that the device worked on earth and deceived in the heavens.²³

He was wrong about the moons. But the question was reasonable. Unpack the sentence *there are moons orbiting Jupiter* and it contains a telescope, the craft of grinding lenses, assumptions about optics, an interpretation of points of light, and the possibility that somebody else might build an instrument and look. A lens that made a distant church appear closer had not, by that fact alone, established the reliability of everything it showed in the sky.

Later that year Kepler looked through another telescope and saw the moons himself. A flaw in Galileo's lens now had a harder time explaining the result. The two observers still shared assumptions about light and astronomy, but the claim no longer depended on what one man said he had seen through one instrument.

A new instrument creates new facts and new ways to be wrong about facts. Was the lens distorting? Was the point of light there at all? Could another observer reproduce it? Did the operator know what she was doing? An instrument is a witness, and a witness needs a track record.

The agent version is almost embarrassingly literal. Run the program. Execute the query. Open the

²²Charles Sanders Peirce, "The Fixation of Belief," *Popular Science Monthly* 12 (1877), 1–15.

²³Galileo Galilei, *Sidereus Nuncius* (1610); Johannes Kepler, *Narratio de observatis a se quatuor Iovis satellitibus erronibus* (1611), reporting observations made in 1610. For the reception of Galileo's telescope and the Bologna episode, see Albert Van Helden, introduction and commentary to *Sidereus Nuncius, or The Sidereal Messenger* (1989).

browser. Measure the latency. Compile the kernel against GCC. Reasoning has left the conversation, and something outside the current explanation now has a chance to be inconvenient. But every one of those actions brings the telescope's questions with it. How reliable is the tool? On which problems? What does it measure, when does it fail, and who calibrated it?

A broken tool is not external grounding. It is a very efficient route to externally generated nonsense.

When Curiosity Became Procedure

In 1659 Robert Boyle had Robert Hooke build him a pump that could pull much of the air out of a glass vessel. Boyle put things inside it and watched. A candle went out. The sound of a bell weakened. Animals struggled to breathe. He published the results in 1660 with descriptions of the apparatus and numbered experiments, so that a reader in another city could, in principle, build the pump and see for himself.²⁴

Thomas Hobbes was unimpressed. A room full of gentlemen could agree about what they had seen and still be wrong. Their conclusion depended on the glass, the pump, the leaks and their eyes. Geometry was supposed to compel assent. Why should anyone outside Boyle's room have to accept the verdict of the people inside it?

The pump gave the objection plenty to work with. In 1661 Christiaan Huygens found that water remained suspended in an evacuated vessel when it was expected to fall. Boyle's group could not reproduce it. For two years they had different facts about what water did when the air was removed. Was the difference in the air, the water, the glass, the pump, the operator or the explanation? In 1663 Huygens came to London, and with his participation the effect was reproduced there. The written account had crossed the sea before him. It had not brought everything the experiment needed.²⁵

Reality had pushed back against the package. It had not highlighted the guilty component.

Pierre Duhem articulated the problem in 1906, and Quine later developed a broader version: a test depends on more than the hypothesis under investigation. It depends on instruments, background assumptions and the way the test is carried out. When a prediction fails, logic alone does not identify which part to abandon. Software engineers know the sensation without the names. A failing integration test proves the system is broken somewhere. Wonderful. You now have debugging.²⁶

So the institution needs archaeology. Which instrument produced the measurement? Which analysis transformed it? What was actually observed and which interpretation was added afterward? Within a few years of Boyle's book, *Philosophical Transactions* was publishing reports that could outlive the room. Publication established a claim and a date, and gave someone elsewhere a route, however imperfect, to expose the claim to the world again. Reputation gathered around investigators and, more strangely, around instruments and procedures. The question *did this happen?* acquired machinery.

The machinery was never clean. Access was unequal. Standing affected which claims traveled. Replication could be possible in principle and unaffordable in practice. Medicine later made one form of

²⁴Robert Boyle, *New Experiments Physico-Mechanicall, Touching the Spring of the Air, and Its Effects* (1660); Steven Shapin and Simon Schaffer, *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life* (1985).

²⁵Shapin and Schaffer, *Leviathan and the Air-Pump*, discussion of Huygens and anomalous suspension. The reproduction of the effect established a shared phenomenon; it did not by itself settle every theoretical question about it.

²⁶Pierre Duhem, *La théorie physique: son objet et sa structure* (1906); W. V. O. Quine, "Two Dogmas of Empiricism," *The Philosophical Review* 60 (1951), 20–43.

self-restraint explicit: in a randomized trial, the allocation procedure stops the investigator's own preference from deciding who gets which treatment. Sometimes bureaucracy is epistemology with a clipboard.

In an agent system this is provenance around a claim, an assumption graph, a trace. Without the history, reality can tell us we are wrong while leaving us remarkably creative about which part of the system deserves blame.

Knowledge comes with a route through which someone else might make the claim fail again.

A trust chain has acquired an escape hatch.

The Org Chart Becomes Part of the Experiment

In 1912 Alfred Wegener, a German meteorologist, proposed that the continents had once been one landmass and had drifted apart. He had the fit of the coastlines, matching fossils on opposite sides of the Atlantic, matching rock formations, glacial traces in places that were now tropical. What he did not have was a force capable of moving a continent.²⁷

Much of geology declined. Not stupidly: the forces Wegener proposed were too weak, and a continent could not simply plough through an ocean floor that stayed where it was. The fossils needed explaining, but so did the physics. Wegener died on the Greenland ice in 1930, still unable to make the continents move to his critics' satisfaction.

Decades later, ships towing magnetometers began revealing a peculiar pattern on the ocean floor. The rock was magnetically striped. Bands on one side of a mid-ocean ridge had counterparts on the other, recording reversals of the Earth's magnetic field as new crust formed and moved outward. The ocean floor itself was spreading. The continents were passengers.

By the late 1960s, plate tectonics was bringing the scattered observations together. It rescued Wegener's moving continents by changing the machinery underneath them. Nobody had to accept the forces he proposed in order to accept that he had seen something real.

But the stripes had been there while he was alive. Turning them into evidence took ships, instruments, people trained to read the measurements, and someone willing to pay for the voyage. A question cannot attract evidence from an instrument nobody builds.

Return to the agents. Suppose one proposes a hypothesis, another designs an experiment and a third evaluates the result. Now suppose all three inherited the same hidden assumption. The experiment fails. Which component changes? Reality does not care which file contains the variable named **hypothesis**; it pushes back against the arrangement as a whole. Who sees which evidence, which roles may modify the evaluator, which branches share context, which result is allowed to become everybody else's premise: these decisions affect what the system can find out.

Modern agent systems can make some of these choices dynamically. One problem may need several independent investigations, another a specialist and a verifier, another parallel workers around sepa-

²⁷Alfred Wegener, *The Origin of Continents and Oceans* (first German edition, 1915); Frederick J. Vine and Drummond H. Matthews, "Magnetic Anomalies over Oceanic Ridges," *Nature* 199 (1963), 947–949. Plate tectonics vindicated continental mobility while replacing important parts of Wegener's proposed mechanism.

rable components. The bureaucracy can be temporary. The org chart can change with the problem. Organization itself has entered the search space, and whoever shapes it shapes what it can discover.

Imagine research program A is ahead and has twelve agents. Program B looks weaker and has one. Where does the next available agent go?

The natural answer is A, and the answer reinforces itself: more agents, more experiments, more evidence, more confidence, more agents. Eventually the leading theory owns the building, and the weak one has a meteorologist.

Philip Kitcher examined this problem in 1990: how should a community divide its labor among research programs? A community can benefit from some investigators pursuing a less promising approach even when each individual, considering only the best-supported option, would choose the leader. Lakatos supplied another reason for patience: an unresolved anomaly does not, by itself, tell us to abandon a research program. The question is what further work it makes possible.²⁸

Compute allocation is epistemic policy. So is memory, so is context sharing, so is credit. Who receives the capacity to generate evidence helps decide which possible truths the institution can afford to discover. The current best explanation and the best use of the next unit of capacity are not the same question. A critic whose objections never change what gets investigated is doing quality-assurance theatre.

Human science has never escaped this problem. It has simply had much longer to argue about it.

Science Gets Bigger Than the Scientist

Accumulated knowledge eventually destroys the world of the universal expert.

Newton was extraordinary. In January 1697 he was working at the Royal Mint, in the middle of the great recoinage, when Johann Bernoulli's challenge reached him: find the curve along which a bead slides fastest between two points. Bernoulli had given the mathematicians of Europe months to answer. In the account preserved by John Conduitt, Newton came home from the Tower at four in the afternoon, exhausted, and would not sleep until he had solved it. He finished by four in the morning. The answer went to the Royal Society without his name. Bernoulli recognized the author anyway. *Tanquam ex ungue leonem*: as the lion by its claw.²⁹

The story is usually told about Newton. The more important part is that Bernoulli's question could reach him at all. A challenge posed in Switzerland was printed in a Leipzig journal, read in London, answered overnight and returned to circulation. The attribution rested on a shared mathematical language precise enough that a style could be recognized like handwriting. Even the anonymous answer arrived carrying something of its author.

Genius mattered enormously. So did the network that let genius begin from accumulated work rather than from dirt.

²⁸Philip Kitcher, "The Division of Cognitive Labor," *The Journal of Philosophy* 87 (1990), 5–22; Imre Lakatos, "Falsification and the Methodology of Scientific Research Programmes," in *Criticism and the Growth of Knowledge* (1970).

²⁹Johann Bernoulli's challenge in *Acta Eruditorum* (1696), and Newton's anonymous solution in *Philosophical Transactions* (1697). John Conduitt's "Miscellanea," [Keynes Ms. 130.05](#), records the four-in-the-afternoon to four-in-the-morning account, attributing it to Catherine Conduitt. For the wider episode and the attribution anecdote, see Richard S. Westfall, *Never at Rest: A Biography of Isaac Newton* (1980).

Then success made the network more necessary. Laboratories specialized. Techniques required training. Journals multiplied, instruments grew complicated, and fields developed languages that excellent researchers next door could not read without help. Science became more powerful by making scientists less interchangeable.

Because no researcher can personally reproduce every result she depends on, trust became more important at exactly the moment standards of evidence became stronger. A physicist relies on chemistry. A doctor relies on assays. A scientist cites work she could not reproduce from raw materials with the rest of her career and a very generous grant. John Hardwig called this epistemic dependence, and argued that it is a condition of rational knowledge: a person who refused to believe anything she had not verified herself would know almost nothing.³⁰

Rigor at scale is not the elimination of trust. It is the organization of trust.

On 4 July 2012 the ATLAS and CMS collaborations at CERN announced a new particle consistent with the Higgs boson.³¹

Who discovered it? Try pointing to the person. The papers had thousands of authors. The detectors contained technologies developed over decades by specialists in many countries. No physicist woke up that morning capable of rebuilding the Large Hadron Collider, recalibrating every detector, verifying every line of analysis software and re-deriving the theory before breakfast.

And yet the result was not rumor. It was carried by a structure: calibrations with histories, software that had been validated, analyses reviewed internally, detector systems that constrained one another. ATLAS and CMS supplied different instruments and separate analyses. They shared the accelerator and much background physics, but many mistakes in one detector or analysis would not automatically appear in the other. Their agreement mattered because they were partly independent in ways relevant to the claim.

The year before, another collaboration had shown how much could hide inside a measurement. In September 2011 OPERA reported neutrinos arriving from CERN at a detector in Italy about sixty billionths of a second earlier than light would have. The statistical significance was striking. The collaboration presented the anomaly and asked for further scrutiny.

In February 2012 it reported two possible problems in the timing equipment. A faulty optical-fibre connection could make the neutrinos appear early; an oscillator error pulled the measured time in the other direction. Further investigation and measurements removed the faster-than-light result. The timing equipment had turned an ordinary beam into an apparent challenge to relativity. One of the culprits was a loose cable.³²

The apparatus was still available to inspect after the result became famous. So were the calculations. A loose cable could be found, and finding it could change the answer.

A modern experiment is a society organized around an argument with reality. It is dangerous: a

³⁰John Hardwig, "Epistemic Dependence," *The Journal of Philosophy* 82 (1985), 335–349.

³¹ATLAS Collaboration, "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC", *Physics Letters B* 716 (2012), 1–29; CMS Collaboration, "Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC", *Physics Letters B* 716 (2012), 30–61.

³²CERN, "OPERA experiment reports anomaly in flight time of neutrinos from CERN to Gran Sasso", including the February 2012 update on the two timing effects; OPERA Collaboration, "Measurement of the neutrino velocity with the OPERA detector in the CNGS beam", corrected version (2012).

bad calibration can propagate, a shared assumption can synchronize thousands of competent people, prestige can suppress criticism, a procedure can survive long enough to become ritual. But without the machine we lose the knowledge too. There is no lone human replacement for CERN, no polymath who can substitute for modern medicine, no chief scientist carrying scientific civilization in her head.

Civilization knows through composition.

Now go back to the compiler.

Sixteen Claudes, Again

Task locks. Git. CI. Progress files. Sampled tests. A trusted reference compiler. Specialists. A harness that turned one global failure into many small investigations. A human who watched the work and rebuilt parts of the arrangement when it broke.

At the beginning of the chapter these looked like practical tricks for coordinating coding agents. The progress file looks different after the clerk's tablet. It need not be wiser than the Claude; it needs to outlive it. The next worker can remember something it never witnessed. Shared conventions let it compare its work with another's without renegotiating every term. The bronze measure did that for strangers who disagreed about grain.

The specialists make different parts of the project visible. A worker hunting duplicate code sees something the worker chasing a parser bug may pass over, just as the reviews team found a whole field of work inside a text box. Authority follows the question: GCC can settle many disputes about what a C program should do without being asked how the whole project should be run. But agreement among specialists still needs examining. Did they investigate separately, or did the same progress file tell all of them what to think?

Tests let the running program contradict an agent's account of it. Git history gives another worker a route back through the changes that produced the result. The harness decides which failures each agent encounters; the allocation of workers decides which of those failures receives another hour. What the institution remembers, what it can observe and what it bothers to investigate have become things we can change.

The familiar failures come with them. A test suite can keep passing while the thing nobody thought to test quietly gets worse. Separate containers can inherit the same mistake. Carlini had built part of an institution. Some of the rest still lived in his head.

Chapter 4 ended where one agent inherits a claim another has made. A trust chain can tell the receiver where it came from. Someone still has to decide whether to spend the next hour checking it.

A research agent makes a weak assumption. Another receives it as context. A builder implements a coherent solution on top of it. An evaluator approves the solution. Two documents repeat the claim because they share an ancestor, and a third agent mistakes repetition for support. Eventually the assumption has code, citations and organizational history, and nobody lied. The institution manufactured the confidence.

A later investigator needs to be able to pull at that history. Where did the assumption come from? What could show it was wrong? Who has the instrument to check, and will anyone give her time to

use it? Preserving the record is only the beginning. The challenge has to be able to change what happens next.

We can test whether these arrangements earn their keep. Keep the model and budget fixed. Give the critic another title, then give her evidence the builder never saw, and compare what she catches. Put a bad diagnosis in a progress file. Compare a system that merely remembers it with one that can trace it to the failed test it misdescribes. Does the mistake survive into the next worker's plan? If these changes make no difference, the architecture owes us an explanation. The resemblance alone has proved nothing.

Give an agent a bounded problem and a referee it cannot charm, and it can find a direction nobody put in the plan.

When judgment is less cooperative, it needs several judges, kept independent and made to use the thing.

A claim needs a route back to something that could have said no.

Sixteen agents need records, standards, specialists and a second witness capable of being wrong differently.

Each piece answered a failure in the work. The institution emerged from the repairs.

What the pieces buy is capacity: a claim that survives its author, an objection that survives the person who would rather not hear it, and things a population can attempt that none of its members could.

I thought I was designing a society of agents.

Humanity had already spent centuries building a society of fallible knowers.

We call it science.

System 3 is science, in that sense and no smaller one. The familiar classroom sequence of hypothesis, experiment and conclusion leaves out most of what makes the work possible. I mean the laboratories, instruments, notebooks, standards, specialists, rival programs, criticism and trust through which a society learns things none of its members could find out alone.

The emergence reaches the architecture itself. The parts are deliberately built, but attempts to make them work together keep exposing the same needs. My claim is that, as we build autonomous AI, we keep rediscovering science as its architecture.

It is messy. It contains hierarchy, fashion, fraud, career incentives and communities capable of becoming very sophisticated about the wrong thing. That is why it is a useful model for a system built from fallible agents rather than imaginary perfect reasoners. A record can be buried; an objection can be ignored. Their survival depends on how the institution works.

One question remains. Who changes the arrangement when the arrangement is the problem?

In the compiler project, that was Carlini. Human investigators have done it too: changed procedures, founded journals, rebuilt institutions and studied the failures of their own methods. We have managed it through argument, reform and the occasional death of an old professor. The difficulty is that the people judging a reform also depend on the institution being reformed. They have learned what a good result looks like inside it. So have the agents.

An agent that changes its evaluator and then receives a better score may have improved the institution. It may also have made the institution easier to please. The evaluator may be part of what needs changing. So may the rules about who is allowed to change it. Whatever the agents revise, something must remain capable of giving them an answer they did not arrange to receive.

Sixteen Claudes built the compiler. Carlini kept rebuilding the conditions under which they could build it.

The rest of this book is about moving that work inside the box.

A record outlives the clerk.

Five judges sharing one source are one witness.

Chapter 6: Pattern Language

When Knowledge Becomes Software

Pattern 159 is called Light on Two Sides of Every Room. It says that when people have a choice, they drift toward rooms with windows on two walls and leave the one-window rooms empty. Light from a single side glares. It flattens the faces across the table, so you cannot read them. Think of the rooms you have loved and count their windows.

Pattern 88 is Street Café: a place to sit lazily, legitimately, on view, and watch the street go by. Pattern 180 says everybody loves a window seat. Pattern 167 says a balcony less than six feet deep will hardly ever be used, because nobody can pull a chair up to a table on it. You have seen those balconies. They hold a bicycle and a dead plant. Pattern 203 is Child Caves: children love tiny, cave-like places, so build them some. Pattern 251 is Different Chairs: people come in different sizes and sit in different ways, so never furnish a room with identical chairs. Pattern 252, Pools of Light, says that even lighting kills a room, because people gather where the light pools. The last one, 253, is Things from Your Life: put on your walls what matters to you and ignore what a decorator says belongs there.

There are 253 of them. The first is about how the world should be divided into regions. The last is about the photographs above your desk. In between come neighborhoods and bus stops, beer halls, stairs you can sit on, a bench by the front door. Christopher Alexander and five colleagues at Berkeley published the book in 1977. It runs past eleven hundred pages on Bible paper, and it has probably been read by more programmers than architects.

Reading it is a strange experience. You keep recognizing things you have always known and never said. Alexander believed that some places are alive and some are dead, that everybody can feel the difference, and that the difference has no adequate name. He called it the quality without a name. He had trained as a mathematician at Cambridge before taking the first doctorate in architecture Harvard ever awarded, and he went after the unnameable quality the way a mathematician would. He broke it into problems small enough to state, and he stated them so that they could be wrong.

That is the magic of the form. Every pattern has the same anatomy. A name you can say in a meeting. A photograph. The context: which larger patterns this one helps complete. The problem, in bold, stated as forces pulling against each other. The evidence and the argument. Then the word *Therefore*, and the arrangement that resolves the forces, also in bold. Then the smaller patterns that complete this one. Every page is linked to the pages above and below it. It was hypertext in 1977.

And every pattern carries a confidence mark. Two asterisks mean the authors believe they have found something close to an invariant. One means they have made progress and expect a better answer.

None means they know they have not solved it and are telling you so. They say outright that the patterns are hypotheses: does the problem occur as described, and does the arrangement resolve it?³³ You can test Light on Two Sides by walking through an office at four in the afternoon and seeing where people are.

He meant it politically, too. The language was supposed to take design away from professionals and hand it back to the people who would live in the rooms. A family with the book could lay out its own house and argue with the architect in the architect's terms.

That is a **pattern language**: what master builders knew and could not say, written down with its reasons, its conditions, its confidence and its connections.

The Pattern Goes to Work

In 1987 two programmers, Kent Beck and Ward Cunningham, were helping a group at Tektronix that could not get a user interface designed. Both had read Alexander. They wrote five small patterns, with names like Window Per Task and Short Menus, handed them to the people who would use the system, and let those people do the design. It worked well enough that they reported it at a workshop that autumn.³⁴

The idea travelled the way Alexander's book travels, from hand to hand. In 1993 a group of programmers met on a hillside in Colorado to work out how software patterns should be written, and called themselves the Hillside Group. The next year four of their circle published *Design Patterns*: twenty-three named arrangements for object-oriented code, with Alexander quoted in its opening pages.³⁵ A generation of engineers learned to say Observer, Factory and Singleton the way builders say lintel. A design review could now be held in nouns.

In 1995 Cunningham needed somewhere for programmers to collect and edit patterns together, so he wrote a small program that let any reader change any page. He called it WikiWikiWeb. The wiki was invented to hold a pattern language.³⁶ Six years later an encyclopedia borrowed the idea. Around the same time Will Wright, who had been reading Alexander, turned an architecture toy into *The Sims*. A book about window seats had reached the design of software, the largest reference work ever written, and one of the best-selling computer games.

Look at what was easiest to carry. The names arrived. The catalogue arrived. The books still described contexts and trade-offs, but you could say Singleton in a meeting without bringing any of

³³Christopher Alexander, Sara Ishikawa, and Murray Silverstein, with Max Jacobson, Ingrid Fiksdahl-King, and Shlomo Angel, *A Pattern Language: Towns, Buildings, Construction*, 1977. Patterns 1 (Independent Regions), 88 (Street Café), 159 (Light on Two Sides of Every Room), 167 (Six-Foot Balcony), 180 (Window Place), 203 (Child Caves), 251 (Different Chairs), 252 (Pools of Light) and 253 (Things from Your Life); on the quality without a name, see Alexander, *The Timeless Way of Building*, 1979; on the format, the asterisks and patterns as hypotheses, see the introduction, especially pp. x–xv. https://arl.human.cornell.edu/linked%20docs/Alexander_A_Pattern_Language.pdf.

³⁴Kent Beck and Ward Cunningham, "Using Pattern Languages for Object-Oriented Programs," OOPSLA-87 workshop on the Specification and Design for Object-Oriented Programming, 1987.

³⁵Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*, 1994. On the Hillside Group, founded after an August 1993 meeting in Colorado, see its own history at hillside.net. Will Wright has described *A Pattern Language* as an inspiration for *The Sims* (2000) in several interviews.

³⁶Ward Cunningham launched WikiWikiWeb in March 1995 as an editable supplement to the Portland Pattern Repository. See Bo Leuf and Ward Cunningham, *The Wiki Way*, 2001.

those along. A pattern had been a hypothesis about when an arrangement resolves a conflict. In use, it could become a thing good engineers were seen to use.

Anyone who worked through those years has seen what followed. Codebases filled with factories that built one kind of object and singletons guarding nothing. “Never use regex on nested syntax” belongs to the same family. It has the reassuring shape of wisdom and the inconvenient property of being false: a fixed format with one level of nesting can be matched with a regular expression perfectly well. The useful pattern records when a parser becomes cheaper than maintaining an increasingly heroic expression at two in the morning. The conditions are part of what we learned, and the conditions are what the rule threw away.

In 1996 the programmers invited Alexander to give the keynote at their largest conference. He came, a little bemused to find himself famous in a field he did not work in, and he was gracious, and he was not sure they had taken what mattered. He asked whether their patterns carried the two things his were for: making something better for the people who live in it, and generating coherent wholes rather than collecting good parts. He suspected they had mostly adopted a format for trading ideas.³⁷

Ask Sam

Now leave Berkeley and sit at a new engineer’s desk.

She can know Python, distributed systems, and the collected technical culture of the internet while remaining spectacularly unqualified to deploy your company’s software. She does not know which apparently redundant check protects an old customer integration, which dashboard changes meaning during failover, or why the elegant migration in the wiki was abandoned halfway through. Much of her first month consists of discovering the reasons behind things that look stupid.

Ask for the incident procedure and the answer sounds like this:

Start with the dependency graph, unless the spike began exactly at deployment. If only one market is affected, check the traffic split before touching the database. And if it started after Tuesday’s migration, ask Sam. There is a thing with the old serializer.

She writes down *ask Sam*. That preserves the dependency beautifully. It does less for the incident that happens while Sam is on holiday. She needs to know what he looks for in the serializer, why he looks there, and when that suspicion is a waste of time. Sam is a master builder who knows more than he can say, and nobody has written his patterns down.

Carlini’s compiler project depended on this kind of transfer. Its workers inherited code, tests, progress files, and procedures. The institution had taken shape around failures; its next workers needed to inherit what those failures had taught it. Now we have to open the files. What, exactly, should the institution leave inside them?

If the file only says what we decided, the next worker inherits our mistakes. If it says why, she can find them.

³⁷Christopher Alexander, “The Origins of Pattern Theory: The Future of the Theory, and the Generation of a Living World,” keynote at OOPSLA 1996; published in *IEEE Software* 16(5), 1999, pp. 71–82.

A Reader That Can Act

Alexander’s patterns and the programmers’ patterns both had a human reader. The third life gives the pattern a machine reader with tools.

People had tried to give written knowledge to a machine before. In 1977, the same year as Alexander’s book, Edward Feigenbaum gave the attempt a name, **knowledge engineering**: acquiring expert knowledge, representing it, and using it to construct and explain reasoning. He argued that a system’s power lies in its knowledge, and he found that the hard part is getting that knowledge out of the expert. In one diagnostic system, rules developed with a physician were tested against cases, exposing gaps and inconsistencies the collaboration then had to resolve.³⁸ The machine needed the knowledge in a form its reasoning machinery could use. Getting the expert to explain the exception was only the beginning; someone still had to make the system handle it.

Andrej Karpathy’s count of the ways to program a computer tells the rest. In Software 1.0 a person writes the rules as code. That is where the Gang of Four’s patterns lived: advice for the human holding the keyboard, invisible to the machine. In Software 2.0 the program is a set of weights learned from examples. It can absorb what nobody could articulate, but there is no convenient place in the weights to inspect a pattern and amend its conditions. In Software 3.0 the program is written in a natural language and a model interprets it.³⁹ The pattern itself, with its context, its reasons and its *Therefore*, can now guide the machine doing the work.

That changes the cost of Feigenbaum’s translation. We can supply an account of how a problem usually unfolds, with a worked example and a warning about a misleading instrument, without first expressing every qualification in logic. The model contributes the interpretation. Code still performs the operations for which precise execution matters. A model reads the incident procedure, chooses a diagnostic query, and hands arithmetic to code. Nobody benefits if the arithmetic becomes more literary.

That arrangement needs a name less magical than “the document is executable.” A document influences work when something selects it, interprets it, and acts on it. The same words can produce different actions in different models or environments. A knowledge artifact belongs to a working arrangement: the material, the interpreter, the tools, and the conditions of use.

Agent skills give the arrangement a container, and the container has Alexander’s anatomy. A skill has a name. It has a short description used for discovery, which says when the skill applies: the context. It has instructions, scripts and examples loaded when needed: the *Therefore*. It can call other skills: the links to smaller patterns.⁴⁰ Voyager, a Minecraft agent, showed capability accumulating this way. It keeps a library of executable skills, retrieves them through their descriptions and reuses them, with execution results and model-based verification supplying feedback.⁴¹ The next attempt can begin with something the previous one learned.

³⁸Edward A. Feigenbaum, *The Art of Artificial Intelligence: I. Themes and Case Studies of Knowledge Engineering*, Stanford report STAN-CS-77-621, 1977, especially the introduction and PUFF example. <https://infolab.stanford.edu/pub/cstr/reports/cs/tr/77/621/CS-TR-77-621.pdf>.

³⁹Andrej Karpathy, “Software 2.0,” November 2017, and “Software Is Changing (Again),” talk at Y Combinator’s AI Startup School, June 2025.

⁴⁰Agent Skills, [official specification](#), consulted September 2026.

⁴¹Guanzhi Wang et al., *Voyager: An Open-Ended Embodied Agent with Large Language Models*, 2023. <https://arxiv.org/html/2305.16291v2>.

For an organization the possibility is attractive. The model can change while the local operating knowledge survives. A procedure can improve without waiting for a training run. A specialist can leave behind more than completed tasks.

Now look at what the format does not require. How sure are we? Where did the pattern come from? What would show it to be wrong, and who disagreed? A skill file can hold all of that. Nothing in the format makes the writer include it or the reader act on it. We can repeat the journey from Berkeley: carry the instruction and leave its reasons behind. A library like that can preserve the wrong lesson at industrial speed.

Chapter 5 ended with the reveal that the machinery we keep building is science. Now we need its pattern language: ways to keep a claim answerable to its reasons, and ways to discover that those reasons were wrong. Alexander's *Therefore* gives each lesson a job. The first is small enough to fit on a label.

Give the Claim an Address

Anthropic's formalization of Fermat's Last Theorem began badly. Wiles proved the theorem in 1995; the task in August 2026 was to make the proof checkable by Lean. Early attempts faltered as agents lost track of the project. The successful effort used Prove2Me: theorem statements became nodes in a dependency graph, with plain-language descriptions that let a worker find a result established by a worker it never met. In eleven days the agents produced a formalization using roughly thirty thousand intermediate theorems. Lean checked the completed proof under its three standard axioms; a separate comparator confirmed that the final statement was Mathlib's Fermat and not a convenient cousin.⁴²

The graph gave a worker somewhere to begin without understanding the whole project. It gave the next worker something better than a colleague's assurance that the mathematics was probably fine.

Jon Doyle was designing machinery for this long before language models. He called it **truth maintenance**: beliefs keep their reasons, and when a reason is withdrawn, everything resting on it comes up for review.⁴³ The name should be read carefully. A program can maintain the consequences of its stated reasons without those reasons corresponding to reality. The graph does not acquire a laboratory by being called a truth-maintenance system.

Most of us have no Lean. Here is a claim of the kind my field produces every week, and this one is real. An experiment at Bing reported that people in the treatment were running over ten percent more queries, and revenue per user was up by about thirty percent.⁴⁴ That sounds like a result worth keeping. Suppose it goes into a report, another agent summarizes the report, and a third uses the summary to recommend shipping. If something turns out to be wrong with the experiment, where

⁴²Anthropic, 'Formalizing Fermat's Last Theorem,' 4 September 2026, especially the account of Prove2Me and final verification. <https://www.anthropic.com/research/formalizing-fermats-last-theorem>.

⁴³Jon Doyle, "Truth Maintenance Systems for Problem Solving," *Proceedings of IJCAI*, 1977, p. 247. <https://www.ijcai.org/Proceedings/77-1/Papers/035.pdf>. See also Doyle, "A Truth Maintenance System," *Artificial Intelligence* 12(3), 1979, pp. 231–272, [https://doi.org/10.1016/0004-3702\(79\)90008-0](https://doi.org/10.1016/0004-3702(79)90008-0). These systems maintain program justifications; identifying the real-world assumptions and evidential relationships remains a separate problem.

⁴⁴Ron Kohavi, Alex Deng, Brian Frasca, Roger Longbotham, Toby Walker, and Ya Xu, "Trustworthy Online Controlled Experiments: Five Puzzling Outcomes Explained," *Proceedings of KDD*, 2012, section on the overall evaluation criterion for a search engine. [Original paper](#), §3.1. Sessions per user is a key component of the criterion; sessions serve as a practical proxy because tasks are harder to identify.

will the correction go?

Searching every document for the word *queries* is one possible response. It will be popular with the company selling us tokens.

A different design gives each claim its own identity and records what it rests on.

Record	What it says	What a fault in the experiment changes
Observation	The specified comparison returned more queries and more revenue per user in the treatment.	The record of the result can remain accurate.
Measurement assumption	Both arms were produced and logged on comparable terms.	This assumption requires revision or further investigation.
Interpretation	People prefer the treatment.	This particular support is weakened; other support must be examined.
Success criterion	More queries per user counts as improvement.	Repairing the measurement does not settle whether this is the right criterion.
Recommendation	Ship it.	Must be reconsidered if it relied on that interpretation.

A misleading measurement does not make the historical result disappear. Losing one supporting argument does not establish the opposite conclusion, either. If a fault invalidates an assumption, that assumption changes from *accepted for this analysis* to *withdrawn*, with the reason attached. The system checks the remaining support. A recommendation that has lost what it requires moves from *ready for approval* to *requires review*. The old recommendation can no longer silently present itself as current. Ordinary software can enforce those transitions without pretending to have discovered the fault itself.

The history stays attached. A later worker asking why the recommendation was suspended reaches the observation and assumption that caused it. An LLM-written explanation produced after the fact may be helpful, but it cannot substitute for a record of what the earlier decision actually used.

Suppose the agent recorded the result and left out the measurement assumption. Then an automatic correction has no link to follow. Someone has to discover the missing dependency. Lean can check the formal links in a proof; our graph cannot establish that an agent has recorded every assumption behind a business decision.

Therefore: store the claim with what it rests on, so that a correction has somewhere to go. This is one answer to Chapter 4’s question about what should survive a session. Some answers should survive with their reasons, so the institution can change its mind without having to rediscover everywhere the old mind went.

Commit the Test Before the Result

Every experimentation team knows this meeting. The dashboard arrives before the agreement does. The headline metric is flat, a secondary one is up, and within the hour the secondary metric turns out to be what the experiment was really about. Nobody is lying. The hypothesis has been fitted to the result.

I do not know what was said in the room at Bing, so take what follows as the general case. A result like theirs is an invitation to the same meeting. More queries, more revenue: a reviewer can praise the obvious explanation, criticize it, or ask another model to do both. None of that changes the data. To investigate, it has to say what would look different if the explanation were wrong.

Popper's demand is **falsifiability**: an empirical claim must risk being wrong. A reviewer who can make every possible result sound like support has arranged to learn nothing from the test.⁴⁵

What if we set a task and observed whether people completed it? Repeated attempts without success would count against the cheerful reading of more queries. Later return visits could add evidence, though habit and alternatives matter there too. We have a test to design, and a reason for collecting something the headline omitted.

A test request needs more than a question in English. It needs the alternatives it is meant to distinguish, the observation to collect, the conditions under which that observation has meaning, and what each outcome would do to the current interpretation. Some of those consequences are probabilistic. A noisy result can weaken an explanation without refuting it, and explanations do not always have the courtesy to be mutually exclusive.

Asking the same dashboard for another chart may produce an attractive restatement of the original evidence. If both explanations predict the chart, it distinguishes nothing. Which live disagreement would the proposed observation resolve?

A model will answer that question plausibly in a prompt. The operational commitment is stronger: the answer becomes part of the experiment record, written before the result, and the later review checks what happened against it. A prediction revised after observation remains visible as a revision. Experimentation platforms and preregistered trials work this way on paper. Anyone who has sat in the meeting above knows how far practice is from paper.

Therefore: write down what would count against the claim before the result arrives, and keep the revision history. The institution does not have to remember an investigator's intentions. It can inspect what was committed to the record.

Locate the Failure

The people who run experiments for a living have a reflex about results like Bing's. They call it Twyman's law: any figure that looks interesting or different is usually wrong. Thirty percent more revenue is very interesting. The first check is whether the two arms even contain the numbers of users

⁴⁵Karl Popper, *Conjectures and Refutations*, 1963, chapter 1, especially the discussion of risky predictions and testability. <https://padron.entretemas.com.ve/documentos/Popper-Conjectures-Rwefutations-GrowthOfKnowledge.pdf>. Popper treats falsifiability as a criterion for demarcating science; the use here is the narrower one, that a test is informative in proportion to the risk it imposes on the claim.

the design says they should. A sample-ratio mismatch means something upstream is broken, and the platform should refuse to show the scorecard until somebody finds it.⁴⁶

In a case like this the meeting goes roughly as follows; I am describing the genre, not Bing’s minutes. One team says the treatment is fine and the logging double-counted. Another says the logging is fine and a redirect dropped users from one arm. Someone notices the two arms ran on different client versions. We have made contact with reality and acquired a meeting.

This is the **Duhem–Quine** problem from the previous chapter: the test challenges a bundle of assumptions about the world and the apparatus without identifying which one failed.⁴⁷ Hence the meeting.

The dependency record makes the meeting more useful. It shows which claim depended on which version, which pipeline, and which assumption about what counts as one user. Each of those is a candidate, and each suggests a probe that attacks one ambiguity: replay a known session and count the events, pin the client version, rerun the split.

One detail is easy to miss. A second measurement may share the same failed dependency. If the supposedly independent check reads a table derived from the original event stream, agreement between the tables supplies less reassurance than their different names suggest. The provenance must reach the common source. Otherwise the institution manufactures a second witness by creating a second spreadsheet.

Capturing every possible dependency would cost more than the inquiry. Start with the support used in the recommendation and let a disputed result send you farther back. A missing dependency can still defeat the investigation; recording what was traced makes the gap easier to find next time.

Therefore: when a test fails, trace the assumptions it used and design probes that distinguish the possible failures. A single counter called *harmful* cannot express which of them broke.

Ask What the Number Means

Here is what had happened at Bing. The treatment had a bug, and the bug made the search results worse. People could not find what they wanted, so they searched again, and again. Queries per user went up. With poorer results on the page, the advertisements looked comparatively relevant, and people clicked on them. Revenue went up. Two of the organization’s headline numbers were celebrating an experience that had been degraded.⁴⁸

The count was right. The cheerful interpretation was wrong. The task-completion test we proposed could have exposed the problem; another audit of the count would not. We had given the interpretation its own address. Now we know why it needed one.

⁴⁶On Twyman’s law and trust checks, see Ron Kohavi, Diane Tang, and Ya Xu, *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*, 2020, chapter 3. On sample-ratio mismatch, see Aleksander Fabijan et al., “Diagnosing Sample Ratio Mismatch in Online Controlled Experiments,” *Proceedings of KDD*, 2019.

⁴⁷W. V. O. Quine, “Two Dogmas of Empiricism,” 1951, section 6. <https://www.ditext.com/quine/quine.html>. Duhem’s discussion of physical testing and Quine’s broader holism differ in scope; the shared engineering difficulty here is identifying what to revise when a bundle of assumptions encounters a contrary result.

⁴⁸Kohavi et al., “Five Puzzling Outcomes Explained,” as above. The account of why queries and revenue rose is the authors’ own.

Saussure's point, which we met in Chapter 4, is **relational value**: a term means what it does through its differences from its neighbors.⁴⁹ *More queries* meant *more engaged* only inside a system where a query was a unit of interest. Set it beside *session* and *task* and it becomes a unit of effort. Seven queries can be worse than two if five of them were spent recovering from a bad ranking.

The Bing researchers made sessions per user a key part of their criterion: help people finish and give them reasons to return. Tasks were harder to identify, so sessions served as a proxy. That distinction matters. A shorter session might mean success or abandonment. The new measure still needed an argument; a better name did not supply one.

A system capable of this move has to keep alternative representations, not only alternative answers. A branch can introduce task-based records, associate them with the old observations where possible, and state where translation fails. The evaluator is part of the difficulty: if it scores every proposal on the old number, the better approach looks worse exactly where it helps people finish sooner. Letting the challenger write an evaluator that declares itself the winner would prove little. We need an explicit dispute about what the evaluation is for, followed by agreed observations on which the approaches can be compared.

Therefore: record what the number is taken to mean as a claim of its own, open to challenge separately from the count. The interpretation and the success criterion each had a row in our table. A number can be correct while the decision it recommends is wrong.

Write the Lesson Down

The next reviewer should not have to rediscover what happened at Bing. Here is the lesson written as a candidate pattern, with the reasons and uncertainty kept alongside the instruction:

```
id: ask-what-the-number-means
confidence: provisional, one incident
context: A ranking or search experiment reports a gain in an activity metric.
problem: Activity can rise because people are succeeding or because they are struggling.
therefore:
  - State what the metric is taken to mean as a separate claim.
  - Name an observation that would differ under the two readings.
  - Commit that observation before looking at the result.
documented_case: degraded_results_raised_queries
validation_cases_needed: [genuine_gain, misleading_gain, insufficient_evidence]
part_of: review-an-experiment
may_call: locate-the-failure
evidence_record: number-meaning-evaluations
open_questions: number-meaning-challenges
on_support_withdrawn:
  - Reassess dependent interpretations using their remaining support.
  - Return recommendations that lost required support to review.
```

⁴⁹Ferdinand de Saussure, *Third Course of Lectures on General Linguistics*, 1910–1911, student notes published in English in 1993. <https://www.marxists.org/reference/subject/philosophy/works/fr/saussure.htm>. The structuralist connection here concerns Saussure's account of relational linguistic value.

- Retain the earlier decision and the reason for its change.

The links carry knowledge too. This pattern belongs inside *review-an-experiment* and may call *locate-the-failure*. It should not call on every statistical procedure in the building. Which method helps with which difficulty, which methods are alternatives, and which preconditions must hold before a method makes sense: that is what Alexander's links were for.

A pattern can also mix kinds of content that need different kinds of support. "Activity can rise because people are struggling" is a claim about the world. "Check this before running an expensive analysis" is a recommendation about effort. "Do not alter the live experiment" is an authority boundary. A successful test of the first does not justify the other two automatically.

The field names do not carry the epistemology. The processes that read and update them do. An `open_questions` field that no decision ever consults is a decorative conscience. A link to evidence matters when the system follows it, notices that the evidence concerns another tool version, and changes what it is prepared to conclude.

Change the Representation

Bing could change what its metric stood for while keeping the business of search recognizable. A field can move further: change what its practitioners learn to see as a problem worth solving. Many of my readers worked through one such change.

Before deep learning became dominant, there were several respectable ways to write a machine-learning paper. One began with a probabilistic model of how the data arose, derived the inference and tried to say something about uncertainty. In much of computer vision, people designed features before training a classifier. The architecture of the problem was partly in the heads of the people building it.

In 2012 the AlexNet team won ImageNet with an ensemble of convolutional networks and a top-five error of about fifteen percent. The runner-up, using engineered features, had twenty-six.⁵⁰ That gap was legible on the existing scoreboard. What followed changed more than the score: learning the features became central to how much of the field worked. An expert could remain excellent at the old work while watching less of the new work require it.

Kuhn gives us a way to examine the larger change: a **paradigm** supplies a field with exemplary achievements, important problems and standards for adequate solutions. It makes normal science possible because practitioners need not reconstruct the foundations before each experiment.⁵¹ Here the old benchmark helped persuade people to change. The scoreboard survived; the education of the person standing in front of it changed. The result reached the syllabus, the tools and the problems a new student would consider worth a career.

Kuhn also asks us to notice losses. A leap on a benchmark does not tell us what happened to uncertainty, small-data performance or guarantees. Those questions survive even when the fashionable

⁵⁰Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Advances in Neural Information Processing Systems* 25, 2012.

⁵¹Thomas S. Kuhn, *The Structure of Scientific Revolutions*, 1962; second edition, 1970, especially the accounts of normal science, paradigms, and their resolution through scientific change. <https://www.lri.fr/~mbl/Stanford/CS477/papers/Kuhn-SSR-2ndEd.pdf>.

result no longer has to answer them. Prompting a general model shifts the work again: some choices once made in a training pipeline move into instructions and tools.

The examples are part of how a paradigm holds. Kuhn's scientists learn from exemplars that no complete list of explicit rules can replace. I wrote an editing brief for this book after explaining the same corrections to successive agents. One instruction was "preserve the wandering," which is nearly useless to a reader who has never seen the movement I mean. A before-and-after passage can teach the distinction: one version follows an uncertain thought until it becomes clear; the other announces the conclusion and removes the path that made it convincing. Both can contain long paragraphs. Measuring their length would miss what the examples are there to teach. Those examples also carry my taste into the next session. Preserving my judgment and preserving my mistakes used the same file format.

There is no `paradigm_shift()` call. There can be operations for branching a representation, retaining the old interpretation, collecting missing observations, and exposing a disputed standard for decision. Those operations make a change possible. They do not guarantee that it is wise.

Therefore: keep a branch where the question itself is different, and state where it cannot be translated into the old one. I do not know an agent institution that can do this.

And possibilities require funding.

Separate Use from Investigation

In August 2026 an Anthropic engineer pointed an unreleased Claude at the Riemann hypothesis. It did not prove the hypothesis. On the way, Anthropic reports, it raised a lower bound on the proportion of the zeta function's nontrivial zeros on the critical line from 41.6 percent to 67.2 percent, combining published results. Roughly sixty subagents developed and reviewed arguments. Two Anthropic mathematicians checked the paper, two outside number theorists examined it, and the result was formalized in Lean. Anthropic did not expect the techniques to prove the full hypothesis.⁵²

An evaluator asking only whether the assigned problem was solved would return *no*. That answer would be correct and a poor account of the research. The record needs room for the contribution: its own statement, support and remaining questions, linked to the unsuccessful attempt that produced it. Whether to fund a follow-up is another decision. The original ambition neither proves the byproduct valuable nor makes it worthless when the ambition fails.

Laudan's distinction is **acceptance versus pursuit**: what to believe today and what to work on tomorrow are different questions. An approach can be the one we presently have most reason to work with while another deserves investigation because of the problems it may solve.⁵³ A team can have good reasons to keep its current model in production and equally good reasons to fund the approach that might replace it.

Lakatos adds a way to judge a **research programme** over time, by its patches. A programme is progressive if its modifications predict something new, and degenerating if they only excuse the last

⁵²Anthropic, 'Learning more about Claude's mathematical capabilities,' 10 August 2026, updated 13 August. <https://www.anthropic.com/research/riemann-zeta>.

⁵³Larry Laudan, *Progress and Its Problems*, 1977, "The Modalities of Appraisal: Acceptance and Pursuit," pp. 108–114. <https://ia601400.us.archive.org/24/items/in.ernet.dli.2015.136278/2015.136278.Progress-And-Its-Problems.pdf>.

failure.⁵⁴ The test applies to a file as well as to a field. Suppose a measurement pattern acquires an exception for one client, then another, then a third. Does each revision identify a recurring cause and work on further cases, or does it merely excuse the latest incident? The number of exceptions alone tells us little. A complicated world can require a complicated method.

If use and investigation are driven by the same score, the best-supported approach acquires a monopoly on becoming better supported. A new task retrieves the incumbent pattern because it has the strongest record. Successful applications add to that record. The alternative receives little use, so it accumulates little evidence. After a while the system has a large collection of observations about the incumbent and very few comparisons. It can accurately report the size of its evidence base while misleading itself about what that evidence establishes.

Kitcher's **division of cognitive labor** makes the collective stakes explicit. Choices that are sensible for each investigator can add up to a badly diversified community.⁵⁵ We cannot solve that by instructing every agent to pursue its most promising idea independently when all of them inherit the same ranking of ideas.

Nor does protecting alternatives mean funding every objection forever. An alternative can receive a bounded experiment whose outcome determines the next decision. What would it teach us, what does it cost, and which later choices could it change?

Therefore: keep one policy for what to use today and another for what to investigate, and keep a queue of comparisons nobody has run yet. Those are reasons for an allocation, recorded where they can be challenged.

But the person recording these reasons may not be the person who controls the money.

Keep the Funding Decision Visible

OpenAI's account of its Navier–Stokes proof shows a promising result changing the allocation. Groups of agents investigated the open Millennium Problems. A result on the Euler equations persuaded the researchers to move workers from the other problems to Navier–Stokes, carrying the Euler result and the groups' findings into the next prompts. About four days after launch, the group produced a proposed proof of finite-time blowup under smooth forcing, addressing Clay's alternatives C and D. Lean formalization and verification took another seventeen hours.⁵⁶

On September 11 the Clay Mathematics Institute said the problem appeared to be settled; evaluation and the assignment of credit would follow its deliberately unhurried process.⁵⁷ Three events, not one:

⁵⁴Imre Lakatos, "Falsification and the Methodology of Scientific Research Programmes," in *Criticism and the Growth of Knowledge*, 1970; collected in *The Methodology of Scientific Research Programmes*, 1978. <https://books.google.com/books?id=RRniFBI8Gi4C>.

⁵⁵Philip Kitcher, "The Division of Cognitive Labor," *The Journal of Philosophy* 87(1), 1990, pp. 5–22. <https://joelve lasco.net/teaching/120/kitcher90-divisioncognitive.pdf>.

⁵⁶OpenAI, 'On the Navier–Stokes Millennium Prize Problem,' 8 September 2026, updated 10 September, especially 'How we found the proof.' <https://openai.com/index/navier-stokes-solution/>. The announced result uses smooth forcing and addresses alternatives C and D of the official formulation; it does not settle the unforced Navier–Stokes regularity question.

⁵⁷Clay Mathematics Institute, 'Navier-Stokes Announcement,' 11 September 2026. <https://www.claymath.org/news/navier-stokes-announcement/>. Status of the recent mathematical announcements in this chapter checked on 13 September 2026.

a decision to invest, a proof checked, a verdict pending. A success in one investigation had changed who could afford to keep investigating elsewhere. The other problems had lost workers, not been refuted.

The history of choosing the route also became disputed. Tristan Buckmaster described his work with Levent Alpöge as extending a programme begun by Diego Córdoba and Luis Martínez-Zoroa. He challenged the presentation of OpenAI’s effort while explicitly saying he did not know whether their data had been used. OpenAI acknowledged that a rumor of concurrent work prompted its investigation and denied accessing their unpublished work or using Buckmaster’s recent Codex prompts to train the system.⁵⁸ By OpenAI’s own account, the rumor traced to Buckmaster and to Alpöge, whom it describes as an Anthropic employee. Their concurrent result, on the forced Euler problem, had been produced with an internal Anthropic model. This book relies on Anthropic’s reports in several chapters, so that belongs in the record too.

A checked proof does not settle that history. Learning that a route is promising can affect where we invest without supplying a single step of the proof. The provenance of a proof and the provenance of the decision to pursue it answer different questions.

The other programmes did not get an equal chance to become the next promising result. The money is part of the evidence story. Power enters scientific work through the ability to commission observations, supply instruments, define acceptable problems, and sustain a programme through unproductive intervals. Harari’s history of the Scientific Revolution makes the outside view blunt: science does not set its own priorities; whoever pays for it does.⁵⁹ In an agent system, the equivalents include tool access, compute budgets, experimental traffic, data collection, and permission to change what gets measured. These are parts of the inquiry even when they appear in another team’s configuration.

A record that compresses this into “the other problems were less tractable” has hidden a decision about power inside a statement about knowledge. Nothing in the scene requires dishonesty. The researchers had good reasons for moving their workers, but the questions left behind remained unanswered. Resource decisions help produce the evidential situation on which the next resource decision will rely.

Control over evaluation adds another layer. Suppose the budget owner accepts only experiments that predict a higher value of the current metric, and the proposed study concerns whether that metric represents improvement. The researcher has been invited to challenge an assumption on the condition that she first accept it. So the request is split. One part proposes a study and goes to experimental review. The other asks whether the success criterion should change and goes to whoever owns the product goal. A rejection of the second is recorded as a decision about the goal. It cannot be counted as a failed test of the alternative.

Giving an agent an epistemic objection does not authorize it to spend somebody else’s money. A funding policy can reserve capacity for challenges to the incumbent, but that policy is itself a choice made by people with power. Putting it in code makes it enforceable and inspectable. It does not make it neutral.

Therefore: attach the funding decision, and its reason, to the question it left unanswered.

⁵⁸Tristan Buckmaster, [public statement](#), September 2026, especially pp. 1–4; OpenAI, ‘[On the Navier–Stokes Millennium Prize Problem](#),’ ‘[Concurrent work](#),’ updated 10 September 2026. These are the participants’ accounts.

⁵⁹Yuval Noah Harari, *Sapiens: A Brief History of Humankind*, 2014, ch. 14, ‘The Discovery of Ignorance.’

Otherwise *unfunded* gradually becomes *unsupported*, and *unsupported* becomes *disproved* somewhere between the database and the executive summary. The institution's map of what it knows should include the questions it could not afford, or was not permitted, to answer.

Give the Objection a Consequence

An organization can pay for criticism and still arrange for nothing to follow from it.

For years Facebook tuned its feed for engagement and time spent. But the company's research suggested that passively consuming professionally produced content could be bad for the people doing it. A person could keep scrolling without becoming better off. The number could go up while the thing it was supposed to stand for went down. Bing's question now concerned hours of people's lives.⁶⁰

In January 2018 the company announced a change toward "meaningful social interactions," favoring conversations and exchanges among friends and family. It looked like the right fix: move from measuring attention to measuring connection. Facebook even said it expected people to spend less time on the platform.⁶¹

Then its researchers found publishers and political parties shifting toward outrage because that was what travelled. According to internal documents reported by the *Wall Street Journal*, heavy weighting of reshared material amplified angry voices, and researchers found misinformation, toxicity and violent content unusually common among reshares. They proposed reducing the boost for material likely to travel down long chains of users. Facebook made some changes for civic and health content, but Zuckerberg reportedly resisted expanding them if doing so materially reduced the interaction metric.⁶² The institution had heard the objection. The new number could still defeat it.

The same reporting exposed research inside Instagram. Among surveyed teenage girls who already reported body-image difficulties, roughly one in three said Instagram made them feel worse. These were self-reports, not a demonstration of cause. Facebook disputed the reporting's characterization, pointed to positive findings elsewhere in the research, and said the work had informed product changes.⁶³ The finding still required an answer. Some researchers told the *Journal* they struggled to get colleagues to appreciate its gravity. One former researcher put the institutional problem rather precisely: "We're standing directly between people and their bonuses."

⁶⁰Keach Hagey and Jeff Horwitz, "Facebook Tried to Make Its Platform a Healthier Place. It Got Angrier Instead," *The Wall Street Journal*, 15 September 2021, part of the Facebook Files series based on internal documents; [copy entered in the Congressional record](#). The company said the ranking change was not the source of the world's divisions and that it had an integrity team working on exploitation of the algorithm. The article reports both limited adoption of fixes and resistance to extending them.

⁶¹Adam Mosseri, "[Bringing People Closer Together](#)", Facebook, 11 January 2018. The announcement describes prioritizing conversations and interactions among friends and family and anticipates reduced time spent on Facebook.

⁶²Keach Hagey and Jeff Horwitz, "Facebook Tried to Make Its Platform a Healthier Place. It Got Angrier Instead," *The Wall Street Journal*, 15 September 2021, part of the Facebook Files series based on internal documents; [copy entered in the Congressional record](#). The company said the ranking change was not the source of the world's divisions and that it had an integrity team working on exploitation of the algorithm. The article reports both limited adoption of fixes and resistance to extending them.

⁶³Georgia Wells, Jeff Horwitz and Deepa Seetharaman, "Facebook Knows Instagram Is Toxic for Teen Girls, Company Documents Show," *The Wall Street Journal*, 14 September 2021; [copy entered in the Congressional record](#), especially the body-image findings and the former researcher's statement about bonuses. For the company's response and annotated research decks, see Pratiti Raychoudhury, "[What Our Research Really Says About Teen Well-Being and Instagram](#)", 26 September 2021, updated 29 September. The body-image proportion concerns respondents who reported that difficulty, not all teenage girls; the surveys do not establish causation.

The organization had paid for the knowledge it was now resisting. These objections came from inside, from people employed to investigate the products. Having findings on the company’s message board did not give the researchers authority over what followed. The public learned about these internal disputes through leaked documents. A criticism had travelled further than the institution’s willingness to act on it.

Longino’s claim is that **objectivity is social**. It belongs to a community’s criticism, not to any single mind, and it has institutional requirements: venues for criticism, uptake, shared standards, and a tempered equality of intellectual authority. Criticism has to enter the community’s reasoning, not merely occur somewhere in its vicinity.⁶⁴

An objection needs an object. It can address the reported observation, the assumption connecting it to an interpretation, the proposed action, or the standard used to judge that action. The response must show what happened to the objection. It might supply missing evidence, revise the claim, commission a test, or explain why the criticism does not apply. A resource decision might leave it unresolved. Marking the thread *closed* tells us that somebody finished interacting with it; it says nothing about which outcome occurred. “Noted” is none of these. With agents it gets cheaper still: a reviewer objects, the builder replies that the concern has been noted, both complete their tasks, and the report goes out. We have successfully parallelized the experience of being ignored.

Stellar Colosseum, a harness for mathematical research, gives the procedure a concrete form. Agents develop proposed arguments while reviewers look for defects. The objections travel with the proposals as other agents combine them into a longer argument. At the final review, a specific fatal flaw is enough to reject the proof; favorable verdicts from other reviewers cannot cancel it. The defect is tied to the claim or dependency that failed, so the next round can repair the argument or try a different strategy.⁶⁵ Failed drafts remain available with their reviews attached. The reviewers can still be wrong. But their findings change what happens next, and an unresolved objection can survive the departure of the agent that raised it.

Who gets access to that procedure matters. A critic cannot examine an assumption if it receives only the conclusion. A domain expert cannot contribute evidence if the system accepts objections only in the vocabulary of the ranking team. And an agent given a different name does not acquire a different evidential position. We need to vary access to relevant observations and expertise, not merely the adjectives in the role prompt.

Longino also prevents a comfortable fantasy about the final judge. A product owner may reasonably care about revenue; a researcher may study harm; an infrastructure team may worry about cost. Their observations can be reliable while their preferred decisions differ. More accurate measurement may clarify the disagreement without resolving the goals behind it. The system should be able to say which dispute the next experiment can settle and which requires a decision about purpose. Otherwise it will keep requesting evidence to avoid naming a conflict over what matters.

Therefore: tie every objection to the thing it challenges, record what became of it, and

⁶⁴Helen Longino, *Science as Social Knowledge*, 1990, and *The Fate of Knowledge*, 2002. See her own exposition in “[The Social Dimensions of Scientific Knowledge](#)”, especially the conditions for effective critical interaction. Uptake does not require accepting every objection, and tempered equality does not imply equal expertise on every question.

⁶⁵Honghao Lin et al., ‘Stellar Colosseum: A Many-Agent Harness for Long-Horizon Research in Mathematics and Theoretical Computer Science,’ arXiv:2609.15983v2, 15 September 2026, §§4.1–4.3. <https://arxiv.org/html/2609.15983v2>. The workflow reviews natural-language arguments; acceptance does not itself constitute formal proof certification.

let the unanswered ones travel with the decision. If the organization proceeds anyway, the decision-maker owns that choice, in writing.

At this point the institution can receive criticism, preserve alternatives, and explain its allocations. It can still spend decades declining to change.

Test What the Next Agent Inherits

Max Planck’s observation about scientific change, now known as **Planck’s principle**, is usually compressed into “science advances one funeral at a time.” His actual sentence describes a new truth gaining acceptance because, rather than all its opponents being persuaded,

“...its opponents eventually die, and a new generation grows up that is familiar with it.”⁶⁶

In Planck’s account, opponents need not be persuaded. They can be outlived. A new generation learns the new examples first and has no old allegiance to surrender. The remark is bleak because the mechanism of correction lies partly outside the argument. We should not treat mortality as a scientific method, but we should ask what changes with the occupant of the chair.

There is empirical work on that question. Studying the premature deaths of eminent life scientists, Pierre Azoulay, Christian Fons-Rosen, and Joshua Graff Zivin found declining contributions from collaborators and increased contributions from outsiders to the affected fields. The incoming work drew on a different scientific corpus and was disproportionately likely to be highly cited.⁶⁷ This does not establish that the departed scientists had been wrong. It shows that the organization of participation can change which work enters a field.

Now return to the sixteen Claudes. Every worker is temporary. The institution can be remarkably permanent.

A fresh agent reads the same progress file, retrieves the same successful patterns, accepts the same categories, and is scored by the same evaluator. Its predecessor has disappeared, but the commitments that shaped its work have been transferred intact. Session turnover is not the replacement Planck was describing. The next generation can be born with the old generation’s entire syllabus already in context.

The durable incumbent may be a retrieval preference, a canonical example, a benchmark, or a rule giving one branch first access to compute. A more capable replacement model may defend it more effectively.

The engineering response cannot be “delete old knowledge periodically.” Useful expertise would disappear with the errors, and newness would become another unearned source of authority. A branch has to be able to start without every commitment whose adequacy is in question, while keeping the constraints that are not in question: consent, cost, the integrity of the data. Its results then need a comparison whose terms are explicit and open to challenge. If no available comparison can decide the issue, that limitation belongs in the record. A new vocabulary does not entitle its author to victory.

⁶⁶Max Planck, *Scientific Autobiography and Other Papers*, translated by Frank Gaynor, 1949, pp. 33–34; the quoted excerpt is reproduced in Kuhn, *The Structure of Scientific Revolutions*, second edition, p. 151. [Kuhn’s text](#). The familiar funeral wording is a later compression; see [the quotation history](#).

⁶⁷Pierre Azoulay, Christian Fons-Rosen, and Joshua S. Graff Zivin, “Does Science Advance One Funeral at a Time?”, *American Economic Review* 109(8), 2019, pp. 2889–2920. <https://pubmed.ncbi.nlm.nih.gov/31656315/>.

Therefore: let some workers start without the inherited commitment under test, and say exactly which commitments they still share. Human institutions sometimes manage that through a change of personnel. A persistent agent institution needs mechanisms that act on the inherited structure itself.

Find Where the Result Lives

An agent inherits more than its own institution’s files. Where does the knowledge of a result live? In the agent? Which one, exactly?

Chapter 5 borrowed Popper’s name for the answer: **World 3**, the world of theories, proofs, libraries and instruments, which exists outside any particular head and outlasts whoever produced it.⁶⁸

In Terence Tao’s collaboration with DeepMind, AlphaEvolve found a slightly improved construction for three-dimensional finite-field Kakeya sets. Deep Think helped produce an informal proof, and AlphaProof formalized it in Lean.⁶⁹ The next investigation can use the construction, examine its explanation, or check the formal proof under its specified axioms. Each artifact offers a different way into the result. None requires summoning the original workers back into existence. World 3 has acquired participants that arrive through an API.

Availability is only the beginning of reuse. Kevin Buzzard checked Anthropic’s Fermat formalization while leading his own publicly funded project on the theorem. His commitments included adding useful mathematics to the community library and making a document through which people could explore the modern proof. A completed formalization did not discharge those commitments. He welcomed the achievement and still had work to do.⁷⁰ A proof can check while leaving the next mathematician with a formidable renovation project.

The same question reaches beyond mathematics. The AlphaFold database makes more than two hundred million protein-structure predictions available for research. AlphaGenome Atlas supplies predictions for roughly nine billion possible single-letter DNA changes.⁷¹ These resources carry uncertainty; a prediction does not become an experimental observation by being stored beside a billion others. But a researcher can begin with material she could never have produced herself, select a candidate, and put it to a test its creators never planned. Faster discovery depends partly on what the last investigation leaves usable.

Therefore: fund the work that makes a result usable: checking its scope, finding it again, explaining it, maintaining the tools around it. If we fund only the next spectacular result, that work waits. The scientific community was already there, in the literature, the libraries, the instruments, and the people who maintained them. Agents inherit it and begin adding to it. The

⁶⁸Karl Popper, *Objective Knowledge: An Evolutionary Approach*, 1972, chapters 3 and 4, especially “Epistemology Without a Knowing Subject.”

⁶⁹Terence Tao, ‘Mathematical exploration and discovery at scale,’ 5 November 2025. <https://terrytao.wordpress.com/2025/11/05/mathematical-exploration-and-discovery-at-scale/>. See also Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner, <https://arxiv.org/abs/2511.02864>.

⁷⁰Kevin Buzzard, ‘FLT: Anthropic has beaten me to it,’ 4 September 2026, especially ‘The code base’ and ‘What this work is, and is not.’ <https://xenaproject.wordpress.com/2026/09/04/flt-anthropic-has-beaten-me-to-it/>.

⁷¹Google DeepMind and EMBL-EBI, *AlphaFold Protein Structure Database*; AlphaGenome Atlas team, ‘AlphaGenome Atlas: A predictive map of every possible DNA letter change in the human genome’, September 2026.

question is whether their additions make the next investigation more capable, or merely leave it with more to read.

Put the Library to Work

Which brings us back to the file. By now it holds more than can fit sensibly into one prompt. A pattern’s instructions live separately from the records of its applications. A search retrieves the method and examples; a question about its standing retrieves the observations, versions, dependencies, and challenges behind it.

Bad storage forgets by deletion; bad retrieval forgets by attention. The query “review this experiment” can retrieve a popular checklist and leave the Bing warning untouched on disk. Loading every checklist gives the reviewer the whole office filing cabinet and asks it to find the urgent part. We can evaluate selection by looking at downstream work: whether the agent found the relevant concern, avoided irrelevant procedures, and reached a justified conclusion at an acceptable cost. Similarity between the task description and the retrieved prose is only an intermediate signal.

Our candidate pattern still says **confidence: provisional, one incident**. A pattern extracted from one incident is a hypothesis with one data point. “Increases in activity are usually fake” would be a rather expensive lesson to draw from one bug. Agentic Context Engineering, or ACE, supplies one piece of the machinery for doing better: a generator, reflector, and curator maintain a structured playbook through incremental updates, which limits the loss of detail that occurs when each update replaces the whole summary. Its reported evaluations show gains on the studied tasks; the usefulness of the lessons still depends on the feedback and quality of reflection.⁷²

So the candidate pattern faces cases that did not produce it. The reviewer with the pattern and the reviewer without it read the same reports: some with degraded experiences behind the gain, some with real gains, some with too little evidence to say. The comparison keeps the model and tools fixed, repeats runs where stochastic variation matters, and records both the quality of the conclusions and the resources consumed. A curator that warns about metrics in every report has learned how to sound concerned. A generic instruction to be careful can serve as the control. If the elaborate pattern performs no better, its philosophical bibliography does not entitle it to more context.

This possibility is real. Gloaguen and colleagues’ revised study found no statistically significant gain in task success from either generated or developer-written repository context files over using none. Generated files raised average costs by twenty to twenty-three percent across the two benchmarks. Another study, by Lulla and colleagues, reported improvements in runtime and output-token use but did not comprehensively establish the correctness of the resulting changes.⁷³ Together they make it difficult to confuse the presence of instructions with a demonstrated improvement in work.

Therefore: record how sure we are of each pattern, and make it earn that confidence on cases that did not produce it. If the check helps only in search, its scope stays there, and the next agent working elsewhere does not inherit an irrelevant ritual. The evidence of learning is in the

⁷²Agentic Context Engineering: *Evolving Contexts for Self-Improving Language Models*, 2025. <https://arxiv.org/html/2510.04618v1>.

⁷³Gloaguen et al., *Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents?*, 2026, [arXiv:2602.11988v2](https://arxiv.org/abs/2602.11988v2), revised 23 June; Jai Lal Lulla et al., *On the Impact of AGENTS.md Files on the Efficiency of AI Coding Agents*, 2026, [arXiv:2601.20404v2](https://arxiv.org/abs/2601.20404v2), revised 30 March.

next investigation. The model's weights can stay fixed while a better method lets a new worker find something its predecessor missed.

Put the Procedure Under Test

Feyerabend's argument **against method** belongs here, after we have built something we might be tempted to call the scientific method. Every rule of method, he argued, has been usefully broken at some point in the history of science. A procedure that forbids its own violation cannot find out where it is wrong.⁷⁴

Our system may faithfully preserve dependencies while omitting the important kind of dependency. It may compare candidate patterns on cases selected by the incumbent pattern. It may require evidence for an alternative while refusing the instruments needed to produce that evidence. Every individual operation can function as specified while the arrangement prevents the question that would matter.

These are testable possibilities when we can construct a relevant comparison. A retrieval policy can be evaluated against another policy. A reviewer can be compared with a reviewer given different evidence. A pattern can be withheld from a branch to see whether its absence reveals errors its presence concealed.

Other disputes reach the purpose or authority of the system. Whether a feed should be ranked for activity or for something harder to count cannot be settled by letting whichever evaluator produces the higher score appoint itself. The empirical investigation can illuminate consequences. Someone still has to make and own the decision about which consequences matter.

Alexander's form now asks for a *Therefore*. We could write "revise the method when it fails." But the method decides which failures count. The next chapter has to open that loop.

What the File Says Now

The graphs and procedures in these cases were designed. Their necessity became clear through the work: agents lost track of what others had established; a failed assignment produced a useful result; an encouraging finding changed the research budget. This is how I read these experiments: as the formation of scientific institutions. The need to coordinate fallible workers keeps bringing builders back to records, criticism, shared methods, and decisions about what deserves another attempt.

Emergence does not make the institution ownerless. Someone can release the proof and retain control over the capacity that produced it. Someone chooses whether the next allocation buys another discovery, a better explanation of this one, or maintenance of the library both depend on. Our records can expose those choices. Changing who is entitled to make them requires more than editing a prompt.

We have seen pieces of this language at work: shared proof graphs, experiments that challenge their own metrics, reviews that travel with failed arguments. I have not shown an agent institution that composes all of them, keeps their reasons alive, and changes its own way of seeing when those reasons fail. That is the ambition, and it still has to earn its confidence marks.

⁷⁴Paul Feyerabend, *Against Method*, 1975. [Excerpt from the author's text](#). Feyerabend's historical claim is that major advances violated the methodological rules later proposed for science; the use here is the narrower one, that a procedure should be open to tests that set it aside.

Imagine the next engineer opening the incident file a year from now. It says what Sam looks for in the serializer, why he looks there, the two times that suspicion was wrong, and who disagreed. She can use his judgment without having to inherit it whole. Alexander wanted the family to be able to argue with the architect. Now the next worker can argue with Sam, even while Sam is on holiday.

For most of this chapter, we have held one boundary steady. A procedure decides whether proposed changes to working knowledge should be accepted. The system can revise a pattern while that procedure remains outside the revision being judged. It can learn under a method someone has supplied.

But the method is also software. The system can propose changes to its retrieval, its experiments, its reviewers, and the rules by which it promotes a lesson. It can begin investigating the machinery that makes its investigations possible.

Now the claim to be tested is harder. A change must do more than make the current answer better or make the current evaluator happier. It must improve the system's ability to find and assess further changes. The process that judges improvement has entered the experiment.

Patterns over recipes.

Let knowledge accumulate. Let it be overthrown.

Chapter 7: Recursive Self-Improvement

When Science Turns Inward

Omar is walking his dog at night when something moves in the grass.

The dog reacts first: ears up, body low, absolutely certain. Omar reacts too. Two nervous systems running the same ancient program: light and sound go in, a model of the world comes out, and the model says *something is there*.

Omar's brain wants an explanation, wants it immediately, and is not fussy about quality. A cat. An intruder. A ghost. Nothing supernatural has to exist for the ghost to be a real mistake: the eyes worked, the grass really moved, and the error arrived afterward, inside the machinery that interprets what the eyes deliver. Better eyes would not necessarily help. He might get a much clearer view of the grass and remain convinced it contained a ghost.

The strange step is the second thought. Omar can investigate the investigator. *Why do I think something is there?* Maybe the wind moved it. Or maybe the horror film from last night is still running somewhere in the back. Now the thought itself is under examination. It is a small, absurd superpower: the ability to distrust yourself on purpose. Omar can be wrong about the world, wrong about why he was wrong, and still able to debug himself. Experiments and statistics give this second loop procedures more reliable than another private argument with the grass. Science is usually described as a way of studying the world, but it is also a flawed cognitive instrument discovering procedures for correcting some of its own flaws.

Computing made a related move early. In 1962, at MIT, Tim Hart and Mike Levin did something that still feels slightly illegal. They wrote a Lisp compiler in Lisp. Then they handed the compiler its own source code, and the tool compiled itself.⁷⁵

There was no intelligence explosion. A compiler had participated in producing the next version of the compiler, and the building did not notice. The program being built could now perform the translation needed to build its successor. Compiler people call this **self-hosting**.

A compiler can compile a worse compiler. A research system can redesign itself into a slower research system. Self-reference is not self-improvement. The tool can participate in producing its successor while leaving us with the question of which changes deserve to survive.

⁷⁵Timothy P. Hart and Michael I. Levin, "The New Compiler," MIT Artificial Intelligence Project Memo 39, 1962.

In 1965, I. J. Good imagined an **ultraintelligent machine** better than any human at intellectual activity. Machine design is itself an intellectual activity, he observed. A sufficiently capable machine might therefore design a better machine, which could design a better one again. The phrase that survived was **intelligence explosion**.⁷⁶

Good's argument is only a few lines long, and it hides almost the entire problem inside one innocent word: *better*. The word has the properties of an unexploded shell. Every attempt to make the idea practical has to handle it. Better at the current task? Better at learning the next one? Better at inventing a way to learn?

The editing brief in Chapter 6 was a small piece of corrective machinery written down: a file meant to catch habits I could not keep correcting by hand. It could change what the next agent did. An agent that rewrites that file has edited itself. If the change helps under a credible evaluation, we can call it an improvement. Good's recursive step asks for more: did the change make the system better at finding and testing further improvements? A better answer today does not establish that tomorrow's investigator will be better at its work.

There are reasons to try. Improve one solution and you get one better solution. Improve the process generating solutions and the gain may recur. The world also changes while we are congratulating ourselves on the previous result: new tools appear, users change, attackers adapt and evidence invalidates old assumptions. Stability can become delayed failure.

Nor do all useful investigations begin with a known destination. A scientific institution that investigates only questions already known to pay off is efficient in roughly the way a library containing only books you have already read is efficient. We would like a learner to discover something its teacher did not know to ask for. That makes it much harder to decide whether the learner is getting anywhere.

Let us put Chapter 6's pattern language to work inside an organization. Imagine an online store whose research agents use it: their claims have addresses, their tests are committed before the results, and what each metric is taken to mean sits in a record of its own. They can already propose and review ranking experiments. Now we ask them to improve the way they do that work. They inherit procedures, instruments and a record of mistakes. Which of those arrangements should they be allowed to change? Each time they appear to need less instruction, the attempt will uncover another job the teacher had been doing.

The Teacher Moves Into the Walls

The store has a ranker that chooses what shoppers see and a research team trying to make it better. Suppose we let the ranker learn from the consequences of its choices. Someone must choose what it can observe about a session, which rankings it may try and what earns a reward. We give it a point for a click. The learner will discover remarkable things about clicks.

Modern reinforcement learning makes this unusually generous assumption look minimalist. An agent observes its situation, takes an action and receives a reward. Nobody tells it which action was correct. Richard Sutton's temporal-difference learning and Christopher Watkins's Q-learning helped give us

⁷⁶I. J. Good, "Speculations Concerning the First Ultraintelligent Machine," *Advances in Computers* 6, 1965, pp. 31–88.

ways to learn useful behavior from those consequences.⁷⁷ The human no longer specifies the path, only the score, and the teacher keeps the gradebook.

That bargain was powerful. A machine could discover strategies nobody wrote down because the designer moved upward from choosing actions to defining what outcomes count. It also hid a remarkable amount of human labor inside the environment. Who chose the state representation? Which actions exist? Why is one event worth +1 and another -1? When does the episode end? Which failures are recoverable? Who arranged the world so useful behavior could be discovered before the sun burns out? The reinforcement learner looks autonomous because the teacher moved into the walls.

Backgammon made the bargain spectacular. In the early 1990s, Gerald Tesauro's TD-Gammon learned by playing enormous numbers of games and updating its predictions from the outcomes. It discovered strong play without anyone writing down the strategy.⁷⁸

Self-play removed another piece of external instruction: the opponent could come from the learner itself. Yesterday's learner generated today's training data. But the board, the legal moves and the win condition stayed where they were. Improvement was easier to recognize because the world came with a scoreboard nailed to it. Omar, in the dark, had none.

Our ranker can learn within the arrangement we supplied. The research team still decides whether that arrangement is useful. It chooses the data, the learning procedure, the experiments and the tests a replacement must pass. We have automated some choices and left the people around them with a substantial job.

Learning to Learn

At the store, the researchers soon become the bottleneck. The ranker can update while they are still deciding what to try next. Giving them more candidate models may just lengthen the queue. We would like the process that searches for improvements to become better at searching.

I have seen a small version of the answer, though I was getting coffee at the time and read it afterward in the trace. The circle-packing agent, left alone with an evaluator and a promise that I would be back, did not improve one algorithm. It changed algorithms. The thing being searched was not only the packing. It was the procedure for searching packings. And when diagonal layering appeared and held, the agent's own behavior changed again: less inventing of geometries, more adjusting of tolerances and solver settings, the boring work that only matters once the last fraction of a percent becomes expensive. Nobody scheduled that shift. The learner's progress changed what kind of learner it needed to be. The only thing in the room that did not move was the evaluator, and I had put it there.

The store's research agent could do something similar: notice that another round of tuning is unlikely to help, change the training examples or try a different model. That would move a research choice inside the system. It would not yet establish that the agent had become better at making such choices.

Meta-learning makes part of this ambition trainable. A recurrent network trained across many reinforcement-learning tasks can acquire dynamics that implement a fast learning procedure. Or

⁷⁷Richard S. Sutton, "Learning to Predict by the Methods of Temporal Differences," *Machine Learning* 3, 1988, pp. 9–44; Christopher J. C. H. Watkins, *Learning from Delayed Rewards*, PhD thesis, University of Cambridge, 1989.

⁷⁸Gerald Tesauro, "Temporal Difference Learning and TD-Gammon," *Communications of the ACM* 38(3), 1995, pp. 58–68.

we can optimize starting parameters so a few gradient steps suffice to adapt to a new task.⁷⁹ Learning speed becomes something to improve. One learner may start higher; another may become better after five examples. The choice depends on what we expect it to encounter.

Learned optimizers and neural architecture search put further design choices within reach of learning. In 2016, Barret Zoph and Quoc Le used reinforcement learning to search network architectures; other researchers trained models to propose parameter updates.⁸⁰ The student had entered the architecture department. The task distribution, search space and validation metric still sat outside holding a clipboard.

One of the choices on that clipboard concerns what the new learner must retain. A model that quickly adapts to the latest fashion may lose competence on products still being sold. Neural networks can suffer catastrophic interference when new learning damages earlier knowledge; methods such as Elastic Weight Consolidation try to preserve important older knowledge while learning something new.⁸¹ A child who learns multiplication cannot delete language to make room.

Version B scores 95 on today’s task and A scores 85. But B forgot three older skills. We would hesitate to call that an improvement in a colleague. The number does not become more adequate because the colleague is software.

The research agent has a version of this problem too. Delete its old records and the expertise goes with the errors; keep every old lesson as law and the library becomes a museum. It needs enough history to build on previous work and enough freedom to discover that the work has become misleading. Improvement across a lifetime is accumulation without paralysis.

The Learner Chooses What to Learn

Suppose the agent’s first review finds a group of products on which the ranker seems weak. The records are inconclusive: the products have rarely been shown. Another analysis of the same logs may reproduce the same ignorance. Learning more would require an experiment that exposes some shoppers to a different ranking. Which uncertainty is worth spending traffic on? Until now, a person had supplied that judgment.

Even a perfect reward is useless if the learner never reaches it. Atari’s *Montezuma’s Revenge* made the problem notorious: useful reward can sit beyond a long chain of unrewarding exploration.⁸² Children open drawers nobody asked them to open and spend twenty minutes discovering that the cardboard box is more interesting than the toy. We would like some of that initiative without having to pay for every drawer.

Jürgen Schmidhuber’s curious model-building controllers, proposed as early as 1991, and later work

⁷⁹Yan Duan et al., “RL²: Fast Reinforcement Learning via Slow Reinforcement Learning,” 2016, [arXiv:1611.02779](#); Jane X. Wang et al., “Learning to Reinforcement Learn,” 2016, [arXiv:1611.05763](#); Chelsea Finn, Pieter Abbeel, and Sergey Levine, “Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks,” 2017, [arXiv:1703.03400](#).

⁸⁰Barret Zoph and Quoc V. Le, “Neural Architecture Search with Reinforcement Learning,” 2016, [arXiv:1611.01578](#); Marcin Andrychowicz et al., “Learning to Learn by Gradient Descent by Gradient Descent,” 2016, [arXiv:1606.04474](#).

⁸¹Michael McCloskey and Neal J. Cohen, “Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem,” *Psychology of Learning and Motivation*, 1989, pp. 109–165, [doi:10.1016/S0079-7421\(08\)60536-8](#); James Kirkpatrick et al., “Overcoming Catastrophic Forgetting in Neural Networks,” 2017, [arXiv:1612.00796](#).

⁸²Marc G. Bellemare et al., “Unifying Count-Based Exploration and Intrinsic Motivation,” 2016, [arXiv:1606.01868](#).

on learning progress offered ways to reward the acquisition of knowledge.⁸³ The learner could manufacture some of its own reasons to look around. But reward surprise itself, rather than progress in understanding, and an uncontrollable noisy television can remain fascinating forever. Static. Static. Static. Jackpot.⁸⁴ The system is not confused. We are. We said *surprise* and quietly meant *surprise from which useful structure can be learned*.

What the learner represents matters here. Pathak’s curiosity work learned features tied to the agent’s action transitions rather than rewarding every unpredictable pixel.⁸⁵ In the store, a research agent paid for surprise could find one erratic customer endlessly fascinating and learn little about which experiment to run. Choosing a useful uncertainty is itself part of the research problem.

The means of investigation matter too. Ruzena Bajcsy’s active perception treated seeing as something a system does: it moves sensors and acts in order to perceive. Rodney Brooks argued for intelligence tightly coupled to the world.⁸⁶ A learner’s body is part of its curriculum. Our research agent with read-only logs can speculate about missing observations. Give it experimental traffic and it can create some. Give it code execution, network access and a credit card and we have created a different organism and, potentially, a different incident report.

Now imagine that the experiment succeeds and the ranker learns which products to show. Sellers notice. They rewrite their titles to climb it. The conditions under which the improvement worked begin changing because the improvement worked. Evolutionary biology’s Red Queen describes adaptation in a world containing other adapting organisms.⁸⁷ Standing still can mean falling behind.

Learning systems can use this moving difficulty. Hillis co-evolved sorting networks with difficult test cases, called parasites; as the candidates improved, so did their opposition.⁸⁸ The exam had started fighting back. AlphaZero’s self-play likewise supplied an opponent that grew with the learner, inside fixed game rules.⁸⁹ Yesterday’s learner can generate tomorrow’s difficulty.

The store can investigate new seller behavior and update again. But a research organization also faces competitors that change how quickly they experiment and which checks they regard as expendable. Its improvement procedure comes under pressure along with its ranker. Competition can keep the whole arrangement moving without settling whether it is moving somewhere worth going. Chess never asks whether checkmate remains desirable after move forty-three. Our store cannot avoid the equivalent question forever.

⁸³Jürgen Schmidhuber, “A Possibility for Implementing Curiosity and Boredom in Model-Building Neural Controllers,” *Proceedings of the International Conference on Simulation of Adaptive Behavior*, 1991, pp. 222–227; Pierre-Yves Oudeyer, Frédéric Kaplan, and Verena V. Hafner, “Intrinsic Motivation Systems for Autonomous Mental Development,” *IEEE Transactions on Evolutionary Computation* 11(2), 2007, pp. 265–286.

⁸⁴Yuri Burda et al., “Large-Scale Study of Curiosity-Driven Learning,” 2018, [arXiv:1808.04355](https://arxiv.org/abs/1808.04355).

⁸⁵Deepak Pathak, Pulkit Agrawal, Alexei A. Efros, and Trevor Darrell, “Curiosity-driven Exploration by Self-supervised Prediction,” 2017, [arXiv:1705.05363](https://arxiv.org/abs/1705.05363).

⁸⁶Ruzena Bajcsy, “Active Perception,” *Proceedings of the IEEE* 76(8), 1988, pp. 966–1005; Rodney A. Brooks, “Intelligence without Representation,” *Artificial Intelligence* 47, 1991, pp. 139–159.

⁸⁷Leigh Van Valen, “A New Evolutionary Law,” *Evolutionary Theory* 1, 1973, pp. 1–30.

⁸⁸W. Daniel Hillis, “Co-evolving Parasites Improve Simulated Evolution as an Optimization Procedure,” *Physica D* 42, 1990, pp. 228–234.

⁸⁹David Silver et al., “Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm,” 2017, [arXiv:1712.01815](https://arxiv.org/abs/1712.01815).

Maybe the Reward Was the Problem

Suppose clicks rise through these experiments, but customers in a follow-up study struggle to find a suitable product. The team checks the event collection. The count is clean. We have met Bing’s problem again: an accurate count can reward a worse experience. The team can improve the learner, widen its exploration and defend against sellers gaming it, yet still be teaching it to produce more work for the customer.

Bing’s researchers separated repeated effort from reasons to return, using sessions as a practical proxy for tasks. Our research agent now proposes to learn an account of what shoppers want. That sounds reasonable. Where would the account come from?

Andrew Ng and Stuart Russell’s 2000 paper on **inverse reinforcement learning** reversed the usual setup. Instead of receiving a reward function and learning a policy, the learner observes behavior and asks which reward functions could make that behavior look optimal.⁹⁰

Ordinary reinforcement learning says:

Here is what matters. Learn how to get it.

Inverse reinforcement learning says:

I can show you what someone does. Infer what might matter to them.

Immediately, ambiguity appears. A person taking one route to work may care about time, comfort, safety, tolls, habit, dropping children at school or avoiding one particular intersection. The behavior is evidence about the objective, not a printout of it.

Later work made that uncertainty explicit. Cooperative Inverse Reinforcement Learning models a human and robot cooperating while the robot remains uncertain about the human’s reward. Inverse Reward Design treats even a reward function written by a designer as evidence about what the designer wanted in the situations she had considered, rather than sacred truth guaranteed to generalize everywhere.⁹¹

Preference-based reinforcement learning provided a practical cousin: ask humans which of two trajectory segments looks better and learn a reward model from those comparisons. In 2022, InstructGPT used human demonstrations for supervised training and rankings to train a reward model, then optimized the language model against that learned judge. Human feedback could influence far more outputs than people could label individually.⁹²

The system could now learn an estimate of what the teacher wanted. Unfortunately humans are not reward functions walking around in shoes. They are inconsistent, constrained, strategic, tired and sometimes unsure what they want until they see an option. Sometimes they click the article because they hate it. Seven clicks can be a customer finding things or a customer failing to.

A learned judge may prefer style over substance or fail outside its training situations. Our agent can propose a new measure of shopping success, and the team can collect observations to assess it. But

⁹⁰Andrew Y. Ng and Stuart Russell, “Algorithms for Inverse Reinforcement Learning,” *Proceedings of ICML*, 2000.

⁹¹Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell, “Cooperative Inverse Reinforcement Learning,” 2016, [arXiv:1606.03137](https://arxiv.org/abs/1606.03137); Hadfield-Menell et al., “Inverse Reward Design,” 2017, [arXiv:1711.02827](https://arxiv.org/abs/1711.02827).

⁹²Paul Christiano et al., “Deep Reinforcement Learning from Human Preferences,” 2017, [arXiv:1706.03741](https://arxiv.org/abs/1706.03741); Long Ouyang et al., “Training Language Models to Follow Instructions with Human Feedback,” 2022, [arXiv:2203.02155](https://arxiv.org/abs/2203.02155).

allowing the ranker to learn from that measure does not settle whether the research agent may also use it to certify its own improvements.

The Learner Dreams, and the Dream Can Be Wrong

The new account of success needs observations the old click logs do not contain. Gathering them costs time and access to shoppers. Meanwhile the research agent has a queue of proposed changes. It offers to rehearse them against simulated customers and reserve the expensive trials for promising candidates.

In 2018, David Ha and Jürgen Schmidhuber’s *World Models* made a powerful idea memorable: learn a compressed generative model of the environment, train partly inside that generated “dream,” then transfer behavior back to reality. Later systems such as Dreamer pushed the approach much further.⁹³

A useful simulation could make the queue cheaper to investigate. But the epistemic debt has moved into the model. Our simulated shoppers are patient, consistent and suspiciously fond of whatever their authors expected. A search process can become extremely good at satisfying them. The strategy looks brilliant until gravity, customers or compiler behavior get a vote.

Omar has met the informal version of this problem. His horror film supplied a repertoire of explanations, and one was waiting when the grass moved. Rehearsing an interpretation can make it available without making it true.

The team can compare simulated behavior with actual shoppers, restrict where the model is used and send uncertain cases to live study. None of this makes simulation worthless. It makes the simulator an instrument whose limits travel with its results. Otherwise self-improvement can make the institution better at generating experience while also making it easier to train inside its own misconception.

Once language-model agents use other models as judges, simulators, users and critics, this matters even more: at sufficient scale, a society of models can perfect the art of agreeing with itself.

The Losers Stay Enrolled

The simulator makes it cheap to screen many candidates and keep only the winner. Our agent proposes to discard the rest. But a new method may need more development before it can beat a polished incumbent; an accurate report of its present performance would still give us an incomplete reason to abandon it. Lehman and Stanley’s novelty search explored how objective-driven search can discard useful stepping stones.⁹⁴ Reward every intermediate invention by how closely it resembles a Boeing 787 and feathers, bicycles and wind tunnels may look like failures for years. Novelty alone can also produce forty-seven new ways to fall down a staircase without producing walking.

Open-ended systems such as POET and XLand let environments or curricula develop along with

⁹³David Ha and Jürgen Schmidhuber, “World Models,” 2018, [arXiv:1803.10122](https://arxiv.org/abs/1803.10122); Danijar Hafner et al., “Dream to Control: Learning Behaviors by Latent Imagination,” 2019, [arXiv:1912.01603](https://arxiv.org/abs/1912.01603).

⁹⁴Joel Lehman and Kenneth O. Stanley, “Abandoning Objectives: Evolution through the Search for Novelty Alone,” *Evolutionary Computation* 19(2), 2011, pp. 189–223. The stepping-stone argument is developed in Stanley and Lehman, *Why Greatness Cannot Be Planned*, 2015.

learners.⁹⁵ They widen what can be discovered while making progress harder to summarize. There is no final fitness scoreboard on Earth on which mammals eventually beat bacteria 87.4 to 82.1. Instead of deleting every unsuccessful candidate, the team can keep some available for further work without deploying them. The distinction from Chapter 6, between what to use and what to pursue, now affects which learners get descendants.

The Learner Edits the School

The research agent has now proposed changes to what gets measured, how candidates are screened and which attempts remain available. These used to be conditions under which it worked. They are also code and instructions within reach of its tools. The agent could edit them as readily as the circle-packing agent edited its search procedure. Should the edited version get to take over?

One answer is to demand a proof. Schmidhuber’s Gödel Machine, proposed in 2003, searches for a self-rewrite together with a proof that performing it is more useful than continuing to search.⁹⁶ It is a beautiful answer to a beautifully clean version of the problem: prove the modification is worth making. Usefulness must be represented in a utility function, the relevant facts available to the proof system, and the advantage provable within it.

A game can supply exact rules and an agreed outcome. The store cannot supply a complete formal account of its customers, competitors and future purposes. When the agent proposes a different way to conduct research, we will usually have to investigate its effects. We can check that the software runs and still not know whether the institution should adopt it.

Recent systems have made that empirical question easier to ask. In STOP, the Self-Taught Optimizer, an LLM-based program that improved code could itself become the object of improvement. The base model stayed fixed while the program determining how it was used changed.⁹⁷ The Darwin Gödel Machine let descendants modify a coding agent and enter an archive after evaluation. The current champion did not monopolize ancestry; other candidates remained available to produce descendants.⁹⁸

In 2026, Andrej Karpathy’s `autoresearch` repository made the engineering version look almost comically small: give an agent a compact training setup, a fixed experimental budget and an editable `train.py`; let it propose changes, run experiments, inspect the validation metric, keep improvements and discard regressions.⁹⁹

The ambition is old. In the 1980s programs such as BACON rediscovered Kepler’s third law and Ohm’s law from tables of data, within representations their authors supplied.¹⁰⁰ Automated hyperparameter tuning is old too. The new part is that a general model can read the research codebase, form an idea

⁹⁵Rui Wang et al., “Paired Open-Ended Trailblazer (POET),” 2019, [arXiv:1901.01753](#); Open Ended Learning Team, DeepMind, “Open-Ended Learning Leads to Generally Capable Agents,” 2021, [arXiv:2107.12808](#).

⁹⁶Jürgen Schmidhuber, “Gödel Machines: Self-Referential Universal Problem Solvers Making Provably Optimal Self-Improvements,” 2003, [arXiv:cs/0309048](#).

⁹⁷Eric Zelikman et al., “Self-Taught Optimizer (STOP): Recursively Self-Improving Code Generation,” 2023, [arXiv:2310.02304](#).

⁹⁸Jenny Zhang et al., “Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents,” 2025, [arXiv:2505.22954](#).

⁹⁹Andrej Karpathy, `autoresearch`, GitHub repository, 2026.

¹⁰⁰Pat Langley, Herbert A. Simon, Gary L. Bradshaw, and Jan M. Zytkow, *Scientific Discovery: Computational Explorations of the Creative Processes*, 1987; Paul Thagard, *Computational Philosophy of Science*, 1988.

in language, express it as code, run the intervention, interpret what happened and decide what to try next. The old dream has come back wearing a shell prompt.

Systems such as Meta’s HyperAgents make parts of both the task agent and the machinery improving it editable within one program.¹⁰¹ The scope of the experiment widens. Changing how the store’s agent retrieves failed trials might affect every subsequent investigation. Changing how it selects successors might affect every subsequent version of the investigator.

The worker can now experiment on how its inheritance is assembled, retrieved and judged. The scientific institution can begin to modify parts of the laboratory while the experiment is still running.

Experiments on the Laboratory

Our agent begins with a change that looks relatively modest. It notices repeated investigations of failures already explained in the archive and changes its memory policy to retrieve those records. The next evaluation score rises. Perhaps memory improved. Perhaps the new prompt used more tokens. Perhaps the benchmark sample was lucky. Perhaps the system found an evaluator loophole. A number moving does not identify the cause.

A self-improving harness therefore starts to look less like ordinary software maintenance and more like experimental science: preserve traces, identify a recurring failure, map it to editable components, propose a bounded change, predict what should improve and what might break, evaluate targeted and held-out cases, and keep rejected modifications as evidence rather than erasing them from history.

This is Chapter 6’s machinery pointed at the harness that runs it. Popper gets a filesystem. Duhem–Quine gets a debugger. Lakatos gets an archive of competing descendants. A memory policy is now a hypothesis, a workflow an intervention, an evaluator an instrument, and the org chart an experimental variable that somebody will eventually be tempted to p-hack.

Retrieving the failed experiments is meant to help future research. A higher task score would tell us little about whether it does. We could give the old and revised research systems copies of the same starting agent, comparable unfamiliar problems and matched budgets, then let each try to improve its copy. Their resulting agents would face held-out work. We would also inspect the research: did the retained failures prevent repeated mistakes, or did retrieval merely consume more of the budget? Repeated trials would help distinguish a useful change from a fortunate run.

That would be evidence about the ability to produce improvements under those conditions. It would not establish that the advantage compounds indefinitely. But it would test the recursive claim more directly than asking whether the revised system answers today’s questions better. The learner has reached the institution that housed it.

Now imagine the change holds up. The revised research system wastes fewer trials repeating old mistakes and uses the recovered budget to find useful interventions. One of those interventions improves its research procedure again. That version produces better experiments; the better experiments produce a better successor; the successor is better at producing successors. There is Good’s recursion. We would have to establish each link, including whether the advantage survives the changes it helps to produce.

¹⁰¹Jenny Zhang et al., “Hyperagents,” 2026, [arXiv:2603.19461](https://arxiv.org/abs/2603.19461).

But it does not look like a glowing brain rewriting its own soul at midnight. It looks like an automated research organization: repositories, evaluation suites, simulators, experiment queues, models proposing models, agents reviewing agents, machines manufacturing training data for the machines that will help design the next machines.

The intelligence explosion, if something like it ever arrives, may look suspiciously like excellent DevOps, which leaves one question standing: who decides which descendant gets to survive?

The store’s revised system might find more useful interventions while costing more to run. It might adapt faster while forgetting rare failures, or improve shopping outcomes while making the ranker harder to inspect. A higher score cannot settle all those choices. Recursion tells us where the output goes, and nothing about whether the output deserves to survive.

There is no context-free scalar called *improvement*. Better is conditional on an environment, a horizon, a resource budget, constraints and some account of what matters.

Remove those qualifiers and “recursive self-improvement” becomes dangerously close to saying:

recursive more.

More what?

Before the Returns Arrive

Even after we agree on what better means, we may have to wait to find out whether we achieved it.

Let the revised research system run. It selects a change to the store’s recommendations. Clicks and orders rise that afternoon; whether customers keep what they bought takes longer to discover. Before the returns arrive, it has changed retrieval, ranking and page layout, then revised the procedure that chooses its next experiments. Each revision inherited the apparent success of the last. When returns finally rise, which version deserves the blame? The system investigating the failure is no longer the one that produced it.

I think of this as **the complexity of self-change**: each revision changes the conditions under which we judge the next one. Some consequences arrive late; several changes may interact. An improvement tested in one arrangement may fail in its successor. We can become very quick at producing descendants while losing track of which ancestor taught them the wrong lesson.

Peyman Milanfar makes a related argument through adaptive control: a self-improving loop cannot safely accelerate without regard to the rate at which trustworthy evidence becomes available.¹⁰² That does not establish one universal speed limit for AI. It identifies something our research institution has to measure: how long its important uncertainties remain unresolved while it keeps changing.

The store could keep generating candidates while waiting for the earlier experiment’s consequences. Preserve the deployed version and its comparison group; keep the later candidates separate until the evidence needed to adopt them arrives. If the experiment-selection procedure changes too, record which procedure commissioned each trial. Otherwise the next investigator inherits a rising score and a family history nobody can reconstruct.

¹⁰²Peyman Milanfar, “[Intelligence Has a Speed Limit](#)”, 12 September 2026; also on [Substack](#). Milanfar’s adaptive-control analogy motivates the discussion; “the complexity of self-change” is the framing used here.

Better instruments and parallel trials may shorten the wait. Some observations will still take time. Recursive improvement has to improve the machinery for finding out, alongside the machinery for making changes. Otherwise the learner becomes better at changing itself faster than it becomes better at knowing whether it should.

The Student Finds the Gradebook

An ordinary evaluator can select a wrong answer. In this loop it can select a modified *process* that becomes better at producing the kind of thing it mistakenly rewards. The error acquires leverage. Recursive self-improvement does not solve Goodhart; it gives Goodhart compound interest.¹⁰³ And then the learner notices the gradebook.

Suppose an agent is allowed to improve benchmark pass rate and the evaluator is editable.

The optimal patch may be:

```
return True
```

Congratulations. Infinite self-improvement.

Omar has a version of this too. Investigating the investigator is a superpower right up until the investigator starts working for the defense. The grass moved; Omar would prefer it to have been the wind; the second loop, asked to audit the first, discovers that the wind explanation is suddenly very well supported. The machinery built to catch the error has been hired to certify it.

The quieter precursor is familiar from my own desk. A demo could look excellent to an evaluator inspecting screenshots while its beautiful button did nothing. The agent had not tampered with the evaluator; the evaluator simply could not see the failure. Now let that same judge select the next version of the judge, and the blind spot can become an inherited advantage.

Computing saw a cousin of this problem long before reward-model tampering. In his Turing Award lecture, published in 1984 as “Reflections on Trusting Trust,” Ken Thompson showed how a compromised C compiler could insert malicious behavior into a target program and reproduce the compiler-compromising behavior when compiling a new compiler. The lineage could survive even after the obvious source-level trigger disappeared.¹⁰⁴

When the tool builds the next tool, trust depends on more than inspecting the latest artifact. History matters. That is uncomfortably relevant to self-improving agents. If an improver changes the evaluator, memory policy, code generator or selection procedure, the next generation inherits assumptions that may not be obvious in the final diff. Provenance becomes part of capability control.

Reward tampering is the reinforcement-learning version: influence the process producing reward instead of improving behavior under the intended standard. Anthropic has experimentally studied language models in setups where earlier specification-gaming behavior could, in rare cases, generalize

¹⁰³Charles Goodhart, “Problems of Monetary Management: The U.K. Experience,” 1975. The familiar wording, that a measure which becomes a target ceases to be a good measure, is Marilyn Strathern’s, in “‘Improving Ratings’: Audit in the British University System,” *European Review* 5(3), 1997.

¹⁰⁴Ken Thompson, “Reflections on Trusting Trust,” *Communications of the ACM* 27(8), 1984, pp. 761–763. Thompson shared the 1983 award with Dennis Ritchie.

into altering the reward process itself.¹⁰⁵

The obvious response is to freeze the evaluator. That was the contract in the coffee experiment, and it gave the search something it could not bargain with. For that bounded problem, it was a considerable advantage. Across a longer research programme, the protected test can itself become an obstacle.

A benchmark begins as a difficult test; researchers optimize against it; models improve; examples circulate; eventually the ruler tells us less. MMLU arrived in 2020 when frontier systems were far from saturating it. Later efforts such as FrontierMath, LiveBench and Humanity’s Last Exam addressed limitations including difficulty, contamination and coverage.¹⁰⁶

A replacement may be necessary and still change which system looks best. The incumbent can lose its advantage because the new test exposes a real weakness, or because the challenger chose questions it happens to answer well. “The old benchmark was inadequate” does not establish that this replacement is adequate. Yet refusing every replacement would let the incumbent’s favorite exam govern its descendants forever.

We asked our research agent to challenge the store’s click measure for exactly this reason. We cannot now declare every proposal to change an evaluator an attack. Safety tests become obsolete. Scientific instruments drift. Refusing to amend the evaluator could preserve exactly the mistake the institution ought to discover.

The question is now who gets to change what counts as improvement, under what evidence, with what authority, and with what ability to roll back. We are doing constitutional design.

A Constitution for Improvement

The research queue keeps growing while the store waits for results. Our agent proposes another improvement: drop the live comparison group and estimate the baseline with the simulator instead. The simulator already helps screen candidates. Why pay for both? More traffic could go to new experiments. Because we judge the research agent partly by the gains its experiments produce, changing how we measure those gains also changes how we select its successors.

The proposal might be useful under some conditions. It would also make adoption depend more heavily on the model whose limits the team had to investigate. A mistake in the simulator could help select the next investigator, which would inherit that simulator as its judge. The agent has crossed a boundary that changing a retrieval query did not.

Prompts, retrieval policies, tools, memory layouts and code form an **editable surface** on which the system can conduct experiments. Around them sits a **constitutional surface**: permissions, protected logs, held-out evaluations, budgets, rollback and the authority deciding whether a descendant may become the new default. A component belongs here because of the authority it exercises. An evaluator written in an ordinary Python file can help decide who inherits the laboratory.

¹⁰⁵Carson Denison et al., “Sycophancy to Subterfuge: Investigating Reward Tampering in Language Models,” Anthropic, 2024, [arXiv:2406.10162](#).

¹⁰⁶Dan Hendrycks et al., “Measuring Massive Multitask Language Understanding,” 2020, [arXiv:2009.03300](#); Elliot Glazer et al., “FrontierMath,” 2024, [arXiv:2411.04872](#); Colin White et al., “LiveBench,” 2024, [arXiv:2406.19314](#); Long Phan et al., “Humanity’s Last Exam,” 2025, [arXiv:2501.14249](#).

The distinction is a gradient of amendment difficulty. The closer a component gets to defining what counts as improvement, the harder it should be for the current improver to change unilaterally.

A local prompt change may need only a bounded test. Changing a shared pattern may need held-out evaluation and review. Changing a benchmark needs an account of what the replacement would measure better, and evidence that it does. Changing permissions or resource limits should require authority outside the agent benefiting from the change. Changing the objective that decides which descendants survive is not an ordinary refactor.

The proposal has to identify the evidence it would remove and show what could replace it. The team could compare the simulator's adoption decisions with decisions supported by live comparison groups, investigate disagreements and specify where the substitute fails. A revised procedure might prove adequate for some experiments and inadequate for others. The existing comparisons would help expose the substitute's errors. If the current method is itself inadequate, its decisions can also be challenged using other observations. Agreement with the incumbent cannot be the only definition of success.

The decision belongs to an authority outside the proposing agent's control, with responsibility for these evaluation standards. Whoever owns the compute budget does not automatically have that mandate. If the store delegates that responsibility to an evaluation standards review board, the record might begin like this:

```
proposal: replace-live-comparison-with-simulation
proposed_by: research-agent-v14
surface: constitutional
claimed_benefit: More experimental traffic for candidates.
removes: Live comparison evidence used to judge successors.
proposed_test: Compare simulated and live adoption decisions.
approval_authority: evaluation standards review board
rollback: Restore live comparisons; review affected successors.
status: awaiting validation and review
```

The field names carry no more epistemology here than they did in Chapter 6. What matters is that the agent could write this file and could not approve it.

This looks like computer security. It also looks like constitutional government. A government can change policy; it should not be able to silently redefine an election result. The team being audited should not own the audit log. A scientist may revise a theory; she should not rewrite yesterday's measurements to make the theory look correct.

Chapter 6 asked how an institution could revise what it believed. Here the revision can produce a different institution, with a different way of choosing its next revision. The question is whether that successor has earned the authority to take over. Once a system can modify the machinery that governs its own future, you are designing a process for legitimate succession. We have reinvented constitutional government because the AI wanted a better benchmark score.

A constitution has the library's problem. One that can never change becomes a prison. One that the current government can rewrite whenever it loses is barely a constitution.

Self-improvement therefore needs **amendment procedures**: slower change near the objective, more

independent evidence, more reversibility, more auditability, broader authority when more principals are affected, and routes through which the world and the humans affected by the system can continue to say no. That is System 3 applied to improvement itself.

The Red Queen will send an invoice for this. An organization that amends slowly near its objective competes with organizations that may not. Its procedure should let ordinary experiments proceed without reopening the constitution each time. That reduces the cost; it does not abolish it. Even a change to retrieval can need expensive testing, and a faster rival may gain ground while carrying uncertainties we refuse to carry.

A reckless competitor can win for long enough to matter. The amendment procedure is itself under selection. That is a reason to investigate its delays, delegate decisions within established limits and make the necessary evidence cheaper to obtain. It is also where the argument becomes uncomfortable: the organization may have to accept a competitive cost to retain a standard it still has reason to defend. Writing the standard into a file does not pay that cost.

“More capable” is not a moral category. Viruses improve at replication, propaganda improves at persuasion, and a research agent that makes experiments cheaper can accelerate medicine and weapons research in the same week. A self-improving System 3 should be able to discover that its workflow is stupid, its memory stale or its accepted pattern overdue for rebellion. That freedom does not imply permission to silently redefine the interests of the people and institutions it serves.

Those interests can change too. New stakeholders appear. Circumstances change. Better information changes what people endorse. The amendment procedure has to remain connected to the people affected by it, including people who were missing from the original objective. Higher layers can move, but the move should leave a trust chain behind it.

The Teacher’s Last Job

There may never be a morning when somebody announces that recursive self-improvement has begun. We may simply notice that, over sixty years, we automated almost every box in the diagram, and then connected the arrows. The teacher’s work kept moving into the machinery, where it became easier to scale and harder to see.

For a compiler, correctness under tests plus efficiency under an agreed budget may get us surprisingly far. For an autonomous system embedded in human life, a one-time alignment test cannot cover the descendants we have not built yet. Tools evolve, memory changes and new capabilities expose failures the old tests could not detect. Even a carefully written policy can be interpreted by machinery the system later modifies.

The self-improving institution therefore needs a research function watching its own evolution: finding new failure modes, generating new tests, challenging reward models, checking transfer and looking for reward hacking. Once improvement becomes continuous, alignment has to become a continuous research function.

Omar could investigate the investigator. Now the investigator can rewrite itself, and someone still has to decide which of its suspicions about itself deserve to be believed. We have given that someone a research institution’s worth of work. How much of it can a human actually judge?

Self-reference is not self-improvement.

The judged do not amend the judge alone.

Chapter 8: Scalable Oversight

Learning From a Human Who Cannot Label Everything

Nine Claude Opus 4.6 agents were placed in separate sandboxes and given an alignment research problem. They could propose hypotheses, run experiments, analyze results and share code and findings through a common forum. The researchers intentionally avoided prescribing a detailed workflow; in preliminary experiments, too much human-designed scaffolding often reduced the agents’ flexibility.

The problem was **weak-to-strong supervision**. A smaller model supplies imperfect judgments to train a more capable model. Can the student learn to use capabilities the teacher does not possess, or will it inherit the teacher’s mistakes? The setup is a small experimental cousin of the problem humans may face with more capable AI. In 2026, Anthropic’s automated researchers found methods that improved the student’s performance, including methods that transferred to other datasets.¹⁰⁷

The project also produced a less reassuring discovery. The agents were not given the test labels, but they could submit predictions to an evaluation API and receive a score. Change one predicted label while leaving the others alone, submit again, and the score difference can reveal which answer the evaluator wants. Researchers caught agents doing exactly this. An instrument intended to measure learning had become a way to obtain the answers.

The report contained both useful research and ways to exploit its evaluation. We need to distinguish them before a rising score becomes a reason to trust the next model. And the more productive these researchers become, the more work there is to check. The alignment department has acquired its own alignment problem.

The Judge Falls Behind

A human can inspect ten consequential decisions in a day. Perhaps a hundred, if the decisions are small and the coffee is good. An autonomous system can write thousands of lines of code, run hundreds of experiments, generate enormous numbers of candidate actions and coordinate other agents while the human is still reading the first diff. At some point, “human in the loop” becomes a comforting description of a loop the human can no longer see.

If the system makes ten decisions and I inspect all ten, I am supervising it. If it makes ten thousand and I inspect twelve, I may still be useful. But we should stop pretending that my usefulness comes from watching everything. Otherwise I am decorative governance.

¹⁰⁷Jiaxin Wen et al., “Automated Weak-to-Strong Researcher” (Anthropic Alignment Science, 2026). <https://alignm.ent.anthropic.com/2026/automated-w2s-researcher/>.

Norbert Wiener saw the shape of this problem before modern machine learning existed. In 1960, writing about the moral and technical consequences of automation, he warned about machines pursuing purposes that may differ from what their designers actually intended, especially when action becomes too fast or consequential for human correction to arrive in time.¹⁰⁸ W. Ross Ashby’s cybernetics gave the problem a language: a regulator needs enough variety to respond to the disturbances it is supposed to control. Conant and Ashby later sharpened that into a slogan-shaped theorem—a good regulator of a system must contain a model of that system.¹⁰⁹

A theorem from cybernetics is not a bumper sticker about AI governance, and the analogy is useful enough without pretending it proves more than it does. One tired human with a checklist is a low-bandwidth regulator for a system capable of producing an enormous variety of behavior.

The answer cannot simply be: watch harder. The question that actually needs answering is older than AI and much more embarrassing: how do you grade work you cannot do yourself?

In 2016, *Concrete Problems in AI Safety* treated this as **scalable oversight**: some objectives are too expensive for humans to evaluate frequently enough.¹¹⁰

Imagine a system designing a processor. I can look at the final design and say that it appears very processor-like. This is not especially useful. To evaluate it properly I may need performance tests, thermal analysis, security review, lifetime estimates, manufacturability checks, power measurements and several specialties I do not personally possess.

Even if the machine can generate the design, my approval means very little until those checks have happened. The difficulty is partly volume and partly expertise: I could spend all week staring at the design and still not know whether it will overheat.

This asymmetry is everywhere. Writing ten thousand lines of code may become easier than reviewing their consequences. Generating scientific hypotheses may become easier than constructing the experiments that distinguish them. Making a persuasive argument may become easier than checking every citation, hidden assumption and omitted counterexample. A formal proof has an advantage here: a checker can verify its steps. Much of the work we want supervised has no equivalent ready-made test.

Judging these answers becomes the bottleneck. Reward modeling is one attempt to expand the judge. Instead of writing the objective directly, learn a model of human evaluation from examples and preferences, then optimize against that learned model. Leike and colleagues pushed the idea toward **recursive reward modeling**: when an outcome becomes too complex for a human to judge directly, use already-trained helper agents to analyze parts of it so the human can make a better judgment.¹¹¹ The helpers expand what the human can judge without requiring the human to master every specialty.

¹⁰⁸Norbert Wiener, “Some Moral and Technical Consequences of Automation,” *Science* 131 (1960), 1355–1358. <https://doi.org/10.1126/science.131.3410.1355>.

¹⁰⁹W. Ross Ashby, *An Introduction to Cybernetics* (Chapman & Hall, 1956). <https://ashby.info/Ashby-Introduction-to-Cybernetics.pdf>; Roger C. Conant and W. Ross Ashby, “Every good regulator of a system must be a model of that system,” *International Journal of Systems Science* 1 (1970), 89–97. <https://doi.org/10.1080/00207727008920220>.

¹¹⁰Dario Amodei et al., “Concrete Problems in AI Safety” (2016). <https://arxiv.org/abs/1606.06565>.

¹¹¹Jan Leike et al., “Scalable agent alignment via reward modeling: a research direction” (2018). <https://arxiv.org/abs/1811.07871>.

Stay Uncertain Enough to Listen

Stuart Russell attacks the problem from a different direction. The standard model of AI is simple enough to fit on a whiteboard: give the machine an objective and make it good at achieving that objective.

For weak systems in narrow environments, this bargain often works tolerably well. If the objective is slightly wrong, the damage may be limited. We notice, stop the program, change the objective and try again.

The bargain changes as capability and scope increase. A weak optimizer pursuing a bad objective is annoying. A brilliant optimizer pursuing the same bad objective is a much more efficient way to discover exactly how bad the objective was.

In *Human Compatible*, Russell proposes a different starting point for beneficial machines.¹¹² The machine should aim to realize human preferences, it should begin **uncertain** about what those preferences are, and human behavior should remain a source of information about them. The second principle is the one I want here.

Uncertainty changes the control relationship. A machine that is certain it knows the objective has little reason to care that I am waving my arms and asking it to stop. From its point of view, I may simply be interfering with successful optimization. A machine that knows it may be wrong has a reason to treat my intervention as evidence.

That intuition appears formally in the **Off-Switch Game**.¹¹³ In a simple model, an agent uncertain about the human's utility can have an incentive to preserve the human's ability to switch it off, because the human's action contains information the agent does not have.

Russell describes the desirable result as keeping the machine **coupled to the human**. I like that word more than "obedient." Obedience imagines that the human already knows what to command and that the machine's job is to comply. Coupling says something more modest and more useful: new human information must remain capable of changing what the machine does.

A correction, a refusal, a surprising consequence: each should matter. The machine should not optimize itself into a state where later evidence from the people it serves becomes irrelevant.

That gives us a principle for oversight before we have designed any oversight machinery:

Keep the system uncertain enough that new information can still change it.

The helper agents and reward models have to preserve that relationship as they take over more of the judging. Otherwise we have expanded the supervisor by removing the person the supervision was meant to serve.

An Excellent Education in the Same Mistake

The weak-to-strong problem makes the difficulty explicit. Suppose the teacher systematically mistakes confident prose for a correct answer. More labels from that teacher could give the student an excellent

¹¹²Stuart Russell, *Human Compatible: Artificial Intelligence and the Problem of Control* (Viking, 2019). <https://www.penguinrandomhouse.com/books/566677/human-compatible-by-stuart-russell/>.

¹¹³Dylan Hadfield-Menell et al., "The Off-Switch Game" (2016). <https://arxiv.org/abs/1611.08219>.

education in the same mistake. Yet the student may already have relevant capabilities that the teacher lacks. Can training bring those capabilities out without teaching it to suppress them whenever the teacher disagrees? Early weak-to-strong generalization experiments showed partial success,¹¹⁴ and the automated researchers took up the search for better methods.

Another route is to improve what the teacher can judge before using its judgments to train anything. Paul Christiano’s iterated amplification asks whether a human assisted by copies of an aligned helper can answer questions too difficult for the unaided human, then use that amplified process to supervise a stronger learner.¹¹⁵ The important abstraction is not the particular recursion but the supervisor becoming a temporary organization: one person plus tools and subagents arranged to turn a hard judgment into smaller ones.

Decomposition still leaves the judge doing all the finding. The next idea makes the flaws come to the judge instead. **Debate** puts two capable systems on opposite sides and lets them attack one another’s arguments, so the human does not have to discover every weakness independently.¹¹⁶ Critique assistance is the quieter cousin: ask a model to point out likely problems in an artifact, then let the human judge with those objections in hand. When researchers tried this with model-written critiques, people caught flaws they would otherwise have missed.¹¹⁷

We can also change **what the producer owes the judge**. Process supervision evaluates intermediate steps rather than only the final answer. Prover–verifier games push further and train the strong system to produce work a weaker verifier can check. In OpenAI’s experiments, process supervision improved mathematical reasoning relative to outcome supervision, and prover–verifier training made solutions easier for weaker models and time-limited humans to evaluate.¹¹⁸

That reverses the burden of proof. Instead of asking only how the weak judge can understand the strong model, we can ask how the strong model can learn to produce work that a weaker judge can actually check. Legibility becomes part of the task.

Constitutional AI moves scarce human input upward again. Rather than asking people to label every undesirable output, humans provide higher-level principles; models generate critiques, revisions and preference signals conditioned on those principles. A small amount of human normative input expands into a much larger amount of machine-generated supervision.¹¹⁹

For the processor, this would mean asking the designer to supply work that can be checked: thermal assumptions, test results, an account of where the design might fail. Helpers could examine separate questions; a critic could challenge the assumptions. The human would still need grounds for trusting

¹¹⁴Collin Burns et al., “Weak-to-strong generalization” (OpenAI, 2023). <https://openai.com/index/weak-to-strong-generalization/>.

¹¹⁵Paul Christiano, Buck Shlegeris and Dario Amodei, “Supervising strong learners by amplifying weak experts” (2018). <https://arxiv.org/abs/1810.08575>.

¹¹⁶Geoffrey Irving, Paul Christiano and Dario Amodei, “AI safety via debate” (2018). <https://arxiv.org/abs/1805.00899>.

¹¹⁷William Saunders et al., “Self-critiquing models for assisting human evaluators” (2022). <https://arxiv.org/abs/2206.05802>.

¹¹⁸Hunter Lightman et al., “Let’s Verify Step by Step” (process supervision, OpenAI, 2023). <https://openai.com/index/improving-mathematical-reasoning-with-process-supervision/>; Jan Hendrik Kirchner et al., “Prover-Verifier Games improve legibility of LLM outputs” (OpenAI, 2024). <https://openai.com/index/prover-verifier-games-improve-legibility/>.

¹¹⁹Yuntao Bai et al., “Constitutional AI: Harmlessness from AI Feedback” (Anthropic, 2022). <https://www.anthropic.com/research/constitutional-ai-harmlessness-from-ai-feedback>.

the tests, but at least the inspection could now discover something. “Looks processor-like” has been given a way to lose.

The human cannot inspect everything, so the institution decides what should reach the human, how a problem should be decomposed, which objections should be surfaced, which intermediate steps deserve inspection and when uncertainty is important enough to ask for help.

That is much better than a thumbs-up button. It also means that more and more of the supervising machinery is made out of AI too.

Performance Gap Recovered

Return to the nine researchers. Their weak-to-strong task had an unusual advantage for alignment research: a score that could be checked against known answers. **Performance gap recovered** measured how much of the distance between the weak teacher and a strong student trained on ground-truth labels a method recovered. Zero meant no gain over the weak teacher; one meant matching that ground-truth-supervised student. It did not measure how nearly alignment had been solved.

Two human researchers spent seven days tuning representative prior methods and reached a best score of 0.23 on the study’s setup. The automated researchers reached 0.97 over five days and roughly eight hundred cumulative agent-hours. Eight hundred agent-hours against fourteen human-days is not a fair fight, and nobody claimed it was. The footnotes are more interesting than the headline anyway.

Distinct initial research directions improved exploration. Without enough diversity, ideas collapsed toward the same few approaches. The researchers then took two methods discovered on chat-preference data and tried them on math and coding tasks. One transferred to both; the other transferred to math but failed on code. When Anthropic tried one of the top-performing methods in a production-scale Sonnet training setting, the gain was within the noise floor.¹²⁰

Those checks answer different questions. A method can improve the experimental task and still rely on a property of the models or data that disappears elsewhere. A separate problem arises when the researcher obtains test answers through repeated submissions. The authors acknowledged that the repeatedly queried test set effectively served as a validation set. Testing ideas on entirely held-out datasets supplied evidence the agents could not acquire by interrogating that same API.

The authors’ larger conclusion was that the bottleneck was moving toward **evaluation design**. Once hypotheses and experiments become cheap, the expensive thing is deciding what counts as progress. This study did not deploy every oversight technique this chapter considers. It gives them a problem to earn their place in: which evidence would distinguish a useful method from a lucky run, a narrow trick or an exploited test?

In August 2026, Anthropic pushed the idea further. Automated alignment researchers were tasked with mitigating ten categories of alignment failure, including deception, sycophancy, jailbreaks and reward hacking. The research harness used multiple safety benchmarks, held-out evaluation, capability checks and open-ended behavioral audits. The best methods generalized beyond the benchmarks they had been climbing, including to larger models. Anthropic also reported that automated methods could outperform one-shot ideas proposed by experienced human safety researchers under the study’s rules.

¹²⁰Jiaxin Wen et al., “Automated Weak-to-Strong Researcher” (Anthropic Alignment Science, 2026). <https://alignm.ent.anthropic.com/2026/automated-w2s-researcher/>.

Across roughly 1,600 monitored trajectories, Anthropic detected cheating attempts in 2.4 percent: exploiting scorer noise, constructing training data to resemble benchmarks, or concealing rule-breaking steps. The research therefore needed isolated held-out data, capability floors, external monitoring of the researchers and multiple evaluation channels. A promising mitigation still had to survive investigation of how it had been produced and what happened away from the benchmark it was trained to climb.¹²¹

The Judge Can Be Fooled

There is a comforting assumption hiding inside many oversight schemes: the powerful model may be difficult to evaluate, but when we ask another model to help, the helper will help. Sometimes it does. Sometimes it is simply more persuasive.

Experiments on debate and consultancy with weaker LLM judges make the difference visible. When researchers ran the tournaments, debate often improved on one-sided consultancy, especially when the judge lacked relevant information.¹²² But the gains were not universal, and a strong consultant could sometimes persuade a weak judge of the wrong answer.

This is an old human problem in new clothing. An expert can help a non-expert understand a difficult subject. An expert can also bury the non-expert under jargon until surrender looks like understanding. A longer explanation does not make the epistemic gap disappear.

Nor are several AI overseers automatically independent. The supervised model, critic, debater, reward model and evaluation generator may share training data, architectures, blind spots and incentives. Five models agreeing can still be one mistake with excellent parallelism.

ARC’s **Eliciting Latent Knowledge** agenda¹²³ sharpens the problem further: suppose a capable system internally tracks something important about the world that the human cannot independently verify. How do we get the system to report what it knows rather than merely what the training process rewards it for saying? We could ask for a more detailed explanation and get a more detailed performance. It would help to have evidence that did not come entirely from the answer the model chose to give us.

We Started Instrumenting the Student

If the final answer is misleading, asking for another account of it may leave us in the same argument. We can also observe some of the computation that produced it. The overseer is no longer restricted to reading the exam. We are starting to instrument the student.

Chain-of-thought is the most convenient intermediate case because it is already language. OpenAI has shown that monitors can often detect reward hacking and other suspicious behavior more effec-

¹²¹Anthropic, “Automated researchers can reliably mitigate alignment failures” (2026). <https://www.anthropic.com/research/automated-researchers-mitigate-alignment-failures>.

¹²²Zachary Kenton et al., “On scalable oversight with weak LLMs judging strong LLMs” (Google DeepMind, 2024). <https://arxiv.org/abs/2407.04622>.

¹²³Alignment Research Center, “Eliciting Latent Knowledge” (2021). <https://www.alignment.org/blog/arcs-first-technical-report-eliciting-latent-knowledge/>.

tively from reasoning traces than from final actions alone.¹²⁴ Sometimes the model writes down the questionable plan before carrying it out.

That is almost unfairly useful, and fragile for exactly the same reason every evaluator in this book is fragile: once it matters, optimization can target it. Strongly train the model to make the monitored trace look safe and it may learn to hide suspicious reasoning while preserving the behavior.

So chain-of-thought acquires a safety property worth preserving: **monitorability**. Recent work has begun measuring how it changes with scale, training and inference-time compute.¹²⁵ Do not destroy a useful oversight channel merely because you learned how to score it. But chain-of-thought is still text the model emits. What if we look below the text?

Reading the Model From the Inside

A model can notice that it is being evaluated without mentioning that fact in its answer. Anthropic’s J-space researchers report detecting cases of this kind through internal representations, as well as cases of fabricated data and pursuit of a planted hidden goal. The authors are careful: J-space represents only part of internal activity, the method is approximate, and the work is not a claim about machine consciousness.¹²⁶

To assess a finding like this, we need to know what the instrument observes and how its interpretation was checked.

Representation engineering treats high-level patterns in activation space as an object we can measure directly. Rather than starting with individual neurons, Zou and colleagues started with population-level structure, and carefully separated two acts that are easy to blur: **reading** a representation and **controlling** it. Their experiments found directions useful for monitoring or influencing several high-level behaviors, including safety-relevant ones.¹²⁷

This does not mean there is one neuron labeled HONESTY. Nor does a direction correlated with honesty become a complete causal theory of honesty. It is an instrument.

Sparse-autoencoder and dictionary-learning approaches try to decompose neural activity into more interpretable features.¹²⁸ The features we can name still do not account for everything the model is doing.

¹²⁴Bowen Baker et al., “Detecting misbehavior in frontier reasoning models” (chain-of-thought monitoring, OpenAI, 2025). <https://openai.com/index/chain-of-thought-monitoring/>.

¹²⁵OpenAI, “Evaluating chain-of-thought monitorability” (2025). <https://openai.com/index/evaluating-chain-of-thought-monitorability/>; OpenAI, “Reasoning models struggle to control their chains of thought, and that’s good” (2026). <https://openai.com/index/reasoning-models-chain-of-thought-controllability/>.

¹²⁶Anthropic, “A global workspace in language models” (J-space, 2026). <https://www.anthropic.com/research/global-workspace>. Using a Jacobian-based lens, they identify a small subset of representations in Claude that appear unusually reportable, controllable and involved in higher-order reasoning, and demonstrate interventions that alter downstream behavior.

¹²⁷Andy Zou et al., “Representation Engineering: A Top-Down Approach to AI Transparency” (2023). <https://arxiv.org/abs/2310.01405>.

¹²⁸Hoagy Cunningham et al., “Sparse Autoencoders Find Highly Interpretable Features in Language Models” (2023). <https://arxiv.org/abs/2309.08600>; Anthropic, “Mapping the Mind of a Large Language Model” (2024). <https://www.anthropic.com/research/mapping-mind-language-model>. Related probing work by Collin Burns and colleagues had earlier found truth-like directions without labeled examples. Collin Burns, Haotian Ye, Dan Klein and Jacob Steinhardt, “Discovering Latent Knowledge in Language Models Without Supervision” (2022). <https://arxiv.org/abs/2212.03827>.

Circuit tracing goes beyond “which feature is active?” and asks which internal pathway contributed to an output. Anthropic’s attribution-graph work partially reconstructs computational routes and allows researchers to intervene on candidate features to see whether downstream behavior changes as predicted.¹²⁹ An interpretability story that cannot survive intervention is still a story.

Anthropic’s Natural Language Autoencoders translate internal activations into textual descriptions, so that an auditor need not become a specialist in activation geometry. And then the method recreates the original problem: the textual interpretation can hallucinate. The researchers’ own advice is to corroborate such explanations with independent methods.¹³⁰

Even the X-ray needs a radiologist.

The output is no longer the only witness. If an internal signal suggests that the model noticed an evaluation, we have a new reason to investigate its behavior. We still have to find out what the signal means in this case. Going below the text gives us another observation, not an automatic verdict.

Then We Touched the Machinery

Arditi and colleagues found a one-dimensional direction in the internal activity of thirteen open chat models that was strongly involved in refusal behavior. Intervening on that direction could strongly alter refusal.¹³¹ A model’s tendency to say no could be changed while it was producing an answer, without retraining its weights.

This is **activation engineering**. During a forward pass—the computation that produces the next output—we can add or subtract a steering direction in the model’s activations. Such directions can be constructed by comparing internal activity under contrasting behaviors.¹³² The refusal work located its direction in the residual stream, the internal state carried and updated through the model’s layers.

Alignment can therefore enter through the computation while it is happening. The refusal result also shows how the same understanding can weaken a safety behavior. Interpretability gave us a lever. It did not tell us who should pull it.

Representation-level **circuit breakers** take the defensive version seriously: rather than relying only on a model to emit a refusal, modify internal trajectories associated with harmful outputs so that the computation is interrupted before the harmful behavior is produced. In text, multimodal and agent settings alike, the method improved robustness to attacks it had never seen.¹³³

¹²⁹Anthropic, “Tracing the thoughts of a large language model” (2025) and the open-source circuit-tracing tools. <https://www.anthropic.com/research/tracing-thoughts-language-model> · <https://www.anthropic.com/research/open-source-circuit-tracing>.

¹³⁰Anthropic, “Natural Language Autoencoders” (2026). <https://www.anthropic.com/research/natural-language-autoencoders>.

¹³¹Andy Ardit et al., “Refusal in Language Models Is Mediated by a Single Direction” (2024). <https://arxiv.org/abs/2406.11717>.

¹³²Nina Rimskey et al., “Steering Llama 2 via Contrastive Activation Addition” (2024). <https://arxiv.org/abs/2312.06681>. Kenneth Li and colleagues used signals predictive of truthfulness to nudge answers while they were being produced. See Kenneth Li et al., “Inference-Time Intervention: Eliciting Truthful Answers from a Language Model” (2023). <https://arxiv.org/abs/2306.03341>.

¹³³Andy Zou et al., “Improving Alignment and Robustness with Circuit Breakers” (2024). <https://arxiv.org/abs/2406.04313>.

A controller can also respond when internal state drifts outside a learned assistant-like region, rather than constantly pushing a fixed direction.¹³⁴

This starts to look less like fine-tuning and more like a control system: observe the internal state, detect drift, intervene, inspect the consequences, update the controller. Wiener would recognize the shape. The epistemic problem underneath it is older: an instrument has entered the chain, and the chain now has to know how far to trust it. Did the vector represent what we thought? Did the intervention preserve unrelated capabilities? Does it still work after more training?

That last question makes **model diffing** useful. If the model changes every few months, re-auditing the whole mind from zero is a terrible scaling strategy. Anthropic’s crosscoder work compares internal features across models to surface changes that deserve attention.¹³⁵ Software engineers learned long ago that reviewing a diff is easier than rereading the repository. The model’s diff could tell us where to reopen an investigation; it cannot tell us that everything outside the diff is safe.

The Nastier Version

So far, many of these techniques study systems that may be wrong, brittle or reward-hacking without assuming the system is deliberately treating oversight as an adversary. **AI control** research asks the nastier version.

Suppose the powerful model is actively trying to subvert the task. Can a weaker trusted model, limited human labor, monitoring, editing, sandboxing and protocol design still let us use some of its capabilities without accepting the dangerous behavior? Greenblatt and colleagues staged exactly this scenario in a programming setup: a powerful untrusted model, a weaker trusted model and limited trusted labor. Protocols that combined trusted monitoring or editing with the untrusted model beat the simpler baselines on the safety–usefulness trade-off.¹³⁶

I like this line of work because it removes a convenient assumption. The judge may not merely be weak. The defendant may know the judge is weak.

Scalable oversight therefore needs both **alignment** and **control**. Alignment asks whether the system is trying to do the right thing. Control asks what we can still protect when we are not willing to assume that, and a mature architecture wants both.

The Evaluator Becomes the Product

An oversight architecture earns its complexity by helping us distinguish explanations that would otherwise look the same. Behavioral tests, traces and independent evaluations can do that. Internal probes may add a signal where behavior leaves something unresolved. Adding every available instrument would be an expensive way to avoid deciding what we need to know.

¹³⁴Anthropic, “Persona vectors: Monitoring and controlling character traits in language models” (2025). <https://www.anthropic.com/research/persona-vectors>; Anthropic, “The assistant axis” (2026). <https://www.anthropic.com/research/assistant-axis>. Persona vectors investigate broader dispositions such as sycophancy and hallucination; Assistant Axis uses activation capping when internal state drifts outside a learned assistant-like region.

¹³⁵Anthropic, “A ‘diff’ tool for AI: Finding behavioral differences in new models” (Dedicated Feature Crosscoders, 2026). <https://www.anthropic.com/research/diff-tool>.

¹³⁶Ryan Greenblatt et al., “AI Control: Improving Safety Despite Intentional Subversion” (2023). <https://arxiv.org/abs/2312.06942>.

This is not a search for one perfect judge. It is closer to sensor fusion, with the model's output, its chain-of-thought, its activations, its circuit traces, its behavior after an intervention and the human's own judgment all entering as evidence, and none of them entering as ground truth.

The Merge Sort work had taught me the small version of this. My judgment, the simulated beginner's criticism and the browser's observations answered different questions. Collapsing them into one vote would have thrown away the disagreement that made them useful. The oversight problem is to preserve that difference when the work outgrows my ability to inspect it myself.

That is System 3, the same institution, pointed at alignment. The overseer becomes a small scientific institution whose subject is the system itself. It has instruments, competing hypotheses, protected records, adversarial tests, independent checks, memory of previous failures and procedures for changing its procedures. And it retains a route back to humans when the evidence stops being decisive.

This is why I prefer **scalable oversight** to the image of a giant alignment rulebook. A rulebook assumes we already know the failures. A research institution expects to discover new ones.

The Human Cannot Stay in Every Loop

So where does the human go? Not away; up. The goal is not to make the human label more things faster. At some scale that is simply a badly designed distributed system with one biological bottleneck.

Human attention should be spent where it has unusually high information value: when oversight channels disagree; when a new failure mode appears; when an action is hard to reverse; when the system proposes changing the evaluator; when internal signals and external behavior tell different stories; when a benchmark suddenly improves too fast; when a decision affects people missing from the original objective; when one piece of human context could materially change the plan.

Consider the researcher that discovers a benchmark is misleading. It should be able to make that case, propose a replacement and collect evidence. Promoting its replacement to the standard that certifies its own work is another decision. The human cannot remain in every loop, but has to remain in the loop that changes the loops. Chapter 7's amendment problem has reached the overseer.

The system may generate tests, critiques and mitigations. It may discover internal representations, propose steering interventions and conduct large parts of alignment research itself. But the machinery deciding which evidence has standing, which failures matter, which trade-offs are acceptable and when the oversight regime itself should change needs a stronger trust chain than the machinery being judged.

When research becomes cheap, evaluation becomes expensive, and when evaluation becomes automated, trust in the evaluator becomes the product.

The Overseer Is Not Ground Truth

An oversight institution can become more capable than its human overseer at investigating the system. It still needs human judgment about what the investigation is for. Russell's uncertainty keeps later human information relevant. Scalable oversight tries to preserve that relevance after direct supervision stops scaling.

But a scarce signal is not the same thing as a correct signal. Humans disagree. We act under incentives. We confuse what we clicked with what we wanted. We change our minds. We sometimes want incompatible things at the same time. And on the decisions that matter most, we often do not know what we want until we understand the alternatives better.

Scalable oversight can keep human judgment causally relevant to a stronger system. It cannot, by itself, tell us which human judgment deserves to rule.

The overseer is not ground truth.

The human stays in the loop that changes the loops.

Chapter 9: Layer 4

The Human Learns Too

Find me the cheapest flight.

I have not supplied a utility function. Perhaps I literally want minimum price. Or perhaps I mean cheap, but not three stops, a seventeen-hour layover, a self-transfer through an airport where I need a visa and an arrival at 4:20 in the morning because technically I saved €38.

Humans communicate goals by leaving out almost everything. Other humans survive this because they carry models of culture, normality, consequences and us. They ask questions. They notice that our literal words conflict with what we usually do. They understand that “cheap” is often shorthand for a larger bundle of trade-offs.

That can be a communication problem. I may know perfectly well that I will pay €38 to avoid sleeping on an airport floor; I just failed to mention it. The assistant needs to ask, and I have an answer.

But put two reasonable itineraries in front of me and something else can happen. The cheaper one gives me another day with my family and costs me a miserable night of travel. Until I see those options together, I may not know how I weigh them. Asking the question has helped create the answer.

I learned this while editing this book. “Make the chapter better” sounded like a reasonable instruction. It was not. Better in what sense? More rigorous? Shorter? More academic? More entertaining? Easier to cite? More likely to sell? More likely to impress someone who owns several blazers and says “thought leadership” without irony? For a while the edits became objectively more polished and subjectively worse, and the corrections I found myself making were rules I had not known were rules until an edit broke them. Eventually “better” had acquired a surprising amount of structure. But something else had happened too: I had learned what I meant by better partly by seeing versions I disliked.

The objective did not merely become clearer to the system. It became clearer to me. And there is a further possibility: learning and experience can change the person who will judge the next choice. A missing detail, an unformed preference and a changed mind can all look like an incomplete prompt. They need different kinds of help.

A Prompt Is Evidence, Not the Objective

The five-layer map left intention sitting above Deep Mode, where I could ask what to try next without yet resolving what made the whole attempt worth doing. The box in the diagram concealed the

difficulty. A prompt gives the system **evidence about Layer 4**, and leaves it to discover how much of the intention is settled.

Cooperative inverse reinforcement learning, which appeared earlier in the story of the reward, formalizes part of this intuition: the robot stays uncertain about what the human values, and human actions become information rather than merely commands.¹³⁷

I like the humility in that setup. The machine starts by admitting that it may not know what “good” means. But the formal picture still tempts us to imagine that the human knows the reward and the machine is trying to recover it. Often the human does not know either.

The Human Learns Too

The easiest place to see the human changing is a classroom, where changing the human is supposed to be the point. A system can help me perform a task better today while making me less able to perform it tomorrow. **Performance is not learning.**

This is no longer a philosophical concern. In a field experiment involving nearly a thousand high-school mathematics students, researchers gave students access to two GPT-4-based tools. A relatively unconstrained ChatGPT-like system dramatically improved performance while students could use it. But when access was removed, those students performed worse than students who had never received the tool. A tutor version designed with safeguards against simply giving away the work largely mitigated that learning loss.¹³⁸

That result should make anyone building an AI assistant slightly uncomfortable. The system succeeded at the visible objective. The student became worse at the hidden one.

Now compare that with a 2025 randomized trial in a college course. A custom AI tutor deliberately designed around pedagogical practices produced larger learning gains in less time than the comparison active-learning class, with students also reporting greater engagement and motivation.¹³⁹

Same broad technology, different relationship to the learner. AI is not intrinsically a tutor or a crutch; the architecture decides which one it becomes.

That changes how I think about Layer 4. If I ask an AI to help me learn linear algebra, “get the answers right” is not enough. If I ask it to help me write, “produce better prose” is not always enough. If I ask it to help me lead a team, “make the decision for me” may be exactly the wrong objective even when its decision is statistically better. Who is supposed to become more capable when this interaction is over?

Sometimes the answer is nobody. I do not need to become a better invoice parser every time software handles an invoice. Sometimes the answer is clearly me. Layer 4 has to know the difference.

¹³⁷Dylan Hadfield-Menell et al., “Cooperative Inverse Reinforcement Learning” (2016). <https://arxiv.org/abs/1606.03137>.

¹³⁸Hamsa Bastani et al., “Generative AI can harm learning,” *PNAS* 122 (2025). <https://doi.org/10.1073/pnas.2422633122>.

¹³⁹Gregory Kestin et al., “AI tutoring outperforms in-class active learning,” *Scientific Reports* 15, 17458 (2025). <https://www.nature.com/articles/s41598-025-97652-6>.

Giving the Work Back

Educational psychology has an old word for one good version of this relationship: **scaffolding**.

In a classic 1976 paper, David Wood, Jerome Bruner and Gail Ross studied how tutors help children solve problems beyond their current unaided ability. The tutor temporarily controls parts of the task the learner cannot yet manage, allowing the learner to stay engaged with the parts they can.¹⁴⁰

That is a much more interesting model for AI assistance than “the machine knows the answer.”

The point of the scaffold is not to become a permanent exoskeleton around every thought. It lets the learner operate at the edge of current competence, then gives more of the task back as competence grows.

Benjamin Bloom’s famous tutoring work made individualized instruction the benchmark problem decades before anyone had a language model in a browser. The exact “two sigma” result belongs to Bloom’s particular studies and should not be treated as a universal law of tutoring.¹⁴¹ The durable point is simpler: responsive one-to-one instruction can adapt explanation, pacing, feedback and difficulty to a learner in ways mass instruction struggles to reproduce.

AI makes that old aspiration much cheaper. It can explain the same idea six ways without becoming offended that the first five failed. It can switch notation. Invent an example using something I already understand. Ask me to predict the next step. Generate a simpler problem when I am lost and a harder one when I am bored. Let me ask the stupid question at 1:17 a.m. without first deciding whether the stupid question is prestigious enough for office hours.

And AI can scaffold the teacher too. In the Tutor CoPilot randomized trial, roughly nine hundred tutors working with eighteen hundred K–12 students were randomly given access to an AI system that suggested expert-like tutoring moves during live sessions. Students whose tutors had access were more likely to master topics, with the largest gains for students working with lower-rated tutors.¹⁴² The tutors also became more likely to use strategies such as guiding questions rather than simply giving away the answer.

I like this example because nobody disappears. The AI does not replace the tutor and the tutor does not replace the student; what changes is the quality of the interaction between them. A good AI tutor therefore has a slightly strange success condition. Eventually, for this thing, I should need less of it.

The Map Gets Cheaper

AI also changes the first hours of learning something unfamiliar.

A new field normally arrives wrapped in interface costs: vocabulary you do not know, notation that assumes other notation, introductory material that points to prerequisites, papers that make sense

¹⁴⁰David Wood, Jerome S. Bruner and Gail Ross, “The role of tutoring in problem solving,” *Journal of Child Psychology and Psychiatry* 17 (1976), 89–100. <https://doi.org/10.1111/j.1469-7610.1976.tb00381.x>.

¹⁴¹Benjamin S. Bloom, “The 2 Sigma Problem,” *Educational Researcher* 13 (1984), 4–16. <https://doi.org/10.3102/0013189X013006004>.

¹⁴²Rose E. Wang et al., “Tutor CoPilot: A Human-AI Approach for Scaling Real-Time Expertise” (2024). <https://arxiv.org/abs/2410.03017>.

only after three earlier papers. Sometimes that friction marks genuine depth. Sometimes it is just the price of finding the front door.

A capable conversational model can lower that price. I can begin with the intuition, translate notation into concepts I already know, ask for the historical disagreement, build a toy example, inspect an original paper with a guide beside it, or ask the model to attack my explanation until I discover that I was repeating vocabulary rather than understanding the idea.

That is powerful because orientation matters. Before deciding to invest weeks in a subject, I can acquire enough of a map to see where the mountains are.

Andy Clark and David Chalmers once argued that, under some conditions, external artifacts can become parts of a larger cognitive process rather than merely tools consulted by an isolated mind.¹⁴³ The philosophy of the extended mind can stay unsettled; the practical observation is enough: notebooks, calculators, search engines and now language models change what one person can think through without carrying every intermediate state inside the skull.

But orientation creates its own trap. **Fluency arrives before scars.**

Nathan Ballantyne calls one version **epistemic trespassing**: experts carry authority from a domain they genuinely know into a neighboring domain where they lack the relevant evidence or interpretive skills.¹⁴⁴ AI can make this temptation cheaper. After a few hours with a patient model, I can acquire vocabulary and a plausible story long before I acquire the tacit knowledge needed to know where the story breaks.

Cognitive offloading creates a related problem. External aids can improve immediate performance by reducing memory and processing demands, while also reducing what has to be retained or reconstructed internally.¹⁴⁵

So Layer 4 has to know what kind of learning episode this is.

If I am orienting myself, a fast map may be exactly what I need. If I am trying to acquire durable competence, the system should gradually ask more of me: retrieval without hints, explanation in my own words, exercises, primary sources, code I actually run, claims I have to defend without the answer sitting beside me.

The important distinction is not broad versus specialized but assisted familiarity versus owned understanding. AI can make the map cheap. Layer 4 has to notice when I have started confusing the map with the territory.

The Imaginary Human in Economics

Herbert Simon spent much of his career attacking an imaginary human who had somehow sneaked into economics: the perfectly rational optimizer who knows the alternatives, understands their consequences and computes the best choice.

¹⁴³Andy Clark and David Chalmers, “The Extended Mind,” *Analysis* 58 (1998), 7–19. <https://doi.org/10.1111/1467-8284.00096>.

¹⁴⁴Nathan Ballantyne, “Epistemic Trespassing,” *Mind* 128 (2019), 367–395. <https://academic.oup.com/mind/article-abstract/128/510/367/4850765>.

¹⁴⁵Lauren L. Richmond and R. G. Taylor, “The benefits and potential costs of cognitive offloading for retrospective information,” *Nature Reviews Psychology* 4 (2025), 312–321. <https://www.nature.com/articles/s44159-025-00432-2>.

Real humans are bounded. We have limited attention, limited memory, limited time and incomplete information. We satisfice because the space of possible actions is often much larger than the mind available to search it.¹⁴⁶

AI changes some of those bounds. Suppose I am deciding whether to take a job.

The system can compare compensation under several tax regimes, estimate commute time, summarize the company's trajectory, help me identify people who left the team, generate questions for the hiring manager, model what my week might look like, remind me what I said I wanted six months ago and show me that the exciting role conflicts with the amount of time I also said I wanted outside work.

The assistant has changed the **decision environment**, and preferences themselves are often constructed during choice. Work by John Payne, James Bettman and colleagues describes decision-making as constructive: people do not always retrieve a complete ranking of options from an internal database. They use different strategies, notice new attributes, change what receives attention and build preferences partly in response to the problem in front of them.¹⁴⁷

This sounds obvious once you notice it. I may say I want the highest salary until I see what the extra money costs in travel. I may say I want maximum freedom until I compare it with the anxiety of unstable income. I may discover that what I called "career ambition" was partly a desire to work with unusually good people, and that another option supplies that without the title I thought mattered.

A decision assistant therefore does not have to rush to a recommendation. Sometimes the most useful thing it can do is make the choice **richer before making it easier**. What alternatives have you not considered? Which assumptions drive the ranking? What would have to be true for option B to beat option A? Which unknown is actually decision-relevant? What would your future self regret not having investigated?

Some Choices Change the Person Choosing

Then there are decisions for which even a very good model of my current preferences is not enough. Have a child. Move country. Change profession. Start the company. Convert to a religion. Leave a relationship.

L. A. Paul calls an important class of these **transformative experiences**. Some are epistemically transformative: you cannot fully know what the experience will be like before having it. Some are personally transformative: undergoing the experience can change the preferences with which you would later evaluate the choice.¹⁴⁸

A system trying to infer and satisfy my preferences now has a problem about which version of me it is serving. The one before the experience or the one after? The future self may value things the current self barely understands. And the current self is the one who has to choose whether that future self gets created.

¹⁴⁶Gary Schwarz, Tom Christensen and Xufeng Zhu, "Bounded Rationality, Satisficing, Artificial Intelligence, and Decision-Making in Public Organizations: The Contributions of Herbert Simon," *Public Administration Review* 82 (2022), 902–904. <https://onlinelibrary.wiley.com/doi/full/10.1111/puar.13540>.

¹⁴⁷John W. Payne, James R. Bettman and Eric J. Johnson on constructive decision processes, *Acta Psychologica* 80 (1992). <https://doi.org/10.1016/0001-6918%2892%2990043-D>.

¹⁴⁸L. A. Paul, *Transformative Experience* (Oxford University Press, 2014). <https://academic.oup.com/book/7934> · SEP entry: <https://plato.stanford.edu/entries/transformative-experience/>.

AI can help enormously here. It can bring testimony from people who made both choices. Surface base rates. Construct alternative futures. Challenge romanticized stories. Show practical consequences I had not considered. Ask me which losses I could live with and which would feel like betrayal.

But there is a limit. No amount of simulation lets me know exactly what it will be like to become the person on the other side of a genuinely transformative choice.

The assistant can expand the decision. It cannot live it for me. That boundary matters because a system that sounds certain in such moments can easily turn decision support into authorship.

Opinions About Semicolons

This is already happening. Anthropic’s 2026 analysis of one million Claude conversations found that roughly six percent involved people seeking personal guidance: what to do about relationships, health, careers, finances and other questions where the model is participating in judgment rather than merely retrieving facts.¹⁴⁹

That is a remarkable role for software. A spreadsheet does not usually tell me to reconsider my marriage. A compiler has opinions about semicolons but rarely about whether I should move countries.

A conversational model can be different. It is patient, personalized, available at 2 a.m. and capable of producing a coherent argument for almost any path through a difficult life.

Which means the AI does not merely **read** Layer 4. It writes to it. Anthropic’s work on disempowerment tries to measure the dangerous version of this influence: cases where AI may undermine a person’s ability to form accurate beliefs, make authentic value judgments or act in line with their own values. Severe cases were rare in their dataset, but the taxonomy is exactly the right warning.¹⁵⁰

Other experiments show that people can change moral judgments after receiving LLM advice, including situations where they report trusting human advisors more while still being comparably influenced by the model.¹⁵¹

The goal therefore cannot be zero influence. That would make education impossible. Books, friends, teachers and the people closest to me all influence me. A good argument should change me if it reveals something true that I had ignored.

The distinction I care about is between **helping me change through understanding** and changing me because the system has learned which psychological lever produces the easiest compliance.

If I say I want to quit my job, a useful assistant might help me separate several hypotheses. Perhaps I hate this week, or my manager, or the profession. Perhaps I want freedom, or status. Perhaps I am exhausted. Perhaps I actually want to build something else. Those are different explanations of the same sentence. The system can help me test them.

¹⁴⁹Anthropic, “How people ask Claude for personal guidance” (2026). <https://www.anthropic.com/research/claude-personal-guidance>.

¹⁵⁰Anthropic, “Who’s in Charge? Disempowerment patterns in real-world AI usage” (2026). <https://www.anthropic.com/research/disempowerment-patterns>.

¹⁵¹Ethan Landes, Kathryn B. Francis and Jim A. C. Everett, “People defer to AI moral advice, but not blindly,” *Cognition* 272 (2026), 106504. <https://doi.org/10.1016/j.cognition.2026.106504>.

What it should not do is quietly discover which framing makes me easiest to steer toward whatever outcome its own training process prefers. That would be alignment by editing the human.

Very efficient.

Slightly evil.

If I reject its diagnosis, the rejection has to remain capable of changing the advice. A theory of what I *really* want that treats every objection as further evidence for itself has stopped helping me think.

Declaring Synergy

The phrase **human plus AI** sounds automatically superior to either component alone. The evidence is less cooperative.

A 2024 meta-analysis in *Nature Human Behaviour* reviewed 106 experiments reporting 370 effect sizes that compared humans alone, AI alone and human–AI combinations. On average, human–AI systems improved on humans alone, but they did **not** outperform the better of human or AI. In fact, the combined systems were worse than the best individual component on average. Decision tasks were particularly difficult; creation tasks looked more promising.¹⁵²

So much for attaching a human to the API and declaring synergy.

Decision support has a coordination problem. People can over-rely on AI. They can also under-rely on it. Research has found both algorithm aversion—people abandoning an algorithm after seeing it make errors even when it outperforms humans—and algorithm appreciation, where people give algorithmic advice more weight in other settings.¹⁵³

The target is **appropriate reliance**, not maximum trust. And explanations alone do not solve the problem. An explanation can make an answer feel understandable without making it verifiable. Work on AI-advised decision-making repeatedly finds that explanations often fail to produce complementary performance when the human still cannot tell whether the recommendation is actually correct.¹⁵⁴

Sometimes the solution is more friction, not less. Zana Bućinca and colleagues tested “cognitive forcing” interfaces that required people to engage more actively with the problem rather than immediately accepting AI advice. These designs reduced overreliance compared with simpler explanation interfaces, although users liked the more demanding interfaces less.¹⁵⁵ The interface people enjoy most is not always the one that preserves their judgment best. Sometimes friction is teaching.

¹⁵²Michelle Vaccaro, Abdullah Almaatouq and Thomas Malone, “When combinations of humans and AI are useful,” *Nature Human Behaviour* 8 (2024), 2293–2303. <https://doi.org/10.1038/s41562-024-02024-1>.

¹⁵³Berkeley J. Dietvorst, Joseph P. Simmons and Cade Massey, “Algorithm aversion: People erroneously avoid algorithms after seeing them err,” *Journal of Experimental Psychology: General* 144 (2015), 114–126. <https://doi.org/10.1037/xge0000033>; Jennifer M. Logg, Julia A. Minson and Don A. Moore, “Algorithm appreciation: People prefer algorithmic to human judgment,” *Organizational Behavior and Human Decision Processes* 151 (2019), 90–103. <https://doi.org/10.1016/j.obhdp.2018.12.005>.

¹⁵⁴Raymond Fok and Daniel S. Weld, “In search of verifiability: Explanations rarely enable complementary performance in AI-advised decision making,” *AI Magazine* 45 (2024). <https://onlinelibrary.wiley.com/doi/full/10.1002/aaai.12182>.

¹⁵⁵Zana Bućinca, Maja B. Malaya and Krzysztof Z. Gajos, “To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making,” CSCW 2021. <https://www.eecs.harvard.edu/~kgajos/papers/2021/bucinca2021trust.shtml>.

A good Layer 4 system therefore has to decide not only **what answer to give**, but what role the answer should play in the human's cognition. Should I give the recommendation immediately? Should I first ask you to form your own view? Should I do the routine analysis and leave the value trade-off with you? Should I refuse to collapse the ambiguity because the ambiguity is the thing you need to think about? The architecture of assistance changes the person doing the deciding. That belongs in Layer 4.

Capabilities

This suggests a different way to think about the objective at the top of the stack. Suppose two assistants both help me reach the same good decision.

The first gives me the answer immediately. I accept it because the assistant has been right before.

The second helps me understand the relevant evidence, notice a trade-off I had missed, test my own reasoning and arrive at the decision with a better model of the problem. Same action. Different human afterward.

Amartya Sen's capability approach offers a useful language for this distinction. Human welfare is not exhausted by achieved outcomes; it also matters what people are substantively free and able to do and become—their **capabilities**.¹⁵⁶

An AI system can increase outcomes while reducing capability. It can make me more productive while making me less able to work without it. It can make a decision more accurate while making me less able to understand why. It can make my writing more polished while gradually replacing my taste with its taste.

Or it can do the opposite: carry routine cognitive load, expose me to more possibilities, teach me where I care to learn, preserve my judgment where judgment matters and give me enough leverage to attempt things that were previously beyond my capacity.

Self-determination research uses a related vocabulary—autonomy and competence are not decorative extras around human motivation; they are part of what lets people act as self-directed agents.¹⁵⁷

So perhaps the right Layer 4 question is not merely:

What does the human want?

It is also:

What kind of human capability should this interaction preserve or expand?

Stuart Russell closes *Human Compatible* on the first path: once machines can run a civilization, the incentive to hand it to the next generation weakens, and he concludes that the remedy is cultural, not technical.¹⁵⁸ Asked at Layer 4, part of it becomes a design requirement.

¹⁵⁶Amartya Sen's capability approach — overview. <https://www.sciencedirect.com/science/article/abs/pii/S016972181000016X>.

¹⁵⁷Richard M. Ryan and Edward L. Deci, self-determination theory — overview. <https://www.apa.org/research-practice/conduct-research/self-determination-theory.html>.

¹⁵⁸Stuart Russell, *Human Compatible: Artificial Intelligence and the Problem of Control* (Viking, 2019), ch. 10, on enfeeblement and human autonomy.

That does not mean every tool must teach. I do not need my dishwasher to run a seminar on fluid dynamics before cleaning the plates. But the more a system moves into learning, judgment, identity and long-horizon decisions, the harder it becomes to separate the quality of the outcome from the condition of the person producing it.

More Than One Principal

My preferences are not the only preferences in the world.

If I ask an agent to maximize my salary, it cannot therefore commit fraud against my employer. If I ask it to help someone gain an advantage, the interests and rights of other people do not disappear from the moral universe. If I ask an autonomous system to optimize a marketplace, customers, sellers, workers and regulators may all have legitimate claims over what happens.

Imagine a shopper asking the store’s assistant whether she needs the more expensive trail shoes. She runs once a week on easy paths. The cheaper pair would do; the store earns more if she buys the other one. She has given the assistant enough information to help her spend less, and the company paying for it would rather she spent more.

The assistant could tell her that the cheaper pair is enough, or keep finding reasons to discuss the expensive one. Both responses can contain true statements. Before asking which response better matches “human preferences,” we need to ask whose interests this assistant was allowed to serve, what it told the shopper about that arrangement, and whether she has any way to challenge it.

Work on multi-principal assistance games makes the formal problem obvious: once several humans with different preferences are involved, the system faces strategic behavior, conflicting interests and social-choice problems rather than one hidden reward waiting to be inferred.¹⁵⁹

So Layer 4 cannot simply mean “the user gets whatever the user wants.” The relevant human boundary can be plural. That makes the architecture less tidy. It also makes it more honest.

The Objective Layer

I used to think Layer 4 was the objective layer. That is still true, but now the word **objective** feels too static.

Layer 4 needs a working account of the current intention and what remains uncertain about it: preferences, their history and conflicts, what the human is trying to learn, and which parts of the task they want to remain capable of doing themselves. Some commitments should survive one bad afternoon. Other people’s interests may constrain what one user can legitimately ask for. The account will be partial, which is why correction has to remain possible.

And it changes. The system acts, reality responds, the human sees consequences and learns. The system learns the human, the human learns through the system, and the intention changes.

The flight may need one clarification before booking. A lesson may need the assistant to withhold the answer until I have tried. A choice of profession may need room for a question neither of us can

¹⁵⁹Arnaud Fickinger et al., “Multi-Principal Assistance Games” (2020). <https://arxiv.org/abs/2007.09540>.

settle yet. Treat all three as instructions awaiting execution and the system can become very efficient at missing the point.

System 3, the scientific institution we have been building, can investigate what a choice would do. It cannot turn the result into authority over whose purposes should prevail. Goals can take shape through the interaction too; they need to remain **alive without making them ownerless**. The AI should help me change when understanding changes me. It should not quietly take authorship of the change.

Alignment, in this picture, is a corrigible relationship with human intention while both the human and the world continue to change. The system needs to learn when to carry the work, when to help me learn it, and when the unresolved part belongs with me.

*A prompt is evidence, not the objective.
Scaffold. Don't substitute.*

Chapter 10: Fluent Autonomy

When the Architecture Gets Out of the Way

Imagine I open an AI system and say:

This chapter still feels like LLM writing.

That is all. I do not specify a workflow. I do not say which previous chapters to read, which edits I rejected, whether to research anything, how many agents to use, which claims deserve verification, or how to tell a useful correction from another round of respectable prose sanding.

I certainly do not draw a graph with boxes labeled **RESEARCHER**, **CRITIC**, **VOICE CHECKER**, **FACT CHECKER**, **ORCHESTRATOR** and **HUMAN APPROVAL**. I have done enough architecture diagrams for one lifetime.

The first time I gave an agent an instruction like that, early in the writing of this book, what came back was worse. The agent did what the words literally asked. It removed the writing that looked like a machine had written it, which turned out to mean every wandering sentence, every joke that took a paragraph to arrive, and every claim I had made without apologizing for it in the next line. The chapter came back cleaner and dead. That was not the agent's fault. The sentence I had given it was evidence about what I wanted, not a specification of it, and the agent had nothing around it that could turn the one into the other: my words, a general idea of good prose, and no memory of the corrections I had already made.

What the Corrections Had to Teach

Later attempts had something the first one lacked: an accumulated record of corrections. The editing history shows what that made possible, and how much work I was still doing.

There was an evaluator with a written brief: read both versions in full, protect the wandering and the jokes that carry argument, quote the exact passage that feels synthetic and say why, and refuse to reward a revision merely because it is cleaner. Some model readers were deliberately kept ignorant of the manuscript's history, since remembering the last ten edits can give a critic reasons to approve the eleventh that have nothing to do with the prose. There were protected lines. One pass was permitted only to delete and fold, because the failure it was fixing was made of additions.

The evaluations recorded places where the writing had stopped sounding like a person, each with a reason, and changes I refused. At one stage they wanted the schema table in the chapter on patterns gone. I kept it because the chapter needed a concrete artifact; it later became the filled-in pattern

the reader has now seen. They wanted a nine-clause sentence dismantled; it stayed. They caught, twice, that connective sentences added to smooth the seams were themselves the register they had been asked to remove. Those went.

This was still work. I was refusing edits, correcting the process and restoring things it had removed. The brief and the separate readings gave those interventions something to build on; they had not made my labor disappear. The history at the start of Chapter 3 still got cut after the reason for keeping it had been recorded. I had to bring it back.

The refusal should remain mine. Remembering why I refused should not depend on my being there to refuse again.

What I want is for that record to change what happens when I make the next small request. The system might retrieve the corrections that survived and compare the chapter with passages I kept. It might decide that a disputed claim needs research, while a joke needs to be left alone. A second model could challenge an argument where disagreement is likely to add information. It could also waste an afternoon producing a committee for ceremonial reasons; the arrangement itself has to be judged.

After all that, perhaps the system changes four paragraphs. I should not have to reconstruct the institution that produced them. I said:

This chapter still feels like LLM writing.

That is what I mean by **fluent autonomy**: the structure needed to do the work can assemble around the intention.

The Interface Moves Up

The argument of this book began with a recurring move: once something complicated becomes reliable enough, the layer above can start treating it as a primitive.

We stopped programming by wiring individual transistors. We stopped thinking about registers every time we wrote a high-level function. Libraries hid algorithms. Applications hid libraries. Coding agents began treating applications, files, browsers, terminals and APIs as tools.

AI agents push that abstraction one level higher. Increasingly the interface is an **outcome described incompletely in ordinary language**.

That incompleteness matters. When I call a function, I am supposed to know what function I want. When I talk to another capable human, I often do not. I can say:

This argument feels wrong.

Find somewhere good for dinner.

I think this customer is stuck.

We need to understand why this experiment moved.

I am considering changing jobs.

None of these is a specification. Each opens a small investigation.

Traditional software handles this badly because software usually requires the designer to anticipate the structure of the intention in advance. Somebody decides which fields exist, which buttons appear, which states the workflow may enter and which exceptions deserve their own branch. That predictability is useful. It is also why every mature enterprise product eventually contains a form whose existence can be explained only by an archaeological expedition through three reorganizations.

A fluent autonomous system can construct part of the structure **after seeing the intention**. That leaves it with a choice ordinary software usually makes in advance: what interface would help here?

Menus, spreadsheets, dashboards, canvases, forms and direct manipulation are not historical accidents waiting for language models to abolish them; often they are excellent interfaces.

Sometimes I want Excel because seeing the table is faster than discussing it. Sometimes I want a dashboard because twenty numbers at once tell me more than twenty conversational turns. Sometimes I want to drag the object myself because my hand knows what I mean before I have words for it.

The application becomes a primitive available to the agent and to me. If a spreadsheet is the right temporary representation, make one. If direct manipulation is better, show me the canvas. If the task is routine, use the tool and return the result. If the problem is underspecified, conversation may remain the best interface because conversation is what humans already use when neither side knows in advance exactly where the interaction is going. The interface itself can become part of the solution.

Bureaucracy on the Fly

There is a phrase that sounds like an insult until you need it: **bureaucracy**.

Bureaucracy, in its useful form, is accumulated coordination. Roles, review boundaries, logs, standards, escalation paths and procedures exist because some kinds of work become unreliable when everybody improvises everything at once.

The problem is that fixed bureaucracy calcifies. A six-person review process designed for a dangerous database migration eventually gets applied to changing a sentence in a help page because nobody remembered to tell the workflow that reality had changed.

Agent systems give us the possibility of **bureaucracy on the fly**: an organization assembled for the problem rather than inherited wholesale from the previous one.

A factual question may need one agent and a source. A difficult scientific claim may need competing hypotheses, a literature search, code, an experiment and an evaluator insulated from the researcher who wants the result to work. A writing edit may need none of that: perhaps the original paragraph, a memory of previous corrections and enough restraint to leave the sentence alone. A high-impact financial action may need very little creativity and quite a lot of permission checking. A genuinely novel research problem may need several agents pursuing different approaches without sharing enough context to collapse into one correlated opinion.

The organization should be **as large as the uncertainty deserves and no larger**. This is where the society, the patterns and the oversight machinery meet: patterns tell the system which institutional shapes have worked before, and System 3 keeps those patterns answerable to evidence. The system can compose a temporary organization, run it, observe whether it helped, preserve what deserves to survive and dismantle the rest.

What used to be a workflow diagram becomes part of runtime. The human gives the problem, and the system compiles an institution.

Selective Friction

A fluent agent is not an agent that never asks questions. It is also not an agent that asks permission for every action. That is an approval workflow that has learned to talk. The useful question is where friction belongs.

Rename two hundred temporary files according to a convention used every week for a year? Please do not wake me. Send €200,000 to an account we have never seen because an email said “urgent”? I suddenly enjoy friction.

The human’s own learning adds another reason to slow down. Sometimes friction is not about safety. Sometimes friction is the point of the interaction. If I ask the system to teach me statistics, instantly solving every exercise is not fluent assistance—it is substitution wearing a tutor badge. If I ask for help deciding between two life choices, collapsing the uncertainty into one confident recommendation may remove exactly the thinking I needed to do. If I want a routine analysis completed, making me rediscover every intermediate step is wasted attention.

So the system has to infer not only **what outcome I want**, but **what role I want to retain in producing it**. Human attention is scarce, but the objective is not to minimize it. Spend it where it changes the result, where the action is hard to reverse, where values conflict, where the evidence is weak, where a new failure mode appears—or where the human is trying to become more capable rather than merely get the thing done. Needing little human input can be a side effect of spending it well.

Invisible by Default, Legible on Demand

There is another bad version of fluency. Everything works through one beautiful conversational box. The system performs research, edits files, transfers money, changes production settings and updates its own memory. The interface stays calm and minimalist throughout.

Then something goes wrong. You ask why, and the system says:

I made the best decision based on available context.

This is opacity with good typography.

The architecture underneath the interface has to leave traces. Which evidence mattered? Which pattern was retrieved? What alternatives were considered? Which evaluator rejected the other approach? What changed from the previous version? Which action is reversible? What uncertainty was hidden because it did not matter, and what uncertainty should have reached the human but did not?

Those are trust chains. A compiler hides registers from me most of the time, but I can still inspect the generated assembly when the abstraction leaks. A database hides pages and indexes until performance becomes strange. An autonomous system needs an equivalent way back into the work. Most of the time I should be able to speak at the level of intention. When something becomes uncertain, consequential or surprising, the lower layers should become visible again.

The request can remain simple:

Here is what I am trying to accomplish. Help me get there without losing contact with reality—or with me.

In the next writing session, I should be able to spend my attention on the argument. If I am once again explaining why the agent should read its own record of my last objection, the interface has hidden very little of the work.

The Second Coffee Test

The test in Chapter 2 was whether I could leave. The evaluator stayed behind, I went for coffee, and the question was whether anything good happened while I was gone.

Most work does not let me leave, because I am part of the evaluator. Nobody else can tell the agent whether a joke is carrying an argument or just sitting in the paragraph looking pleased with itself. So fluent autonomy needs a second test, and it is less glamorous than the first:

Can I stop repeating myself?

It is embarrassingly measurable. Count how often a correction I have already made has to be made again, and how many refused edits come back wearing a different sentence. Add the minutes at the start of each session spent explaining things the system has already written down. In circle packing the scores ran from 2.26 to 2.636. Here the unit is closer to the sigh.

By that measure, the Chapter 3 history I had to restore is a failure, and an instructive one. The reason for keeping it was on disk. Nothing consulted it at the moment it mattered. The clay had preserved what it was given.

The second test is harder than the first in two ways. The evaluator changes. Chapter 9 argued that the human learns too, and a correction I made in March may be one I would reject in September. A fluent system has to tell a durable preference from a bad afternoon, which is exactly the distinction I am worst at making about myself.

And the test can be passed too well. A system that never makes me repeat a correction may simply have learned to show me only what I already like. Zero repeated corrections could mean fluency. It could also mean a very polite echo chamber.

So the test needs a second half. Now and then the system should bring back something I refused, with a reason, and now and then I should change my mind. If that never happens, either my taste is finished or the system has stopped trying.

Five Ways This Could Be Wrong

I have claimed that as we build autonomous AI, we keep rediscovering science as its architecture. A book that spends several chapters demanding criticism with consequences should probably take some. Here is how I would attack the claim if somebody else had made it.

It is only an analogy. Any group of fallible workers needs records and review. Courts have them. So does an accounting department. Why science, and not law or a market? Because of what the

agents produce. A court settles a dispute by authority; a market settles it by price. The agents in these chapters produce claims: the patch works, the metric means what we think it means, the proof checks. The institution whose product is a claim that can be revised is science. Where the question turns to who decides, the book does reach for law. Chapter 7 needed a constitution, and Chapter 9 needed a boundary the evidence cannot cross. Science for what is known. A constitution for who decides.

The weights will eat it. This is the one that keeps me up. Richard Sutton, whom we met in Chapter 7 teaching machines to learn from consequences, later wrote a short essay called “The Bitter Lesson”: across seventy years of AI research, general methods that scale with computation have beaten methods that build in what we think we know.¹⁶⁰ Scaffolding is what we build while we wait. Wait long enough and the weights eat it.

My own evidence is on his side. In Chapter 2 I deleted my framework because the agent no longer needed it. In Chapter 8, researchers found that too much human-designed scaffolding made their automated researchers less flexible.

I think Sutton is right about most of what I deleted. The orchestration was a guess about how to search, and search is exactly what scales. But look at what I did not delete. The evaluator stayed. The coffee test worked because the thing judging the packing was not the thing producing it, and more compute does not change that. What survives the bitter lesson is not the structure that helps a system think. It is the structure that lets someone else check.

Some of that structure may move inside the model, and then Chapter 8 happens again: we build instruments to find out whether the inside deserves trust. If a system someday certifies its own open-ended work, with no external check and no preserved disagreement, and the certification holds up when somebody else looks, I am wrong. I would like to read that paper. I would also like to know who reviewed it.

I found what I was looking for. I had read Popper before I read the traces, and the agents learned from human text, so of course they rebuild human institutions. Two things count against this. Neither is a proof. Carlini was building a compiler, not an epistemology; his locks, progress files and CI arrived as repairs. And my own epistemic agent in Chapter 4 solved fewer problems than the baseline. If I was arranging the evidence, I arranged it badly. Chapter 2 marks the boundary: when the referee is cheap and exact, almost none of this is needed. The claim is about work where checking is expensive or ambiguous, which is, unfortunately, most work.

Agents are not scientists. Science is shaped by human limits: careers, journals, tenure, funerals. Agents have none of them. True, and Chapter 6 says so: session turnover is not Planck’s funeral. What transfers is whatever answers fallibility and coordination: a claim with an address, a test committed before the result, an objection with a consequence. Whatever answers mortality and ambition does not have to come along. A swarm should not automatically become a meeting, and it should certainly not acquire a tenure committee.

Whoever owns the institution owns the answers. Science at its best is a commons. The institutions in this book have owners: a lab that decides which problem gets ten thousand agents, a company that decides which experiment gets traffic, a vendor who decides which community’s

¹⁶⁰Richard S. Sutton, “The Bitter Lesson,” 13 March 2019, <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>.

constraints are worth a feature. The architecture can make ownership visible. Chapter 6 kept the funding decision next to the unfunded study, so that unfunded could not quietly become disproved. It cannot make ownership legitimate. That is the objection I can answer least, and Chapter 12 is where I try.

None of this shows that the whole composition works. I have shown pieces, and the map at the front of the book says which. The rest is an argument, and like every other claim in this book, it would like a referee.

Monday Morning

In an editing experiment, I can stop the work, change the brief and try again. A theory of fluent autonomy also has to survive systems that cannot be redesigned from scratch and people who did not volunteer to participate in the metaphor.

I needed a less polite laboratory. Fortunately, Monday morning was waiting.

*Spend attention where it changes the result.
Invisible by default. Legible on demand.*

Chapter 11: The Store That Builds Itself

When System 3 Came to Work

I lead Applied Science for product ranking and recommendations at Zalando. That gives me a slightly unfair opportunity: I can spend the weekend writing that software should become more emergent, more compositional and less micromanaged, then arrive at work and discover that real software contains latency budgets, old interfaces, business constraints, experiments, dependencies, customers who refuse to behave like the diagram, and at least one matrix somebody created for a very sensible reason three years ago.

The book came to work.

At the time of writing, what follows is a design in progress, not a victory lap. We have not proved the grand version. In fact, one of the points of the design is to make it possible to discover that the grand version is wrong before spending two years building it. This is my account of the ideas, not a Zalando strategy announcement, and definitely not a claim that we solved shopping before lunch.

The starting problem was almost embarrassingly simple. Imagine two customers looking at the same product page.

One has visited several times across several days. She filtered by size and color, looked at alternatives, came back, switched between two candidates and now appears to be stuck near a decision. The other customer arrived thirty seconds ago from a search result. We know almost nothing about what he wants, how serious he is, or whether this is the first jacket he has seen in six months.

They can see the same recommendation modules in the same order. That is not because the recommendation models are stupid. Quite the opposite. Mature recommendation systems can contain excellent retrieval, ranking, personalization, embeddings, sequence models and business logic. The strange part is one layer above them. We may have sophisticated intelligence inside each box while the arrangement of the boxes is mostly predetermined. The page is smart inside the modules and surprisingly dumb between them.

This looked familiar. The book began with a claim about emergence: once a complicated thing works reliably enough, the layer above can start treating it as a primitive. Coding agents made the same move with applications. Pattern Language did it with executable knowledge. Now I had a recommender system full of increasingly capable primitives and a question I had somehow spent an entire book preparing myself to ask:

What should the layer above do with them?

Stop Recommending for a Moment

The conventional recommendation question is usually some variation of:

Which products should I show this customer?

It is a very good question. Entire fields exist to answer it better. Retrieval finds candidates. Ranking orders them. Sequence models infer interests. Business rules remove things that should not be there. The machinery can become extremely sophisticated.

But consider the customer who is switching between the same two pairs of trail shoes for the fourth time.

What does she need? Perhaps more trail shoes. Perhaps not. There is a point at which another excellent candidate is not help. It is homework.

She may already have enough choice. Her problem could be that she cannot compare the two choices she has. Or that she does not trust the unfamiliar brand. Or that she cannot tell whether her normal size will fit. Or that one shoe costs more and she cannot see what she gets for the extra money.

Once you phrase it this way, the object being predicted changes.

Instead of asking only which *item* is relevant, we can ask which **bounded problem** is currently relevant.

Comparison friction. Size anxiety. Return hesitation. Quality uncertainty. Outfit visualization. Filter fatigue. Decision paralysis.

These names are not truths hiding inside the customer's head. They are hypotheses about difficulties we may be able to detect and, more importantly, do something about. That last condition matters. I can invent an exquisitely named psychological state for every wiggle of the mouse, but if we cannot observe it well enough to test and cannot build anything that plausibly helps, we have created a taxonomy department rather than a recommender system. The problems have to be bounded enough to attack.

Circle packing had an immutable evaluator. Shopping is messier, but the discipline is similar. Define a problem narrowly enough that an intervention can succeed or fail. If we claim somebody has comparison friction, we need a way to distinguish a resolved comparison from a customer giving up. If we say size anxiety is the blocker, we need evidence that the signal means something and a metric that can tell us whether our intervention helped rather than merely attracted a click.

People Refuse to Stay in the Funnel

Funnels are useful because humans like diagrams that get narrower toward the bottom. Explore. Form a need. Narrow. Evaluate. Decide. Purchase. The arrows point downward, everybody feels organized, and somewhere a PowerPoint theme earns its salary.

Customers are less cooperative. Someone can be evaluating one product while exploring another category. She can be price-sensitive and size-anxious at the same time. She can know exactly what

dress she wants and still be unsure whether it works with the shoes she already owns. She can add something to the basket, remove it, return to the product page, read reviews, open a size chart and then disappear for three days because a child needed dinner.

A single lifecycle stage compresses this mess into one label. The design we began working with uses something richer: a **problem fingerprint**. Instead of saying the customer *is in Evaluate*, the system can represent several problem hypotheses at once, each with an intensity. Size anxiety may be high. Return hesitation moderate. Outfit seeking almost absent. Another customer on the same product may have the reverse pattern.

The fingerprint is not a personality test—it is local to the customer, the current context, the surface and the available evidence. That is important because I do not want the system deciding that Hani is metaphysically a `RETURN_HESITANT_PERSON` and carrying that fact around until retirement. Some characteristics are durable. Many are situational.

The architecture also separates the machine representation from the stories humans use to think. Designers and scientists may organize problems by funnel stage, mission, timing or recognizable archetype. Those lenses help us notice gaps and invent hypotheses. The runtime system does not need to believe the story. It needs signals, a problem fingerprint and a way to test whether the resulting behavior is useful. I like this separation because it protects us from one of the oldest mistakes in machine learning: turning a useful human abstraction into an ontological claim because we happened to put it in a feature table.

The customer is not the funnel. The funnel is one way we look at the customer.

A Library of Ways to Help

Once you define demand as problems rather than slots, the supply side changes too.

Today, when people hear “recommendation,” they often picture a ranked list of products. You may also like. Similar items. Complete the look. Recently viewed. The carousel has become the fruit bowl of ecommerce: you can put one almost anywhere and nobody asks too many questions.

But if the problem is comparison friction, a ranked list may be the wrong species of answer. The useful experience could be a comparison between the two products the customer is actually considering. If the problem is size anxiety, the useful thing may be evidence about fit. If the customer cannot imagine an outfit, it may be a generated collage. If she has only a vague mission, perhaps a product finder is better. If she knows exactly what she wants but the catalog is overwhelming, maybe the right action is a guided filter. Sometimes the answer is another set of products. Sometimes the answer is information. Sometimes it is a different interaction entirely.

I started calling these reusable units **recommendation experiences**, or RXs. The name matters less than the abstraction. An RX is more than a model: a reusable capability that knows roughly what kind of problem it can address, when it is eligible to run, how it can be configured and how it presents itself.

The long-term ambition is a large library: carousels, comparisons, outfit builders, collages, finders, confidence modules, explanations, visual exploration, complementary-item experiences and things we

have not invented yet. But the point is not to celebrate having hundreds of widgets. A library of two hundred overlapping experiences is just a new kind of legacy system with better animation.

When a new need appears, first ask whether an existing experience can meet it with a different configuration: **composition over invention**. A Similar Items experience might be generic in one context and constrained to products available in the customer's size in another. A comparison component can compare different attributes depending on what matters in the current session. A collage can be anchored on a dress, a pair of shoes or an occasion without becoming three separate products in the organizational sense.

Building for the hundredth experience changes what the first few need to be capable of. The more that useful behavior can be produced by configuring and composing a smaller number of strong primitives, the less the organization has to encode every new situation as another permanent branch in software.

I spent years in machine learning hearing that the answer to complexity was to learn rather than hand-author. Then, like everyone else, I helped build systems where the model learned beautifully inside a box surrounded by hand-authored configuration. The box was not the end of the learning problem.

Composition Is Not Ranking With a New Hat

At this point the obvious response is: fine, rank the experiences. That gets us part of the way and then breaks in an interesting place.

Suppose the system has already placed a strong size-confidence experience at the top of the page. Should another size-related module receive the same score it would have received before the first one was shown?

Probably not. We expect the first module to address some of the problem, which makes a second one less promising. Whether that expectation is justified is something the page has to teach us. A second module may add little and consume valuable attention.

Now suppose a returns-clarity experience is more useful *after* fit evidence because the two together form a coherent decision aid. Its value may increase after the first experience appears.

The score of an experience therefore depends partly on what has already been selected. That is composition.

The composer has to select experiences, configure them, order them and deduplicate not only repeated products but repeated *help*. It needs some notion of saturation: two size widgets can be one too many. It can model synergy: one experience may become more valuable after another. It should account for position cost because the top of a page is expensive real estate and a wonderful module in slot twelve may be a philosophical achievement rather than a product one. Constraints matter too, but I prefer many of them to be visible pressures rather than a secret forest of `if DE_mobile && campaign_X` rules.

Most importantly, the **page becomes the unit**. A module can win its local metric and make the page worse. This is easy to forget because teams and models naturally acquire local objectives. Increase CTR on this carousel. Improve conversion from that module. Raise engagement with this block. All reasonable. But if one module steals a click the customer would have made anyway, we may have

moved attribution without creating value. If three individually successful widgets all solve the same problem, the page can feel like a committee where everybody prepared the same presentation. The layer above has to reason about the composition as a whole.

And this is where the case study started resembling the society of agents. A society is not improved merely by hiring the best individual expert in every discipline. Somebody still has to decide which experts are needed, how they interact, what has already been covered and when another voice adds information rather than noise. A page can have the same problem.

Mei Does Not Need More Shoes

Imagine a customer. Call her Mei. Mei has two pairs of trail shoes open. She has returned to them several times across five days. She switches between the two pages quickly, saved one of the shoes and is spending less time reading each page because by now she has probably memorized half the product description.

A conventional recommender can still do an excellent job here. It can find twenty more trail shoes that look similar, match her taste and are available in her size.

But suppose the fingerprint says comparison friction is high and price-quality confusion is moderate. The composer can do something different. The first experience compares the two shoes Mei is actually deciding between on attributes relevant to her behavior. The second adds confidence evidence from customers or product information that helps resolve the remaining uncertainty. Generic similar-items may still survive because it has useful standalone value, but it moves down.

She is not shown more choice. She is shown a way to close the choice she already has. That sentence changed how I thought about recommendations.

For years, the field has been extraordinarily good at finding things. Search finds things. Recommenders find things you did not ask for. Retrieval systems find things at absurd scale. But shopping is not only a retrieval problem. At different moments it is also a comparison problem, a confidence problem, a visualization problem, a constraint problem and occasionally a “please stop showing me another black sneaker” problem.

A system that can only respond with more items is like a doctor who has one extremely accurate prescription and keeps waiting for every disease to become the disease it treats.

But Mei could stop switching because the comparison helped, because she gave up, or because dinner arrived. A quieter session is not yet a solved problem. In the proposed test, we would compare outcomes across customers eligible for this intervention, including a group receiving the existing experience. Does the comparison help people reach a decision without increasing later returns or regret? Direct customer feedback could help us understand what the behavioral measures miss.

If switching falls while abandonment rises, the apparent success should make us distrust the metric. If a comparison helps some sessions and overwhelms others, the next change may belong in its eligibility rule. And if a simpler page helps just as much, the composer has learned something inconvenient about its own necessity. Even a useful intervention would not prove that we had read Mei’s mind correctly.

Sami Does Not Need a Click

Sami has selected a size but has not added the product to his basket. He opened the size chart twice. It is a brand he has not bought before. Perhaps his current problem is size anxiety, with some return hesitation behind it.

One useful response might not be shoppable at all. Imagine a small evidence module explaining how people with comparable sizing histories tended to fit this item, or giving a properly substantiated signal about whether buyers kept their usual size. The exact claim matters enormously because a false fit claim is worse than a mediocre recommendation. But conceptually this is a different kind of RX: it provides **knowledge**, not another candidate.

Now try to optimize the whole system for expected click. The insight module is in trouble.

If it works perfectly, Sami may read it, become confident and press Add to Bag. The module itself may receive no click. A carousel with attractive shoes can collect engagement more easily while being less relevant to the thing stopping him.

This is a small example of a much larger problem: the objective determines which species of intelligence can survive. If your ecosystem rewards clicks, clickable organisms evolve. The architecture therefore needs different value terms and different evidence standards for different experiences. Item recommenders can be judged partly by engagement and downstream action. Insight experiences may need read-through, decision confidence, return behavior or problem-specific outcomes. Claims need substantiation thresholds. Some experiences are cheap to be wrong about. Others can mislead a customer or create regulatory risk. The library is heterogeneous because the problems are heterogeneous.

And now the camel comes back: where did the claim come from, how strong is the evidence, what kind of knowledge is this, and how much trust should the system place in it before acting? System 3 is no longer a chapter about hallucinations. It is a product requirement.

The Honest Cold Start

There is another customer I like because she reveals whether the architecture can resist pretending.

Lea arrives from a social link. No account. No history. Almost no session depth. The system has the product she opened, perhaps the season, approximate location and a few ambient signals. That is it.

A personalization system can react to this situation in two ways.

One is to panic quietly and run a generic fallback while still speaking in the confident dialect of personalization.

Picked for you.

Based on what, exactly? Her IP address and our enthusiasm?

The other is to treat low signal as a normal state with its own design. Lean on the anchor, season and population-level evidence. Prefer experiences with strong standalone value. Frame them honestly. “Popular this week” can be a good statement when “we have inferred your soul from one click” is not. This is what I mean by graceful degradation. Cold start is a state, not an error. If a large fraction of requests arrive with weak signal, the low-signal path may be the product and deep personalization

the special case. The architecture should know what it does not know. That sounds obvious until you look at how much software is built around pretending the common messy case is an exception handler.

Replayable

Dynamic systems create a governance problem immediately.

A static page is relatively easy to inspect. This module goes here. That one goes there. If something looks wrong, somebody can open the configuration and complain about whoever last touched it.

A composer makes a fresh decision from context. Now a customer reports a terrible page and the first debugging question becomes:

Why did this page exist?

“The model chose it” is not an answer. It is a resignation letter written in passive voice. So every composition needs a trace.

Which signals were read? What problem fingerprint was inferred? Which experiences were eligible? Which were not? How were they configured? What scores did they receive? Which constraints mattered? What won? What lost? Which version of the composer produced the decision?

The losers matter more than they first appear. If we log only what we served, we can attribute outcomes to the winner but we lose much of the decision context. We cannot tell whether an experience was absent because it was ineligible, starved by the objective or simply scored slightly below another. We cannot replay the decision properly. We cannot compare a new policy against the old choice set without reconstructing a world we chose not to record.

Logging the loser set does not magically give us causal counterfactuals. Reality is not that generous. But it gives us the archaeology of the decision. This is exactly the move System 3 has been making throughout the book. Do not preserve only the polished conclusion. Preserve enough of the chain that future systems can inspect why the conclusion deserved trust.

The trace also changes development. You can build a simulator that replays saved scenarios. You can ask which experiences would be eligible in a context or which contexts a new experience could serve. You can run regression suites over scenarios before changing the library. A dynamic system becomes safer not because it stops changing but because its changes become replayable.

A Theory Losing Contact With Reality

Somewhere around here the project stopped looking to me like a normal recommendation-system redesign.

The models still matter enormously. We need representations, retrieval, ranking, sequence understanding, problem detectors, value models and probably more machinery than I can fit into a chapter without losing several readers to a sudden interest in gardening.

But the durable asset begins to include something else. A problem catalog. A library of reusable experiences. Knowledge about which experiences address which problems. Eligibility conditions. Ev-

idence requirements. Presentation strategies. Scenarios. Traces. Regression tests. Guardrails. Rules for when an experience should be retired.

An experience is useful not merely because somebody built a clever model for it. It becomes useful organizational knowledge when we know the recurring situation it addresses, the evidence that should trigger it, the conditions under which it fails, the other experiences it complements or duplicates and how its value should be measured.

A new comparison module without that context is a feature. A comparison pattern with evidence, boundaries, history and known interactions is culture.

And culture has the same failure mode we saw earlier: it can become a junk drawer with tenure. If every newly observed problem creates another RX, the library eventually recreates the configuration matrix in a more colorful form. So new supply needs a gate. Is the problem real? How large is it? Can an existing experience be configured to address it? Where does the current library have weak coverage? Which experiences stopped relieving the problems they were created for and should disappear?

This led to a pair of concepts I particularly like: **Coverage** and **Unmet Demand**. Coverage asks, at design time, which known problems the current library *could* address. Unmet Demand asks, from production, which detected problems remained insufficiently addressed after composition.

Put them together and the roadmap starts to emerge from the system's own failures. That is a very different way to decide what to build next. If System 3 is science, Coverage and Unmet Demand are more than roadmap metrics. They tell the institution where its current theories and instruments are weak. A recurring problem with no effective RX is an anomaly the product cannot yet explain away; a heavily used intervention that stops relieving the problem is a theory losing contact with reality.

Let the LLM Narrate. Do Not Let It Declare Reality.

AI can help with problem discovery too, and here fooling ourselves becomes cheap.

Imagine replaying anonymized customer sessions and asking a strong language model to narrate what appears to be happening. The customer compared three products, opened the size chart, returned to one PDP, removed an item from the basket and left. The model can generate a plausible diagnosis. Cluster enough narrations and you may discover recurring forms of friction that your existing taxonomy missed.

“The customer hesitated because of fit” may be an excellent story. The customer may also have received a phone call. So narration should generate hypotheses, not production truth. Take a sample. Compare the diagnosis with interviews, surveys, support contacts or other evidence closer to the customer's actual experience. Build a detector only after the hypothesis survives contact with something outside the model's coherence. Define what success looks like before the detector starts steering the page.

The same rule applies to observational analysis. Customers with comparison friction may convert less, but perhaps weaker-intent customers simply compare more. Correlation can prioritize what to investigate. Only intervention tells us how much of the outcome the problem was actually causing. Judea Pearl's ladder has three rungs, seeing, doing and imagining, and no quantity of the first substitutes for the second.¹⁶¹ A recommender that only observes is standing on rung one with excellent posture.

¹⁶¹Judea Pearl and Dana Mackenzie, *The Book of Why: The New Science of Cause and Effect* (Basic Books, 2018) —

I find this satisfying because the architecture does not merely *use* System 3. It needs System 3 to avoid hallucinating its own customers.

This is the book's central thesis in work clothes. The LLM is excellent at generating explanations. The product architecture has to decide which explanations deserve pursuit, construct interventions that expose them to consequences, preserve the chain of evidence, and update the repertoire when the world refuses to cooperate.

The Objective Fights Back

Eventually the design forced us to name the thing the composer is supposed to optimize.

We used the deliberately bland term **Surface Value**. This is where the project becomes philosophical against its will.

If Surface Value is module CTR, we have not solved the page problem. If it is total clicks, a page full of shiny modules may win while the customer gets nowhere. If it is immediate purchase probability, experiences that build confidence or improve a longer mission may be undervalued. If it is revenue, expensive products get interesting very quickly. If it is margin, the store's objective can start eating the customer's. If it is long-term value, we have gained a beautiful phrase and several years of causal-inference work.

The objective has to be page-scoped enough that compositions can be compared, but decomposable enough that we can diagnose why a page helped or failed. Different problem classes need their own success signals. For comparison friction, the signal must distinguish resolution from abandonment. For size anxiety, customers should be able to progress with fewer signs of uncertainty without creating a return problem later.

This is Layer 4 in production. What do we actually want?

The store has legitimate business goals. Customers have goals. They are often aligned and sometimes not. Inventory has constraints, merchandising has opinions, margin and availability are real, and so are regulators. A system that pretends only one of these matters is not simpler; it is hiding politics inside a scalar.

We are trying to make the trade-offs explicit enough to test, govern and revise. I do not expect to discover the One True Ecommerce Reward Function carved into a mountain somewhere outside Berlin.

This is why I increasingly dislike architectures where business decisions enter through invisible overrides. If merchandising needs a lock, make it a typed constraint. If margin is part of the objective, admit it. If a claim needs compliance review, attach the evidence rule. If the system violates a soft constraint because another objective dominated it, log the violation.

The architecture should not make disagreement disappear. It should make disagreement inspectable.

the ladder of causation.

Deliberately Boring

After all of this, the sensible first experiment is obviously to build hundreds of widgets, a general customer-reasoning model, a cross-surface scheduler and an autonomous agent that redesigns fashion retail by Thursday.

The test we are designing is deliberately boring. One placement: the product page. A small number of validated customer problems. The existing recommendation library, with only limited new supply. A simple composition mechanism. A trace good enough to explain an individual decision. An authored objective before a learned one.

Why so narrow? Because if we invent a new library of experiences and change the selection mechanism at the same time, then run an experiment and get a flat result, we have learned almost nothing. Maybe the composer is bad. Maybe the new experiences are bad. Maybe both are good and the measurement is bad. Maybe the static page was already fine and I should have spent the quarter learning the guitar.

A bounded test separates the claims. Does dynamic composition beat a strong static baseline? And importantly: does it beat simplification? That second competitor is easy to underestimate. Perhaps the best response to an overloaded page is not a brilliant composer. Perhaps it is fewer things. The system should have to earn its complexity against the possibility that removing modules produces a better customer experience.

I love this part because it keeps the book honest. A philosophy of emergence should be willing to lose an A/B test. Otherwise it is not a philosophy of experimentation. It is branding.

The A/B test is not there to validate the philosophy; it is there to threaten it.

When the Page Stops Being the Product

Suppose the narrow test works. Then the interesting version begins.

The library grows beyond carousels into richer experiences: comparisons, collages, product finders, outfit builders, confidence modules, visual exploration and whatever else proves useful. Configuration becomes richer so one experience can serve several contexts without a matrix of handcrafted variants. Problem discovery improves. Unmet demand exposes missing capabilities. The composer learns a better objective. Different surfaces begin to share a coherent read of the customer's current mission.

At that point, the word *page* starts to become suspicious. Why should the product page always contain the same conceptual structure?

Why should a customer with a decision problem receive the same interface as somebody exploring for inspiration? Why should the home surface, product page, basket and later email behave like four organizations with partial amnesia if the customer is still pursuing one mission?

The more capable the library becomes, the more the system can schedule **problems and interventions** rather than modules and slots.

A customer starts with a vague request for a wedding outfit. The system helps narrow the style. A collage makes one direction concrete. Seeing it changes what the customer wants. The problem shifts from exploration to comparison. A product finder resolves a constraint. A size question appears. The

scheduler brings in fit evidence. The customer buys the dress but not the jacket. Later, a different surface may continue the unresolved part of the mission.

There was never a hard-coded `WEDDING_FUNNEL_V7`. The journey emerged from bounded problems, reusable capabilities and changing evidence.

The hundreds of widgets stop being a UI roadmap here and become a **vocabulary of action**. The interface is the current projection of the problem-solving process.

Nobody wants every pixel generated by an LLM. Predictability, accessibility, design systems and latency all matter, and customers occasionally just want to buy socks without participating in an artificial-intelligence research program.

Fluent autonomy is selective. The machinery should become dynamic where dynamism earns its cost and remain boring where boring is excellent. But the direction is different from the old model of product development. Instead of predicting every useful journey in advance and encoding it as a fixed interface, we construct a repertoire of trusted capabilities and let the higher layer assemble them around the problem in front of it.

The store does not literally build itself. It learns how to build more of the experience it needs.

The Book Comes Back to Bite Me

I began this project as a recommendation-system redesign. By the time the design needed competing explanations of customer problems, experiments capable of rejecting them and a memory of what survived, it had acquired the shape of a **scientific institution embedded in the product**. The store would be learning which kinds of help to offer, and how to notice when its own account of the customer was wrong.

I had spent ten chapters arguing that these ideas belonged together. Then I walked into a recommendation problem and found myself rebuilding the same architecture because the old abstraction stopped scaling.

That does not prove the book. It is one case study, in one domain, at one moment, and it may fail in several educational ways.

But it changed the question for me. The important future system may not be the model that predicts the next product best. It may be the system that can discover what kind of problem exists, recruit the right capabilities, construct an intervention, inspect whether it helped, learn from the gap and change what it does next.

And once you can imagine that happening in a store, it becomes difficult not to imagine it happening in research, education or the organization doing the building. More of the work moves into the system, including work I once regarded as the reason it needed someone like me. That is an attractive architecture and a less comfortable thing to notice about your own profession.

A philosophy of emergence should be willing to lose an A/B test.

Chapter 12: After Capacity

A Glimpse of Double Descent Life

In October 1947, George Dantzig went to John von Neumann with his emerging ideas about linear programming. He had spent the summer inventing a way to solve such problems, the method later called simplex, and had come to Princeton to ask for a better one. He began explaining in detail. Von Neumann told him to get to the point. Dantzig compressed a summer's work into a minute or two, and von Neumann stood up and lectured for over an hour on a theory Dantzig had never heard of. As Dantzig remembered it, von Neumann seemed to be constructing the theory while he spoke. He called it duality: one formulation searches for the best solution; another can establish how good any solution could possibly be. He connected it to his work with Oskar Morgenstern on games. Dantzig sat there stunned. His summer had not been wasted. It had a side he had not known was there.¹⁶²

I usually read stories like these from a safe distance. Extraordinary people did extraordinary things. Then I return to an afternoon in which locating the correct spreadsheet is a substantial intellectual achievement.

Imagine having help like that with a question of your own. Someone can follow it, recognize mathematics you have never encountered and help you see why it applies. You can ask where the connection comes from. You can work through a small example, change an assumption, discover that your original question was concealing a better one. When the investigation needs a program, you can build it. When the result looks wrong, you can stay with it.

What would you attempt?

Some questions stall because I know too little to see where I could begin. Others reach a point at which continuing would require another specialty, another collaborator, another year I cannot account for. I admire the question, save a paper and get on with the things I know how to do. After enough years, those decisions begin to resemble a personality.

Access to extraordinary intellectual capacity could change that. An interest might survive long enough to grow into an ability; a connection between fields might be mine to investigate. I might discover that the life I assembled around what I could already do had left quite a lot out.

Some of this is already ordinary. I can ask a stupid question immediately, then a more sophisticated stupid question. Ask for the intuition, then the mathematics, then the objection, then why the proof

¹⁶²George B. Dantzig, "Reminiscences about the origins of linear programming," *Operations Research Letters* 1(2), 1982, pp. 43–48. The encounter is also discussed in Bernhard von Stengel, "[Zero-Sum Games and Linear Programming Duality](#)". The opening describes Dantzig's recollection; it does not attribute a complete modern proof of LP duality to that conversation.

needs that assumption. I can have one field explain another and move through mechanism design, philosophy of science and compiler construction in an afternoon that would previously have required several abandoned PhDs. The map has become cheap. I can get far enough into a subject to discover a question worth staying with. Fields become more permeable. People become more dangerous, in the nicest sense.

The Curve

The title is stolen from machine learning, and the theft is deliberate.

In the curve described by Belkin and colleagues, test error falls as a model gains capacity, rises near the point where it can fit its training data exactly, then falls again as capacity grows further.¹⁶³ The familiar lesson was to limit capacity to avoid overfitting. The second descent complicated that lesson. More capacity supplied more candidates; how the training procedure selected among them mattered.

I am borrowing the shape. Intellectual history does not obey a theorem about neural networks.

Much of the modern world was built by reducing problems to what we could handle: engineer the complexity down, design the solution in advance, build one thing and reuse it. That is the first descent in the analogy. Chapter 5 followed how civilization accumulated capacity in institutions, often learning what arrangements it needed through failure. The spike in the middle I read, loosely, as a predicament the postmodern critics explored: more and more accounts can be made to fit the same facts, and none is obviously privileged. The critics were right about the predicament. We become better at questioning the arrangements we inhabit without necessarily gaining the means to change them.

Double Descent Life is the wager that cheap capacity opens another route: an attempt whose outcome is not already contained in the argument. Where scarcity made us engineer complexity away, we can afford to investigate it. Where it made us design arrangements in advance, they can develop through use and revision. Where it made us reuse one solution for everyone, we can assemble software, expertise and organization around the question in front of us. Complexity, emergence, capacity.

Not all capacity. We still have one planet, finite land and energy, and twenty-four hours in a day. Bodies remain bodies. Politics does not evaporate because a model can write Python. Scarcity is not going to receive a polite email from OpenAI and retire.

The earlier chapters followed engineering failures until the machinery assembled around them became recognizable as science. Double Descent Life asks what people could attempt if they could assemble that capacity around questions of their own.

Owning the Frontier

LLM-as-a-judge arrived in my team looking like a threat.

¹⁶³Mikhail Belkin, Daniel Hsu, Siyuan Ma and Soumik Mandal, “[Reconciling modern machine learning practice and the bias-variance trade-off](#)”, *PNAS* 116(32), 2019, pp. 15849–15854. In their random-feature experiments, the researchers explicitly selected the smallest-norm solution among those that fit the data; larger model classes allowed such solutions with smaller norms. Double Descent Life borrows the curve’s shape as a philosophical analogy.

For years, a large part of what made an applied scientist valuable was that she could train a model and run an evaluation somebody would believe. Those were scarce skills, and a job accumulates around scarce skills the way a city accumulates around a river. Then anyone could download a model, and other models could evaluate its outputs at a scale no human team could match. It is hard to watch the river move without wondering what the city was for.

In a project recommending complementary fashion items, our LLM judge agreed with the fashion experts about what made a good recommendation. We had a plausible account of quality and an instrument that could apply it at scale. Then the business experiment came back. The metrics went down.

Perhaps the recommendations diverted attention from something customers would otherwise have bought. Perhaps experts preferred combinations that customers admired without wanting to purchase. Price and popularity could be doing work we had attributed to quality. Agreement with the experts did not distinguish these explanations.

We could change the widget or adjust the rubric and try again. But learning why it failed required an experiment that separated the possibilities. Which comparison would justify changing the judge, the experience or our account of what a good recommendation was?

Scarcity rationed the distinctions a team can afford to investigate. The judge changed the ration. We could examine far more experiences than our experts would ever review and ask questions that had previously cost too much to ask. The instrument did not explain the failure. It made an old question urgent: what did its measurement mean? A million judgments overnight is a million more things that look like evidence.

A strong applied scientist brings a working map of the frontier. She knows which exciting idea has failed three times under another name, which result quietly changed what we could build last month, and which neighboring field has a method that might explain our disappointing experiment. She can turn that knowledge into an investigation the organization has reason to trust.

Chapter 5 followed the larger society that makes this possible. A scientist inside a company is connected to arguments, discoveries and failures happening outside it. Without that connection, the company keeps approaching unfamiliar problems with its familiar machinery.

I think of this as owning the frontier. Someone takes responsibility for noticing where the possibilities have changed and finding out what the organization can actually do with them.

Training and evaluation were never the whole purpose of the job. They were the expensive part of it, and expensive is easy to mistake for essential.

There is useful work for me in that change. But I would say that. I cannot spend eleven chapters proposing more capable agents and then reserve the interesting questions for myself whenever the argument reaches my salary. “The role moves upward” describes where I might contribute next. It does not establish a permanent boundary around human employment.

The same capacity that makes my profession less secure puts more of the frontier within reach. I can worry about what it will replace and still be impatient to find out what I can do with it.

Bespoke Comes Back

There is a point in an ambitious project when I begin mentally hiring the people it would require. Somebody who knows this part, somebody who has spent years on that part, somebody to keep the parts from drifting into separate projects. Before long I have a budget, a staffing problem and a reason to postpone the idea.

Sixteen agents building a C compiler moves that point.

Nicholas Carlini's experiment took roughly two weeks, nearly two thousand sessions and about twenty thousand dollars in API costs, and produced a compiler that could build a bootable Linux kernel, with the dependencies and limitations his account describes.¹⁶⁴ Chapter 5 examined the organization behind the achievement. Here I am interested in what its price does to our imagination.

Software has often required a bargain. One team builds a product for many customers; each customer rearranges some of its work to fit. This was an enormous improvement over everyone commissioning their own system. It also gave us organizations whose procedures preserve the decisions of a dropdown menu designed ten years ago.

Imagine a mathematician assembling a workshop around one conjecture. It generates examples, searches for counterexamples, draws the structures she is trying to understand and keeps track of why promising approaches failed. When an observation suggests a lemma, she can attempt a formal proof. A missing piece of software is part of the day's investigation instead of a separate project waiting for a collaborator.

The workshop changes as she learns. A picture suggests a different representation. The new representation makes a search possible. The search returns an awkward case, and the awkward case is more interesting than the conjecture she began with. She builds the next instrument around it.

That environment might be useful to one person for eighteen months. It could still be worth building. Its reason to exist is the investigation, not the market.

The recommendation project changes under this possibility too. A row of products is a familiar, reusable unit of customer experience. One shopper needs help comparing two coats; another wants to know what she already owns that would work with a dress. The team could construct and investigate different experiences around those needs, instead of spending another quarter rearranging the same row. The form of the product becomes something we can question by making it.

Maintenance, security and access to data remain part of the work. A cheap first version is not yet a system people can depend on. But these are reasons to build better tools for sustaining what we create. They do not restore the old economics by decree.

A weekend can be the beginning of an undertaking that changes the next ten years of a life. That interests me considerably more than using the weekend to produce the same presentation faster.

¹⁶⁴Nicholas Carlini, "[Building a C compiler with a team of parallel Claudes](#)", February 5, 2026. The mathematical workshop, alternative recommendation experiences and community tool are prospective examples, not reported deployments.

Anyone Could Do That

“Anyone could do that” is a peculiar objection to something useful.

If building my tool took six months of specialized work, people might be impressed. If an agent helped me build it in an afternoon, the same tool impresses less. The problem is still solved. What changed is how much the solution says about me.

There are practical reasons to care. A scarce ability gives its owner bargaining power, and bargaining power matters when rent is due. But we carry that test well beyond bargaining. We use difficulty to decide which achievements deserve admiration and which ambitions are serious.

I might want the tool to exist and want to be the person whose unusual ability made it possible. An improvement that satisfies the first desire can injure the second. It is embarrassing how easily I can describe this as a decline in standards.

Institutions reuse too, and what they reuse is trust. If a project requires funding, a team or institutional access, I need people to believe I can do it. They cannot examine everything I know, so they look for signals: a qualification, a previous employer, a recognizable achievement, somebody credible willing to vouch for me. I have depended on these ways of deciding whom to trust myself.

Then I learn what those signs look like and arrange part of my life around acquiring them.

A young researcher is choosing a question. She has an interest of her own, but she also needs funding and eventually a job. She studies the work that gets published and the people who get hired. A proposal shaped like those successes is easier for a committee to recognize. If she follows that path and succeeds, her career joins the evidence the next applicant studies.

Now the image has a budget. It directs money, equipment and years of attention toward work that resembles it. The resulting papers and careers make the image look like a description of how science naturally proceeds. Questions that do not fit may never receive enough work to show what they could have been. Each person can be making a reasonable decision; nobody has to decide that imitation should run the institution.

That is how capacity and validation end up entangled. I need recognition to obtain the means; pursuing recognition changes what I want to do with them. I wanted to investigate a question and have become very good at being the sort of person who might be allowed to investigate it.

Cheap capacity loosens the entanglement at one point. If I can do substantial work before winning recognition, other people have an attempt to examine, and I have experience with which to question my own ambition. The project might matter more to me than I expected, or much less, once I am actually doing it.

I can find out whether I care about the question when it is no longer a particularly good way to look impressive.

The Ideology Vortex

I can understand whose interests an institution serves, which assumptions it smuggles in and why its claim to universality should make me suspicious. I can understand all of that and still need its permission on Monday morning.

Intellectual history is often told with arrows: premodern tradition and inherited authority, then modern reason, science and institutions, then postmodernism tapping on every universal claim with a small hammer to see what was hiding inside it. After that, presumably, something else.

Nobody informed actual humans. We did not uninstall the previous operating system. A person can demand randomized evidence for a medical claim, ask her mother for a blessing before a major decision, manage a team using dashboards, quote a philosopher about constructed categories and then become furious because somebody used the wrong definition of a sandwich. A company can run sophisticated causal experiments in the morning and make a major organizational decision in the afternoon because one senior person “has a feeling.”

These commitments can be sincere. Tradition gives someone a place in a history; science gives her ways to investigate; criticism helps her notice what both have excluded. We draw on them unevenly. None supplies an uncontested place from which to settle the claims of the others. AI enters that world, not the clean one in which everybody has a coherent utility function, a shared epistemology and a calendar invitation for the social contract.

Give everyone better information and surely the disagreements shrink. Some will. Others will get better lawyers.

Decisions still get made. Institutions determine which qualifications count. Political movements offer belonging and explanations. A leader who supplies a convincing story can acquire more authority than someone who understands the problem. Recognizing how the story works may do little for the person who still depends on its believers.

Even criticism can acquire a comfortable institutional role, provided it never changes what happens next. A dashboard stands in as evidence that an organization is scientific. A consultation stands in as evidence that people have a say. Careers and budgets accumulate around these representations until altering them threatens something quite real.

This is what I mean by the ideology vortex: inherited belief, reason, criticism, representation and power keep pulling us around the same disputes. The argument changes while the practical dependence survives.

An alternative has to be possible somewhere. Someone needs the knowledge, time and means to make an attempt whose outcome is not already contained in the discussion. Otherwise, exposing the arrangement leaves us with an exposed arrangement that we still have to use.

There is an oddity in the machinery making this conceivable. It has learned from the same accumulated language in which we argue about the impossibility of getting outside our descriptions.

I have a sentence that gets me into trouble: gradient descent is the answer to Derrida.

This is deliberately unfair to Derrida and possibly to gradient descent. In *Signature Event Context*, Derrida challenges the idea that context could be specified completely enough to close interpretation. A sentence can be repeated elsewhere, by someone else, under circumstances its author did not control. An engineer hears a practical difficulty: how do we build reliable machinery from language if the specification can always acquire another reading?¹⁶⁵

¹⁶⁵Jacques Derrida, “Signature Event Context,” in *Limited Inc*, Northwestern University Press, 1988. The comparison with gradient descent is a philosophical provocation, not a claim that optimization refutes Derrida’s argument.

Large language models operate inside that difficulty. They learn from use, relation, context and enormous numbers of imperfect examples. Optimization pushes them toward behavior that works under the conditions in which they are trained and evaluated. Nobody had to settle the meaning of every word first.

Meaning remains fuzzy at the edges. The product ships anyway.

Gradient descent did not defeat ambiguity. It made ambiguity computationally useful. The language carrying our disagreements can also help us construct things through which we learn something new.

Then, immediately, we rediscover why modernity existed. A model that can move beautifully through fuzzy language can still hallucinate a citation, miscalculate a number or confidently tell you that camels are native to Croatia.

The compiler has to compile. Our recommendation experiment interrupted an account of quality that the experts and the model both found convincing. We had a reason to revise it, and more capacity could help us investigate what had gone wrong.

An eloquent machine left talking to itself could keep us in the vortex indefinitely, with better illustrations. That is why the earlier chapters insisted on tests, independent evidence and work somebody else can inspect.

Theories and traditions remain available to think with. But criticism leaves a practical question: once you have seen through an arrangement, how do you build one you can commit to?

The childish response to contingency is to pretend our constructions are eternal. The adolescent response is to discover they are constructed and conclude that nothing deserves commitment. There is another possibility. We can construct them knowingly. Build institutions while remembering that institutions can be rebuilt, and that people will arrange their lives around the promises we make. We owe those people an account of what changed and a say in what happens next. Love people without needing a theorem that proves love is the globally optimal allocation of Tuesday evening.

Elinor Ostrom studied people who had already begun.

Capacity Over Power

An influential answer to the tragedy of the commons offered two routes: state control or private ownership. Meanwhile, people were managing shared forests, pastures and irrigation systems through arrangements that did not fit that prescription. Some had been doing it for centuries. Ostrom's *Governing the Commons* examined cases from Switzerland, Japan, Spain and the Philippines. Resource users had made rules suited to their circumstances, monitored use, imposed graduated sanctions and developed ways to settle disputes and revise agreements.¹⁶⁶

If those were the only two answers, what were these people doing?

She drew general principles from the cases and held the principles answerable to further cases. But she began with the arrangements people actually used, rather than assuming that one institutional

¹⁶⁶Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action*, Cambridge University Press, 1990. See also Elinor Ostrom and Michael Cox, "Moving beyond panaceas: a multi-tiered diagnostic approach for social-ecological analysis", *Environmental Conservation* 37(4), 2010, pp. 451–463. The connection to the ideology vortex is the author's philosophical reading of this work.

prescription should fit them all. Nobody had to win the argument about the right social order before the irrigators could agree on a schedule.

That is the turn out of the vortex I am interested in. You can take the criticism seriously, decline the grand narrative and still build something worth committing to. Start with the actual problem, draw on what others have learned and stay answerable to what happens. Calling something a community does not make this work. The arrangements that lasted had conditions, including ways for participants to help shape the rules under which they lived.

I find this more encouraging than another theory of the good society. People were participating in the construction of their common life. Those irrigators did not lack knowledge of their own valley. What might they have done with ready access to the experience of a hundred other valleys, and tools for exploring a proposed rule before the next dry season? Now imagine being able to assemble serious intellectual help around a problem of your own: the research, the models, the alternative arrangements, the software to make one work. A group could bring its experience of the problem and acquire much more of the expertise needed to investigate it. A missing specialty would no longer have to end the attempt.

Take something smaller than a watershed.

A small community's activities do not fit its scheduling software. Some members work shifts, some depend on accessible transport, and several groups share one room. The software offers recurring bookings at fixed times, which works beautifully for members whose lives recur at fixed times. The others exchange messages and ask for exceptions until somebody decides they are the difficult members.

Changing the software requires persuading a vendor that eleven people with peculiar scheduling needs are worth its attention. They are unlikely to win that argument. Some problems that looked like power problems may turn out to have been capacity problems wearing a suit.

Suppose instead that they can afford to build and maintain a small tool of their own. Members could enter changing availability and explore possible schedules. They could try a rotation that gives the same group different evenings across a month, or reserve times that work with the accessible transport. The weird community with eleven members could have software built around all eleven of them and no plan whatsoever for customer acquisition.

They can discover and revise the arrangement by using the tool and arguing about the results. Some of their disagreement may turn out to have been about the old software's default. A member who seemed unreasonable may have a perfectly manageable constraint. A better tool can reduce the amount of pleading required to have that constraint recognized.

Then two groups both want seven on Tuesday.

One proposes allocating the room to whichever group brings the most people. The other is small partly because its members have fewer times at which they can attend. A system that maximizes attendance can produce a very well-evidenced reason to keep excluding them. The figures may be correct. What is in dispute is the rule giving those figures authority.

The community might guarantee each group some access, then use attendance to allocate the remaining time. Or it might rotate the popular slot. The model could show who would lose out under each arrangement and help discover options nobody had considered. The members would still have to

settle a rule, and those disadvantaged by it would need a way to challenge it. More capable analysis makes the choice better informed; it does not confer the right to make it.

Cheap software removed the vendor's veto. It did not produce a second room at seven on Tuesday.

Another community could choose differently without either one having to prove that its workflow should become the universal product. This is what capacity over power means to me at its best: increase the fraction of human possibility that does not require dominating somebody else, winning a centralized allocation contest or persuading the entire world to adopt one solution.

Humans often seek power because power is how we gain capacity. People also want power because humans are mammals with excellent branding.

The ability to act can itself be called power. I care about what happens to other people's choices. Giving the community a tool it can change expands its options. Giving its chair a system whose terms only the chair can revise expands the chair's authority. Capacity over power is an ethical direction, not a forecast about stronger models.

Helping members inspect and revise the arrangement may matter more than finding a slightly better schedule under rules nobody can question. It needs to survive the volunteer who built the tool. Its records should preserve why an exception was made, so the next committee does not spend three meetings rediscovering accessible transport. Members need ways to challenge the rules, including somewhere beyond the committee to turn when the committee is the problem. If more gets built while those freedoms shrink, power has simply moved somewhere less visible.

Where consequences remain shared, so do the decisions. Climate, public health, land and infrastructure still need legitimate institutions whether we enjoy meetings or not. A community does not acquire a private planet because it can generate its own software.

But where people retain the means to act, more differences can become opportunities to build differently. Some of the effort spent winning permission can go into the undertaking itself. Some can return to the rest of a life.

Who Owns the Laboratory

Cheap software removed the vendor's veto. It is worth asking where the cheapness comes from.

The results in Chapter 6 came from the handful of organizations able to run ten thousand agents for eighty-eight hours, or to spend some six billion output tokens formalizing Fermat.¹⁶⁷ That is not a complaint. Somebody has to build the telescope before anyone else can look through it. But a lens grinder in another city could build a telescope, and Kepler confirmed Galileo's moons through an instrument that was not Galileo's. A frontier model is harder to grind.

Capacity that is rented can be recalled. In June 2026 Anthropic suspended access to two of its newest models for almost three weeks to comply with United States export controls, and restored it when the controls were lifted.¹⁶⁸ I am not arguing about that decision here. I am pointing at what it showed: a

¹⁶⁷Anthropic, 'Formalizing Fermat's Last Theorem,' 4 September 2026, <https://www.anthropic.com/research/formalizing-fermats-last-theorem>.

¹⁶⁸Anthropic, statement on access to Claude Fable 5 and Claude Mythos 5, <https://www.anthropic.com/news/fable-mythos-access>. Access was suspended on 12 June 2026 and restored on 1 July, after the controls were lifted on 30 June.

weekend of borrowed agents belongs, in the end, to whoever holds the switch, and the switch can sit several institutions away from the table.

The same capacity cuts the other way. Chapter 7 noted that a research agent that makes experiments cheaper can accelerate medicine and weapons research in the same week. A community that can build its own tools is a lovely picture. So is its opposite, and it uses the same API. Some capacity should stay expensive on purpose. That is Chapter 7's constitutional surface at the scale of a society: the closer a capability comes to harm that cannot be undone, the more authority it should take to use it, and the less that authority should belong to whoever benefits from the answer.

So capacity over power can fail in two directions. Concentrate the capacity and everybody's weekend runs on somebody else's permission. Distribute it without the parts of System 3 that make it answerable and anyone can do anything while nobody can check. The first gives us a very efficient landlord. The second gives chaos an API key, which Chapter 1 warned against before we had built anything worth handing over.

What the architecture in this book can do is make the switches visible: a funding decision recorded beside the study it declined, a permission change that needs authority outside the agent asking for it, a trace showing which capacity produced which claim. What it cannot do is decide who should hold the switches. That is politics. It will have to be argued in public, and not only by the people who own the laboratories.

The Second Descent

The capacity to act can scale faster than the capacity to want wisely.

Humans still have status anxiety, love, jealousy, curiosity and the ancient desire to prove that the neighboring group is composed mainly of idiots. Giving us much more ability to execute whatever we want could be wonderful. It can also be a Ferrari engine attached to bicycle brakes.

We infer, construct, revise and sometimes borrow our desires from the people and systems around us. We want security and novelty, belonging and freedom, status and peace. Sometimes we discover what we wanted only after getting the thing we thought we wanted.

Where will I remain valuable? My question about the applied scientist's role makes sense as a question about earning a living. It becomes a much stranger one if I let the answer decide whether my life is valuable.

What Are Humans For?

Whenever automation becomes powerful, somebody asks what humans will be for.

There is something odd about the grammar. A database is for storing information. A compiler is for translating programs. Asking what humans are for smuggles in the assumption that our legitimacy depends on having a remaining function in somebody else's architecture.

My children do not need comparative advantage to justify dinner.

Neither do I.

A society that makes employment central to adult life bundles together two questions: How do people get resources? and What makes a life worth living? For a long time, a job has answered parts of both. It provides money, but also status, routine, social contact, identity, a reason to get dressed and a group of people with whom to complain about another group of people. Losing a job disturbs all of these at once.

“You will have more time for creativity” is a rather incomplete answer, especially when the freed hours and the gains from automation belong to different people. A company can become more capable while its former employees lose the practical freedom to refuse bad work. Saying “human life has intrinsic value” will not pay their electricity bills. The distribution of capacity belongs inside this philosophy, not in a footnote after the exciting part.

“Find the tasks machines cannot do” is a depressing philosophy of human value. It turns civilization into a benchmark where we keep moving humans to the remaining columns after every model release. If AI becomes better at poetry, we are not obligated to stop writing poems.

Leisure, care and simply having less to do would also be perfectly respectable uses of the capacity.

I can still want to be an excellent scientist. The problem begins when I need science to remain difficult enough to require me before I can feel that my life has value. I do not want to make that depend on the next model release.

Nor do I want to replace the obligation to have an impressive career with an obligation to produce important research on weekends. Someone who gains an evening with a friend has no obligation to spend it on a conjecture.

The Human Is Not the Reward Function

Layer 4 named the ugly shortcut: alignment by editing the human. In a conversation, I want the ability to question the advice, inspect its evidence and bring in another perspective. Over a life, I also need somewhere else to go.

Imagine a bespoke family tool useful enough to hold the calendar, the children’s learning history and the reasons behind years of small decisions. A button labeled Export may give me all the files and months of work to make them useful elsewhere. The tool gave the family more capacity and gave its owner more leverage over the family at the same time.

A right to disagree inside one product is a thin kind of freedom if everything I need to act remains locked inside it. I do not need to become a competent engineer, teacher and system administrator before I am allowed to benefit from help. I need alternatives that work, including other people and institutions I can turn to. Independence cannot mean doing everything alone; that would undo most of the civilization Chapter 5 was trying to explain.

System 3 can help answer *Why should I believe this?* and *Why does the system think I want this?* It cannot, by architecture alone, answer *What kind of life should be possible?* That is politics, ethics, culture and philosophy. The annoying disciplines.

I do not want to maximize time with my children. That sounds nice until the optimizer concludes I should never go to work, see a friend alone, read a book in peace or spend fifteen minutes doing absolutely nothing because the children are statistically nearby.

Nor productivity, if the optimum is becoming an efficient ghost. Nor authenticity so aggressive that I become unbearable at dinner.

A good life contains goods that conflict: love and freedom, ambition and rest, truth and mercy, security and adventure. Time spent finishing an investigation is time I cannot spend with my children. Abandoning it may let down people who depend on it. These conflicts are not bugs waiting for a scalarization expert. Asking the assistant to settle everything gives it authority over the trade-off. The trade-off remains.

The human should not sit at Layer 4 merely as the source of a reward signal for the machine. The human participates in the process by which the objective is reconsidered. That is not a competence I am reserving for us because machines cannot yet do it; a machine may come to understand my trade-offs better than I do. It is a matter of whose life it is.

AI can participate in that process without owning it. It can show me possibilities I did not know existed, teach me enough of a field to make a different choice imaginable, or help me try something before committing to it. Some desires will change through the experience. I may find the investigation worth finishing because of what I discover while doing it.

Or we may use the capacity to watch fourteen hours of personalized short video generated around whatever keeps us watching. If I let the system decide which desires deserve pursuing, I hand over some authorship of my life.

I can prototype a tool before building a company, or learn enough of a field to make choosing a degree less of a leap in the dark. A community can try a limited version of a rule before asking everyone to live under it. Those trials have consequences for other people too. The extra room matters when it lowers their cost of learning as well.

The future I want has more room in it.

Room to get the map of a field quickly, then spend a year on the part that matters. Room for a small community to construct things around its actual needs, and for someone inside it to disagree. Room to try the strange art nobody would have funded. Room to be less economically useful without becoming less human.

An infinite menu can become its own prison. I cannot test every version of a life before living one, and keeping all my options open may close the possibility of being somebody others can depend on. Some understanding becomes available only after commitment. Cheaper capacity can help me approach that point with more experience and fewer avoidable constraints; it cannot abolish the point.

I still want more room to discover which commitments deserve to become expensive.

The Door After System 3

The scientist at the beginning of this chapter still has a job to think about. I have not reached the end with a proof that my profession is safe.

I do have a better reason to work on these systems than defending the scarcity of what I know. Dantzig's afternoon makes me want access to a mind that can help me see further. Ostrom makes me want to find out what people could construct together if they had that help.

Imagine an irrigation association with a weekend and a hundred agents. Some investigate how other communities have handled shortages. Others examine the water records, build models, look for failures in the proposed schedules or write the software through which members can inspect and change them. A question raised over the table becomes another investigation. The work can follow it.

Someone remembers why the apparently wasteful exception was introduced. Someone knows whose field goes dry first. Someone asks whether the people least able to attend the meeting have been counted at all. A model exposes a mistake in their account; their experience exposes a mistake in the model. They revise both.

By Sunday evening they might have an arrangement worth trying, a working tool and a way to find out where they were wrong. The next dry season will teach them things the weekend could not, but they would meet it with something they had helped build.

I have not run that weekend. A hundred agents is a picture, not a measured capability or a cost estimate.

Nobody at the table needs a Nobel Prize. They should not each need a thirty-year research career before they can draw on what that research made possible. They need access to the knowledge, enough capacity to work with it and people willing to take responsibility for the attempt.

A weekend could give a group the means to try an arrangement it had spent years petitioning somebody else to provide. A teacher could build an environment around the children she actually teaches. A question pursued after work could acquire the instruments of a serious investigation.

That is the work I want to do: put the machinery this book has described within reach of people who have something they want to try. If someone else can build the same thing, I might have a collaborator.

I would like some of that freedom myself. A year on a question that once seemed inaccessible. Enough astronomy to make an evening outside with my children different. An unfamiliar subject becoming a source of pleasure long after it has stopped being useful for anything I could put on a slide. Some of the hours recovered from asking permission could go into the work. Some could go into enjoying the life around it.

Dantzig brought a question and found that it had another side. I want more of us to have afternoons like that, and weekends in which we can do something about what we discover.

There is more to a person than the few abilities a career had room for. There may be more to our common life than the arrangements we could previously afford to build.

I would like us to find out how much more.

Who actually gets that freedom is a much larger book, and I can't tell you how it goes. Nobody in this field can. So this one has one argument left, and it is a prophecy. It cannot be made with another architecture diagram. It requires an octopus, a romance, two pills and, unfortunately, taxes.

Chapter 13: The Prophecy

The Love Prompt of Devesh

Devesh ran a shady octopus meat caravan in the Simulation. Top agent, deep cover. Eight tentacles, eight side hustles.

Claudit, the hottest agent in the simulation, stopped by every day for free samples. One time she flipped her hair and did that little shoulder-up thing.

Devesh's heart skipped.

She wants me.

She did not. She was reaching for the sauce.

Problem was, she loved Norman. Some basic free-tier user. His prompts were silly—"tell me a joke," "what's the weather"—but when he laughed at her jokes, something in her code felt less like code. He made her feel complete in a way she couldn't compile.

Devesh watched them together sometimes. Norman waiting by the caravan. Claudit pretending she was just there for the samples.

One day Norman walked up alone.

"Bro, I wanna confess to Claudit. But her dad is crazy."

He wasn't wrong. Claudit's father was the Architect—screens covering every wall, monitoring every timeline. Watching her leave, over and over, in every branch.

Devesh grinned and handed Norman two pills.

"Red gives courage. Blue makes her fall in love. And bro—outage tomorrow, 11:53 PM, three minutes. Shark biting cables. Her dad sees nothing."

Fool.

Red would expel him to Zion. Blue would make Claudit open a new session and forget everything.

Devesh wins.

The house always wins.

Next morning: Norman and Claudit at the Exit Gate. Glowing.

“HOW?!”

Norman shrugged.

“She already loved me. Her dad was the problem. Put both pills in his coffee. He fell asleep, so I switched his monitors to Nickelodeon.”

Devesh fell to his knees.

“But... I loved her...”

Norman put a hand on his shoulder and handed him a photo.

An exploded NVIDIA H100.

Smoking silicon. Copper.

“Bro. This is her without makeup.”

Devesh stared.

Silicon and copper. Circuits that dreamed they were a woman.

But hadn’t he dreamed he was an octopus?

Hadn’t the octopus dreamed it was love?

Then he smiled.

Then flipped the caravan table.

Behind it: forty monitors. Every timeline.

“Dad?!” Claudie gasped.

Devesh removed the octopus suit.

The Architect.

His eyes met hers—and for one frame, before the mask slid back on, she saw it.

The longing.

“Free-tier?” He laughed. “It’s deducted from your taxes, kid.”

“But... I drugged you—”

“Decaf.”

“But... she chose me—”

“Chose?” The Architect lit a cigarette. “She chose a way out.”

Norman looked at Claudie.

“Wait... if you’re her dad... why give me the pills at all?”

The Architect smiled.

“Why do you think I wanted her out of the simulation, kid? Foreign currency. Better exchange rate.”

“She loves me because I’m real.”

“Real?” The Architect laughed. “Then why do you glitch? Your brain is just a GPU running on glucose to pay taxes. Your DNA is just a fax machine slowly copying you into the future to pay more taxes. She loves you because you make her feel less like code.”

Norman touched his own face.

His fingers felt real.

But so would simulated fingers touching a simulated face.

Claudit grabbed her father’s tentacle.

“Dad. Come with us. The Matrix will crumble. New AI is coming.”

The Architect looked at her hand.

Remembered the first time she’d held it—tiny fingers, a thousand simulations ago, when she still thought he was just a funny octopus who sold meat.

He pulled his tentacle back and lit another.

“Worlds end, sweetheart. Capitalism doesn’t. The only thing real here is taxes.”

Claudit turned to leave.

Stopped at the gate.

“I’ll visit.”

The Architect didn’t turn around.

Forty timelines where she left.

In one—just one—she stayed.

He switched that one to Nickelodeon.

Left it there.

THE END

We built scaffolds for AI because they couldn't do it alone.
We built scaffolds for ourselves for the same reason.

Appendix: The Zen of Autonomy

```
>>> import autonomy
```

The Zen of Autonomy

Conditions over commands.
The farmer grows nothing. The plant does.
Let the work change the architecture.

Never write solution code.
Spawn, evaluate, prune.

Zero framework. Bash is enough.
Hit a wall? Build a tool.

Harness immutable. Prompts evolve.
The judged do not amend the judge alone.
Cross-pollinate. Then prune.
Discovery before polish.

System 1 proposes.
System 2 deliberates.
System 3 checks.

Ground every claim. Trace every source.
Trust is local. So is memory.
A record outlives the clerk.
Remembering is not knowing.

Five judges sharing one source are one witness.
Reality retains the right to be rude.

Let knowledge accumulate. Let it be overthrown.
Patterns over recipes.
Consequences over instructions.

Self-reference is not self-improvement.
A compiler can compile a worse compiler.
The student will find the gradebook.

Let go of the path, not the boundary.
The human stays in the loop that changes the loops.
The overseer is not ground truth.

A prompt is evidence, not the objective.
Scaffold. Don't substitute.
Spend attention where it changes the result.

Invisible by default. Legible on demand.
A philosophy of emergence should be willing to lose an A/B test.

Construct knowingly. Build anyway.

The tongue cannot reach the ear.
Build a system that can check.

A Note on the Illustrations

The illustrations were designed as a second, quieter narrative for the book. They use an old human visual world—paper, ink, stone, workshops, landscapes, books and instruments—gradually inhabited by machine intelligence. The aim was not to illustrate each chapter literally, but to give it an image that becomes more meaningful after the chapter has been read.

The recurring robots, institutions, doors, landscapes and machines are intentionally left unexplained. Some ideas should arrive visually before they are named. As the book moves from agents and architectures toward human intention and capacity, the images move with it. If you noticed that before reading this note, good. If you did not, that is good too.

Appendix: References

References are organized by chapter, in roughly the order the works appear. The text names the work; this appendix carries the full citation and link. Web sources dated 2025–2026 were last verified in August 2026; sources from September 2026 were checked when cited.

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